Concept momentum and investor attention: Evidence from China's A-share market

Abstract. Overlapping concepts link stocks across industry boundaries in China's A-share market. Starting from the relationship between concepts' past and subsequent returns across horizons, we examine whether concept strength and stock–peer synchronization help select members with higher subsequent short-horizon returns, and whether investor attention can explain this return pattern. Using 5,228 stocks from December 2016 through August 2026, we combine recent peer returns with stock–peer return correlations in a ranking with transparent rules. Across 108 combinations of formation windows, holding horizons and list sizes, the joint ranking yields positive mean gross return spreads over the corresponding research pools; 65 remain significant after Holm correction across the declared 540 comparisons. Under the reference specification of five-day strength, ten-day synchronization and the top 20 stocks, the mean three-day spread from the next open is 0.2832 percentage points; the industry–size-matched spread is 0.1681 percentage points (95% interval, 0.1039–0.2311), also significant after multiplicity correction. Adding synchronization improves strength-only selection, whereas the increment over synchronization alone is not statistically established. Concept features improve some prediction tasks beyond conventional characteristics; the increment for the reference task is not statistically confirmed. Across the tested parameter range, the member-selection results support short-horizon return continuation associated with concepts. In the causal machine-learning diagnostics, neither the adjusted stock-popularity–return coefficients nor the paired specification contrasts survive the declared multiplicity correction. Return evidence from member selection does not identify an attention mechanism.

Keywords: Concept momentum; Return predictability; Investor attention; Quantitative investment strategies; Causal machine learning
JEL classification: G11; G12; G14; G41

1 Introduction

China's A-share market groups companies into concepts associated with technologies, policies and economic narratives. These labels cross industry boundaries, and a company can belong to several concepts at once. A strong concept does not identify which of its members will subsequently earn higher returns. We examine whether a stock's recent alignment with its concept peers helps resolve this selection problem.

Related-security predictability motivates this question at two levels. At the category level, evidence allowing for overlapping memberships documents monthly Chinese concept momentum, stronger momentum among less-attended concepts, and stock-level lead–lag relations within concepts[1]; stock momentum also varies with comovement with size and value styles[2]. Across firms, customer–supplier links predict returns[3], while evidence from China compares news-co-mention momentum with economic links that include concepts[4]. Separating peer-average characteristics from a firm's deviations within its peer group provides a complementary approach to return prediction[5]. These studies connect group information to individual returns; our question is how that information guides short-horizon selection among overlapping concept members.

To answer this question, we distinguish the information in recent synchronization, the rules used to aggregate it and the increment over conventional predictors. The holding horizons examined range from 1 to 250 trading days, with returns from historical trading simulations serving as a reference for detailed comparisons. This differs from the three- to twelve-month stock momentum studied by Jegadeesh and Titman[6] and the short-lag dependence in monthly returns documented by Jegadeesh[7]. Concept momentum here refers to testing the return continuation of strong concepts through their selected member stocks.

Investor attention offers an economic interpretation of concept-based trading (Fig. 1c). Category-based investment and limited-attention models link grouping to comovement and predictability[8,9], while laboratory evidence shows that attention constraints can shift information acquisition toward categories[10]. Observed attention measures, however, capture particular activities: searches[11] and platform-related buying[12] need not have the same return implications. We therefore treat Wind's individual-stock popularity as a separate observable quantity and examine its associations with trading participation, returns and concept comovement. These associations are used to assess whether attention can explain the observed selection pattern.

The empirical design follows the selection problem. Recent peer returns measure concept strength, and stock–peer return correlations measure synchronization. Their rank-based combination uses no weights fitted to future returns. Peer portfolios use a leave-one-out construction to prevent a stock from mechanically entering its own peer return; measuring returns from the next open separates score formation from outcome evaluation in time. Industry–size matching tests whether composition accounts for the return spread, while component removal and alternative aggregation distinguish the roles of input information and combination rules. Fixed model comparisons assess the increment over conventional characteristics, drawing on machine-learning prediction in empirical asset pricing[13]. Popularity analyses examine the attention interpretation through separate measurements; retrospective accounts show allocation outcomes under a different implementation framework.

2 Data and empirical design

2.1 Sample and empirical specifications

The historical sample contains 5,228 stocks and 8,835,000 stock–date observations on 2,353 signal dates, from 22 December 2016 through 31 August 2026. The daily research pool grows from about 1,900 stocks in 2017 to 5,100 in 2026. Daily membership snapshots contain about 150 concept labels in 2017 and 385 in 2026; pool stocks have a median of four scored memberships (Supplementary Table S1). The pool reflects research-data availability rather than a verified historical investable universe.

The specification grid crosses six F/W window pairs, three holding horizons and six list sizes, giving 108 specifications for each ranking rule and testing sensitivity to these choices. The other comparisons use five-day concept strength and ten-day stock–concept synchronization, denoted F5/W10, together with a Top20 list as a common reference specification. Returns are evaluated over three complete trading days from the next market-day open. Within each paired comparison, competing rules share eligible stocks, signal dates and return endpoints.

2.2 Theme relations and the ranking rule

Concept strength F compounds recent daily returns of a stock's concept peers. Synchronization W is the correlation between the stock's daily returns and those of its peers, with the focal stock excluded from the peer portfolios. Both quantities are ranked across valid stock–concept membership records. FW multiplies the ranks within each membership and averages the products across a stock's memberships (Fig. 1a). A high score thus identifies stocks connected to strong concepts whose recent movements they have followed closely.

2.3 Information timing and attention measures

On signal day t, we use the latest membership snapshot dated no later than the preceding market day and prices through the t close. The ranked list is fixed before returns are attached, with no replacement selections when future returns are missing. We measure an H-day return from the adjusted open at t + 1 to the adjusted open at t + H + 1 (Fig. 1b), so the illustrative three-day reference ends at the t + 4 open; this design is denoted γ. Although the score uses lagged membership records, eligibility for the common comparison pool still requires valid scores under synchronous membership and both formation-window conventions (Supplementary Section S3). All source observations and prescribed outcome endpoints must fall on or before 31 August 2026; Supplementary Table S2 gives the mature sample by horizon. Recorded snapshot dates are not equivalent to the time of first public availability (Supplementary Section S1).

Table 1 separates membership timing from return timing while retaining prices through t close in every cell. The within-clock comparisons isolate the membership convention; the within-membership comparisons isolate the return clock. Close-based cells are diagnostic references: because the score already uses the signal-day closing price, trades cannot be assumed executable at that same close.

Wind total popularity (T), watchlist popularity (S) and attention stickiness (K) record individual-stock prominence within the vendor's measurement system but do not directly observe theme-level attention. Their transformations separate persistent levels from more recent changes, with samples determined by field coverage and historical-input requirements (Supplementary Sections S12–S13). Concept-level profiles and implementation accounts have their own construction rules and samples, described in Supplementary Sections S5 and S17.

3 Results

3.1 Short-horizon returns within overlapping concepts

Under the F5/W10 reference specification, the Top20 list has a mean three-day gross return of 0.3969% across 2,349 mature signal dates, compared with 0.1136% for the equally weighted research pool, a difference of 0.2832 percentage points (Fig. 2a).

To assess how the result depends on the formation windows, the sensitivity grid crosses six F/W pairs, F3/W5, F3/W20, F5/W10, F10/W20, F20/W10 and F20/W50, on the market-trading-day calendar. Within each pair's common eligible pool, we compare five scoring rules across three holding horizons, H3, H5 and H10, and six list sizes, Top5, Top10, Top20, Top30, Top50 and Top100. All 108 FW-minus-pool means are positive, and 65 pass Holm correction in the declared 540-comparison family (Fig. 2). These are related sensitivity comparisons: settings share stocks and dates, and longer windows can change the eligible pool. Concept-level formation–holding profiles provide a complementary view of its time scale (Supplementary Fig. S1 and Supplementary Data S9).

3.2 Industry, size and variation over time

If selected stocks differ systematically from the research pool in industry and size composition, the spread may partly reflect those differences. We match each frozen Top5, Top10 and Top20 list to weighted group means that reproduce its industry composition, capitalization-decile composition or both (Fig. 3). The procedure retains all ten size groups and includes selected stocks in their comparison groups; it changes the benchmark rather than the selections.

With industry and size matched jointly, the mean spreads are 0.2053, 0.1882 and 0.1681 percentage points for Top5, Top10 and Top20, respectively. The Top20 spread has a 95% time-block interval of [0.1039, 0.2311] percentage points and a Holm-adjusted P value of 0.000900. All nine industry-, size- and jointly matched comparisons are positive and survive adjustment within this declared family. For Top20, the industry-only and size-only spreads are 0.2075 and 0.2522 percentage points. The spread therefore does not arise entirely from the observed industry and size composition.

Random portfolios matched on industry and size give a similar result. For Top20, the actual period mean exceeds the median of 1,000 matched portfolio means by 0.1680 percentage points. Retaining the intended list denominator and assigning zero to missing outcomes also leaves the joint-matched spread at 0.1681 percentage points. The random-portfolio distribution describes variation across matched draws; the confidence interval above describes uncertainty over time.

The spread varies across calendar years (Supplementary Fig. S2 and Table S6). Among the nine complete years from 2017 to 2025, the pool-relative mean is positive in seven and the joint industry–size mean in eight. Pool-relative spreads are −0.0045 percentage points in 2022 and −0.0704 in 2024; the joint-matched spread is −0.0317 in 2024.

In the fixed 2017–2021 and 2022–31 August 2026 periods, pool-relative spreads average 0.3364 and 0.2270 percentage points, and joint industry–size spreads average 0.1919 and 0.1426. All four contrasts pass their separate Holm4 families with 20- and 40-date blocks. Disjoint three-day formation offsets retain positive historical means, and the matched spread also remains positive in the common attention-data window. Supplementary Section S6 reports the date support, longer-block results, partial-year means and common-window estimates.

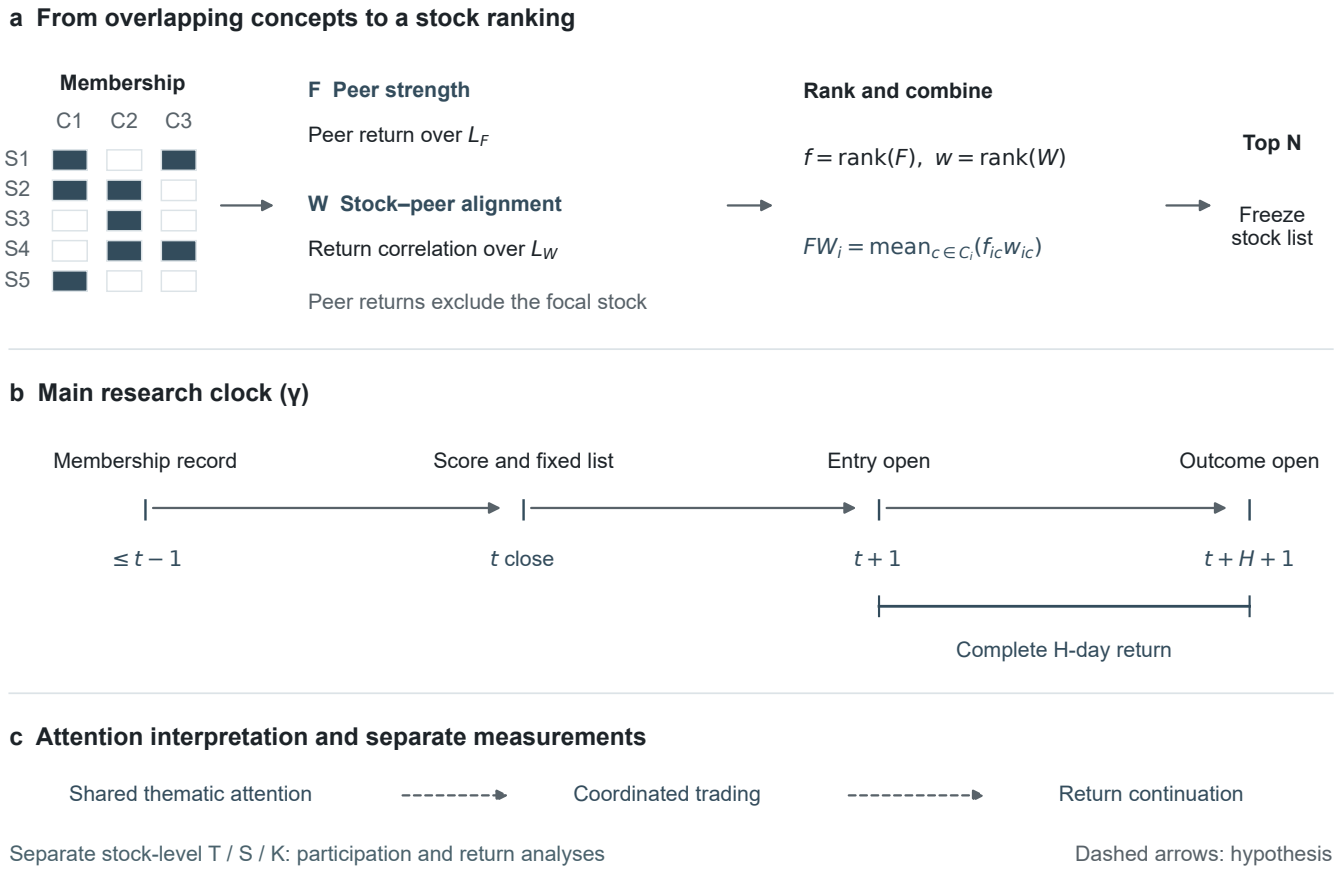
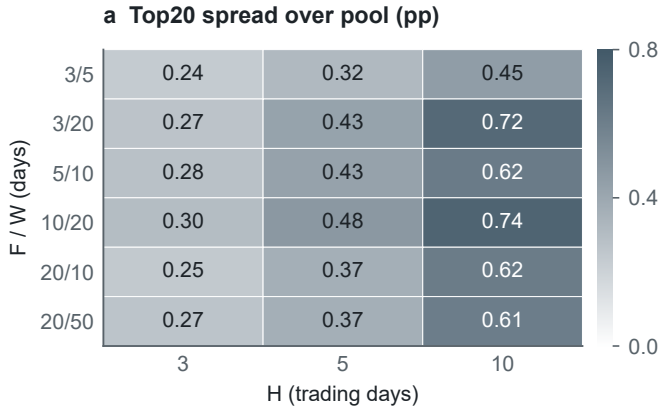


Figure 1. Concept-based selection, information timing and the attention hypothesis. a, C1–C3 and S1–S5 illustrate overlapping concepts and their member stocks. F compounds daily peer returns over L_F days; W correlates stock and peer daily returns over L_W days. Peer portfolios exclude the focal stock. Normalized ranks across membership records are multiplied within each record and averaged within each stock to obtain FW; the Top N list is then frozen. b, The main γ design uses membership records dated no later than the preceding market day and prices through the signal-day close. Complete H-day returns run from adjusted open t + 1 to adjusted open t + H + 1, with endpoints no later than 31 August 2026. c, Dashed arrows denote the proposed attention mechanism. Recorded total popularity T, watchlist popularity S and attention stickiness K are analysed separately and do not enter the price-based score. Solid arrows in a–b denote computation or chronology.

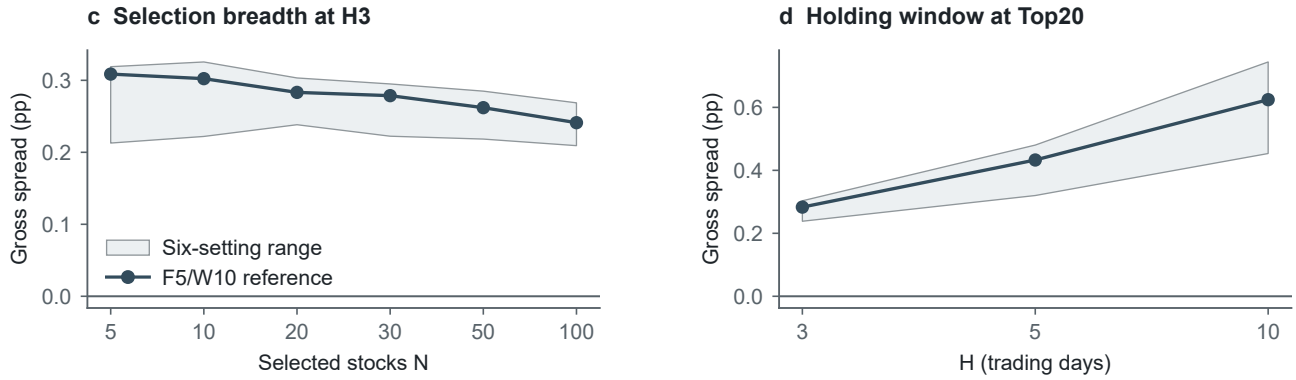
Table 1. Membership timing and return timing. Here t is a signal date on the market calendar; H counts market trading days. O and C denote adjusted open and close prices. A recorded snapshot date is distinguished from its unverified first public availability.

Design	Membership record used for scoring	Score price information	Return interval	Comparison role
α	Date no later than t	Through close t	$C_t \to C_{t+H}$	Same-day-membership, closing-return reference
β	Date no later than t	Through close t	$O_{t+1} \to O_{t+H+1}$	Same-day-membership, next-open evaluation
γ	Date no later than the preceding market day	Through close t	$O_{t+1} \to O_{t+H+1}$	Lagged-membership, next-open evaluation
δ	Date no later than the preceding market day	Through close t	$C_t \to C_{t+H}$	Lagged-membership, closing-return reference



b H3 / Top20: direct differences (pp)

	Pool	F only	W only	A	P
3/5	0.238*	0.193*	0.079	-0.002	-0.002
3/20	0.273*	0.224*	0.121	-0.004	0.004
5/10	0.283*	0.229*	0.157	0.001	0.006
10/20	0.303*	0.209	0.150	-0.004	-0.007
20/10	0.250	0.218	0.120	-0.009	-0.002
20/50	0.269	0.240	0.082	-0.001	-0.006



* Holm540 P < 0.05. Envelopes: minimum–maximum across six settings.

Figure 2. Short-horizon parameter breadth under lagged membership. Six F/W pairs (3/5, 3/20, 5/10, 10/20, 20/10 and 20/50) are evaluated at H=3,5,10 and N=5,10,20,30,50,100. a, Complete-H Top20 FW-minus-pool gross spreads in percentage points, on a white-to-deep-blue sequential scale fixed from 0 to 0.8 pp; shading encodes magnitude. b, Each number identifies the direct H3/Top20 comparison for its F/W row; stars denote Holm540 P<0.05. c,d, F5/W10 is the solid blue reference line; grey envelopes are the minimum–maximum over all six settings, not confidence intervals. Category colours in the line panels identify series, not numerical sign. Each F/W setting has its own historically eligible pool; envelopes include eligibility changes. All 108 pool-relative means are positive and 65 pass Holm540 correction. Inference uses 100,000 paired circular 20-date-block replicates. Observations end on 31 August 2026. Values are complete-H gross differences, not daily returns or net portfolio performance.

3.3 Membership timing and return measurement

The timing of membership records and the return measurement clock may each affect the spread; the four information-timing designs separate the two. With market-calendar scores, the F5/W10 H3 spreads over the eligible pool are 0.3170, 0.2766, 0.2832 and 0.3213 percentage points for α, β, γ and δ, respectively (Table 1). All four use 2,349 common mature signal dates from 22 December 2016 to 25 August 2026. Within each membership convention, the close and next-open outcomes evaluate exactly the same intended selections.

The synchronous-minus-lagged membership contrasts are −0.0067 percentage points under the open clock and −0.0043 under the close clock. Neither passes the correction. By contrast, the close-reference spread exceeds its next-open counterpart by 0.0404 percentage points with synchronous labels and 0.0380 with lagged labels. Their pointwise 95% intervals with 20-day blocks are [0.0190, 0.0618] and [0.0165, 0.0597], with Holm20 P values of 0.0050 and 0.0076. Both clock contrasts also pass with 60-day blocks and when missing returns are set to zero at the original denominators. The interaction between membership timing and the return clock remains unconfirmed (Supplementary Tables S4 and S5).

Spreads are larger under the close clock. Because the score already uses signal-day closing prices, the close-based results are measurement diagnostics and do not provide evidence of execution at that close. The main selection result remains positive with lagged membership and next-open outcomes.

3.4 Information in the inputs and their combination

The FW spread may arise primarily from one of its inputs. The two-input FW spread is 0.2832 percentage points, while strength-only (F) and synchronization-only (W) selection have reference pool-relative spreads of 0.0544 and 0.1267 percentage points, respectively. FW exceeds strength-only selection in all 108 parameter cells, with 81 corrected rejections. Its difference from synchronization-only selection is positive throughout, but none survives the same correction. The supported increment therefore concerns adding synchronization to strength; the increment from adding strength to synchronization is not statistically supported. The contrast between these two significance decisions is not itself a test of the difference between the two increments[14]. The fixed-denominator sensitivity gives the same pattern of decisions (Supplementary Section S3).

The two inputs can be combined without the membership-level product. Holding both inputs and valid memberships

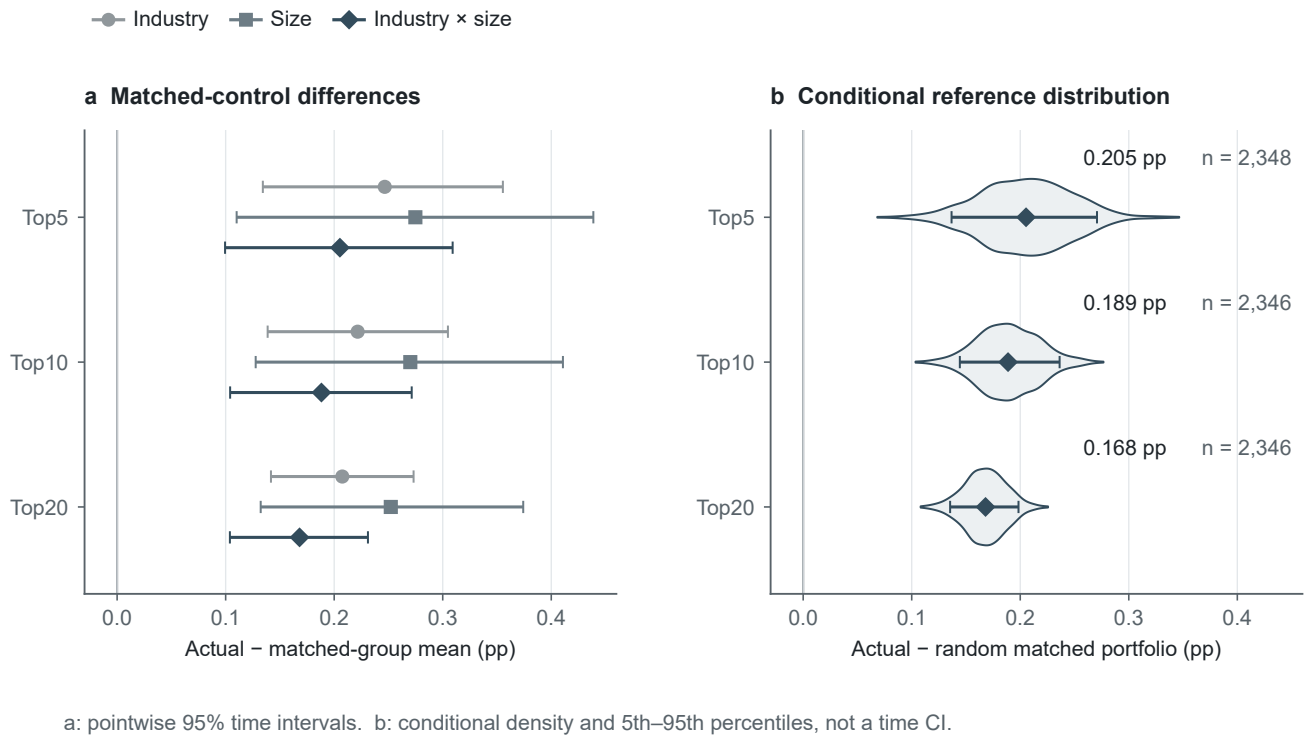


Figure 3. Gross spreads against industry- and size-matched controls. a, Mean paired H3 gross-return differences for Top5, Top10 and Top20; whiskers are pointwise 95% time-series confidence intervals from a circular 20-day moving-block bootstrap with 10,000 replicates. The nine comparisons form one Holm family. b, Kernel-density violins show the full-history actual-minus-random mean gap across all 1,000 saved industry-by-size-matched conditional portfolio draws per selection size, using Scott bandwidth. Diamonds mark medians and inner whiskers mark empirical 5th–95th percentiles. Violin widths are normalized within each selection size and do not encode sample size. These are conditional reference distributions, not time confidence intervals. Joint-matched samples contain 2,348/2,346/2,346 paired dates for Top5/10/20. Actual lists were frozen before outcomes; random controls preserve each list's group counts and may include selected stocks. Differences are gross returns, not risk-adjusted alpha.

fixed, A averages the separate stock-level strength and synchronization scores, while P multiplies their separate stock-level means. If f and w denote membership ranks, the distinction between P and FW is exact:

$$FW_{i,t} = P_{i,t} + \mathrm{Cov}_c(f_{ic,t}, w_{ic,t}).$$

The covariance uses the number of memberships as its denominator. It records whether a stock's relatively strong concept memberships are also those with relatively high synchronization. This is the information preserved by multiplying before averaging rather than afterwards. Accordingly, comparing FW with P asks whether retaining that alignment changes the selection enough to improve outcomes; comparing FW with A additionally asks whether multiplication is needed to combine the two ranked inputs.

The aggregation choice changes few selections. Across all 2,353 signal dates, mean Top20 overlap is 96.74% between FW and A and 97.97% between FW and P (Fig. 4). On mature H3 windows, mean return differences are 0.000604 and 0.006048 percentage points, respectively (Supplementary Table S7). Both pointwise intervals span zero under the main and longer resampling blocks. In other words, the three rules select nearly the same concentrated portfolio from the two inputs. The six-pair F/W grid also shows no corrected aggregation advantage (Supplementary Section S3).

3.5 Prediction conditional on conventional characteristics

Concept strength can coincide with recent stock returns, while synchronization can accompany differences in volatility or trading activity. Paired prediction models therefore test whether concept features provide incremental information beyond these conventional characteristics. Ridge regression and histogram gradient boosting use 19 characteristics and a missingness feature; enhanced versions add FW, F-only and W-only. Paired models share training rows, settings and evaluation pools. Annual predictions from 2020 to 2026 use the preceding three calendar years and matured outcomes (Methods).

The evaluation distinguishes two tasks: ranking the full research pool and selecting its highest-ranked stocks. Daily rank correlation between predictions and future returns, the information coefficient (IC), measures the former; Top20 and Top100 gross returns measure the latter. Better ordering across the pool need not improve its selected tail (Supplementary Fig. S6). Figure 5 reports enhanced-minus-conventional contrasts within each task.

At the reference three-day horizon, both learners have positive Top20 return and raw pool-IC increments, but all four Holm-adjusted P values equal 1. Using every eligible H3 training date also yields no corrected Top20, Top100 or raw pool-IC increment. Boosting's industry-conditioned IC increases by 0.005414 and passes correction under 20- and 60-day blocks (Supplementary Section S9 and Table S8).

At H5, ridge's Top100 return increment is 0.1397 percentage points, compared with 0.1837 for Top20. The direct Top100-

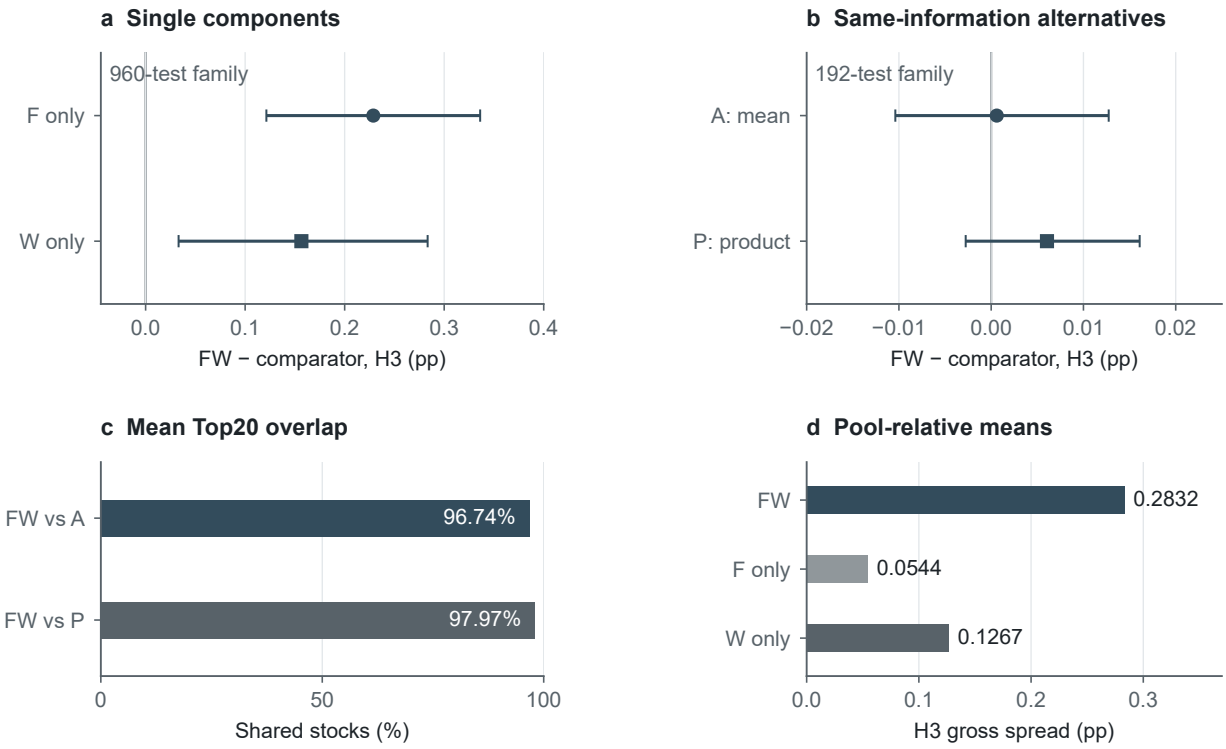


Figure 4. Direct contrasts and the limits of the aggregation formula. a, FW minus either individual component. b, FW minus the arithmetic mean A and product of means P, displayed on a separately labelled tighter x scale. Points are paired H3 mean gross-return differences; whiskers are pointwise 95% intervals from 100,000 circular 20-date-block resamples. The component and same-information comparisons belong to separate 960- and 192-test families, and the intervals are not simultaneous. a/b use 2,349 mature signal dates under the market-day γ, F5/W10, H3, Top20 reference, with every endpoint on or before 31 August 2026. c, Mean Top20 selection overlap across all 2,353 frozen-list dates. d, Descriptive pool-relative gross means for FW, F-only and W-only over 2,349 mature dates; no uncertainty intervals are shown.

minus-Top20 increment is −0.0440 percentage points (pointwise 95% interval with 20-date blocks, −0.1307 to 0.0395; Holm6 P = 1). The direct comparison does not establish a larger increment for the broader list. H5/Top100 support persists across longer blocks and the simulation-precision check; the Top20 decision varies with block length, missingness and simulation precision. Supplementary Section S8 reports the paired list-width and precision analyses.

After subtracting industry mean ranks from predictions and outcomes, three IC increments of 0.004114–0.004903 pass the original 10,000-replicate correction. Ridge H1 remains confirmed under both longer block choices; ridge H5 and boosting H20 depend on block length (Supplementary Section S8).

The component comparison does not confirm an increment from strength beyond synchronization. This motivates a conditional comparison: H3/Top20 boosting models fitted on all training dates retain W and conventional characteristics in both models and add F and FW jointly. The mean increment is −0.0225 percentage points, with Holm6 P = 1 under L20 and L60, leaving a positive increment from this block unconfirmed under the fixed specification (Supplementary Section S10 and Table S9).

Synchronization can reflect exposure to common returns or the magnitude of residual variation. A four-model H3/Top20 comparison adds W to conventional predictors and F, then repeats that contrast with lagged market beta and residual volatility also included. The W increments are 0.0742 and 0.0032 percentage points, respectively. Neither increment nor their direct difference passes the shared six-test Holm correction (Supplementary Section S11 and Table S10). Thus, the improvement from synchronization over strength-only ranking is not confirmed conditional on this richer predictor set.

Daily adaptations of published concept-momentum and comovement rules provide another set of benchmarks. FW has higher mean returns than all four specified daily rules on their matched pool. The monthly mapping gives a different pattern across 103 mature formation months, at most 17 nominal six-month blocks (Supplementary Figs. S4 and S5). These adaptations change the horizon, sorting and portfolio construction of the published strategies [1,2].

3.6 Recorded popularity, participation and concept comovement

To examine the attention interpretation, we turn from concept-member selection to Wind's stock-level popularity measures and compare their temporal relationships with returns and trading participation, then examine concept comovement before and after popularity is recorded. The three fields have distinct profiles (Fig. 6): watchlist popularity is persistent, stickiness decays, and total-popularity innovations show no cor-

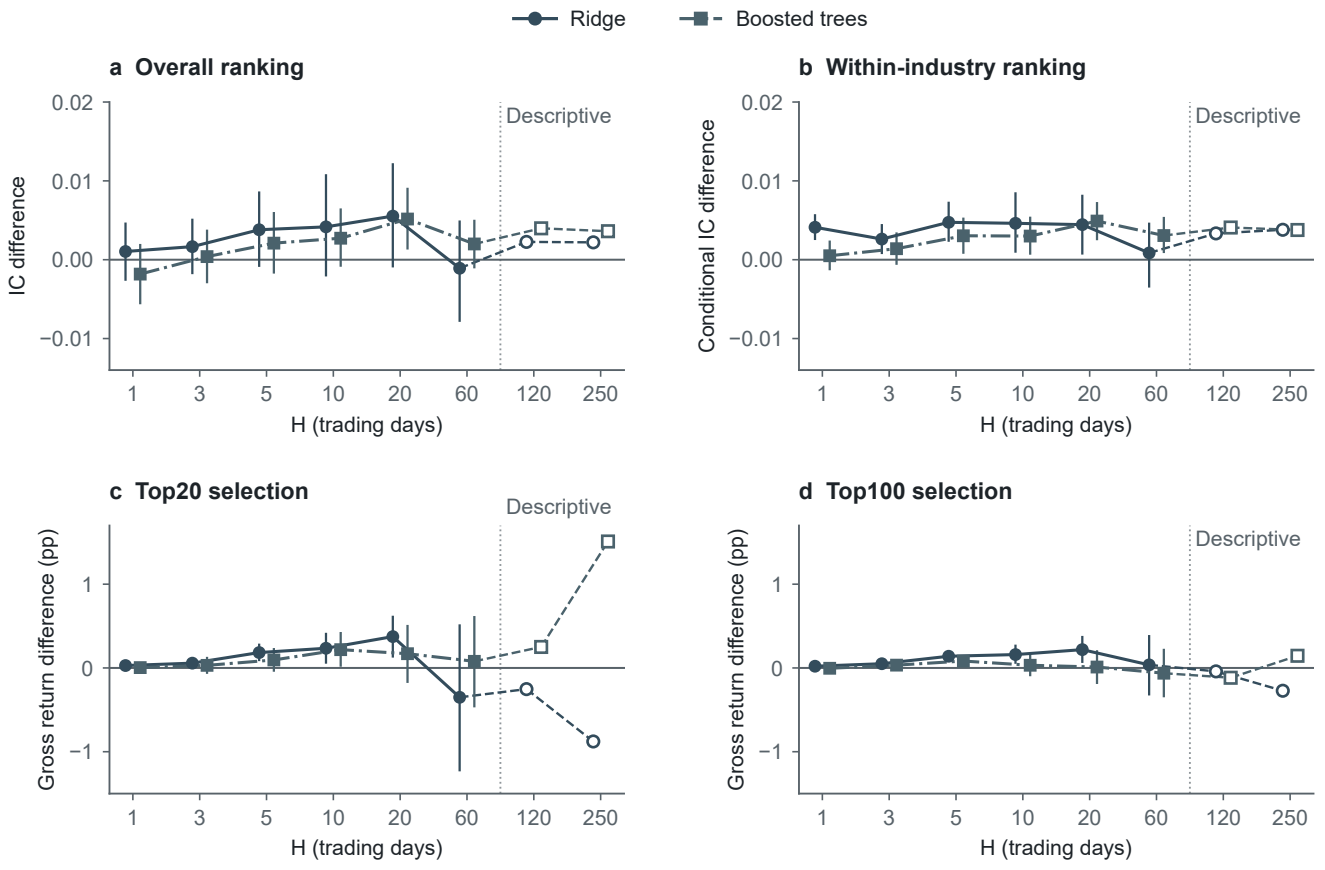


Figure 5. Conditional gains depend on the prediction task. Enhanced minus conventional versions of the same fixed model: a, full-pool rank correlation; b, within-industry rank correlation; c, Top20 complete-H gross return; d, Top100 complete-H gross return. Circles and squares identify ridge regression and boosted trees. Pointwise 95% intervals are drawn as vertical whiskers at supported horizons using the saved 10,000 circular block-bootstrap results with block length max(20,H); intervals are not simultaneous and do not encode the decisions from the 96-test Holm family. Small horizontal offsets separate models at each categorical horizon. Open H120/H250 points and dashed continuations retain results with fewer than 20 nominal blocks for descriptive presentation. Return units are percentage points; IC panels are correlation units. Under this original 10,000-replicate analysis, no primary Top20 or raw full-pool IC increment passes the family correction. Supplementary Section S8 reports the subsequent simulation-precision sensitivity.

rected day-1 or day-3 endpoint response. The endpoint coefficients in Supplementary Section S12 describe persistence of the recorded fields.

The joint temporal criterion is met by eight of 72 attention–trading–horizon combinations, all involving turnover or trading value at 3–20 days (Fig. 6). It requires associations in both directions, lead–lag asymmetry, and within-stratum stock-identity permutations and 60-trading-day historical-misalignment controls. The passing associations have different signs across measures and horizons. None of the return combinations meets this joint criterion, although three one-way attention-to-return paths pass their familywise test: S at one and three days is negative, and T at three days is positive. Supplementary Section S14 reports the full path family and the sensitivity to signal-day trading controls.

The popularity–return comparison also involves two specification choices: the observations used for adjustment and the timing of control measurement. We compare four specifications on the same γ/H3 sample of 217,831 stock-days and 185 dates. Starting from the original role-specific fitting samples, the comparisons align auxiliary support and outcome-label maturity, remove four concept controls, and move the remaining 22 characteristics and industry to records strictly before the attention observation. No coefficient or paired specification difference passes its diagnostic Holm family: 36 coefficient tests and 27 contrast tests cover three declared HAC lengths (Supplementary Section S13.1, Fig. S12 and Table S12). These comparisons assess the sensitivity of conditional popularity–return associations to specification choices, rather than testing specification equivalence or decomposing the FW selection spread.

Concept comovement provides another temporal diagnostic: if attention drives concept-level trading, increases in popularity should be followed by stronger comovement between stocks and their concept peers. The analysis uses 123,530 complete stock-days across 202 dates, with industry benchmarks estimated from a disjoint stock subset and the focal stock excluded from its concept peers. On the same observations, mean correlations with real peers are 0.19861 in the earlier window and 0.19796 in the future window; the pseudo-network means are 0.04782 and 0.04983 (Fig. 7).

The pseudo network preserves coarse strata, and approximately 99.1% of nodes are exchangeable, but it fails the required paired size, degree and liquidity matching on all 202 dates. Its contrasts therefore remain descriptive diagnostics (Supplementary Section S15). The observed comovement pre-

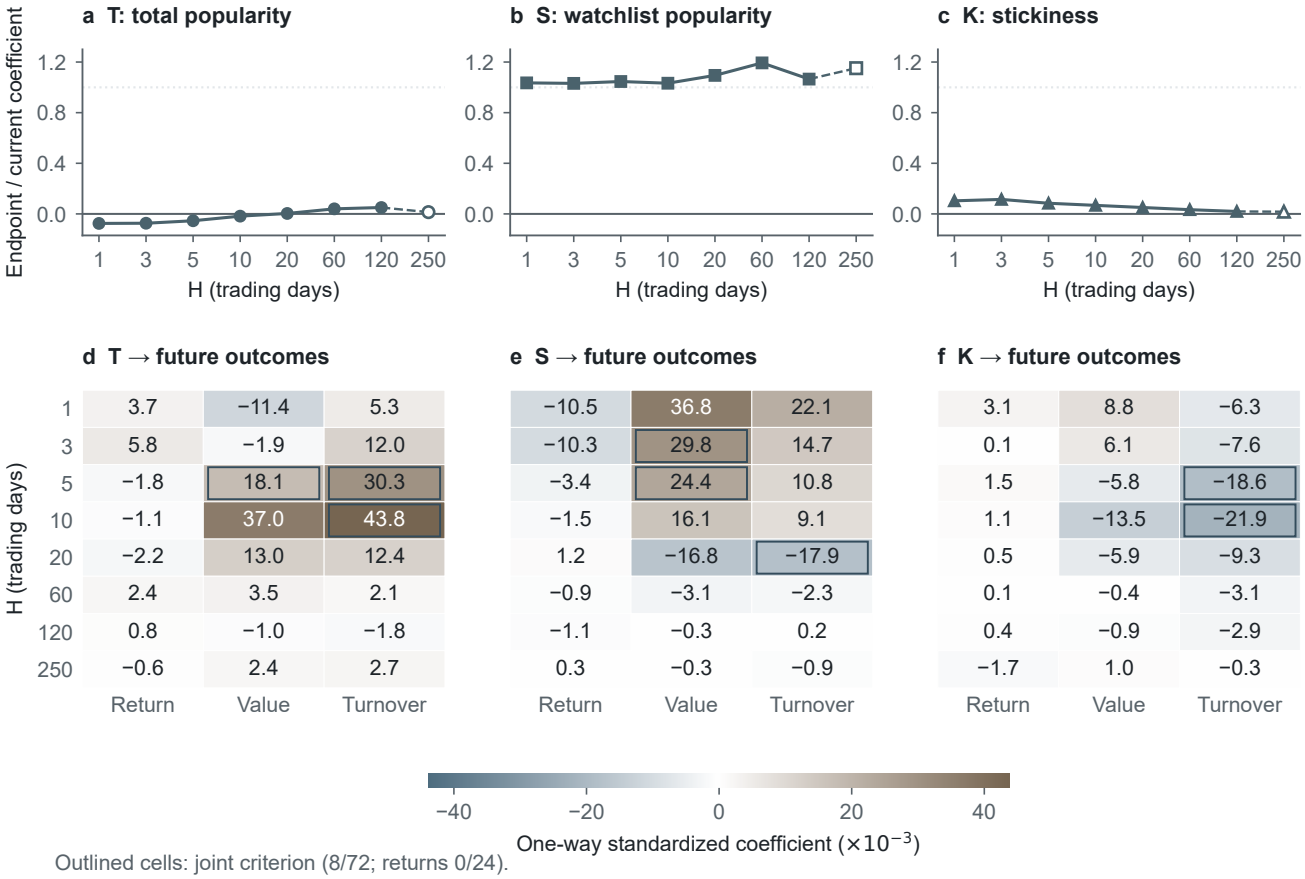


Figure 6. Attention-state persistence and trading participation. a–c show future-endpoint/current-response coefficient ratios for T, S and K on the same horizon-specific support. They are coefficient ratios, not user-retention rates or tests that the ratio equals one. Filled markers identify supported horizons; open H250 points are descriptive under the specified reporting rule. Negative T endpoint coefficients at H1/H3 do not pass Holm24; the positive H120 endpoint does. Primary endpoint inference uses 1,000 circular 120-date-block bootstrap draws and Holm correction across 24 declared cells. d–f show standardized one-way attention-to-future coefficients, with holding horizons in rows and returns, trading value and turnover in columns. Numbers and the colour bar use units of 10^{-3}, with an extra decimal for small values. Outlines identify 8 of 72 combinations meeting the joint bidirectional, temporal-asymmetry and negative-control requirements; they do not denote significance of the displayed coefficient alone. 0 of 24 return combinations meets the joint criterion, while 3 one-way return paths pass the complete-family max-|t| test. All source observations and prescribed endpoints are no later than 31 August 2026; the endpoint analysis uses t+H+1 attention levels, while the participation analysis uses its t+H trading endpoints. Reverse-direction coefficients remain in the source tables. Blue denotes negative values, white zero and muted bronze positive values; slate lines identify attention quantities.

cedes as well as follows the popularity measurement, leaving its source unresolved.

3.7 Retrospective account simulations

The two-input idea is also evaluated in 48 retrospective account simulations from 1 December 2016 to 31 August 2026 (Fig. 8 and Supplementary Tables S13 and S14). The framework simulates trading at 09:55 through staggered sub-portfolios, with concept-return and ranking details that differ from the research construction. Its post-selection filter uses full-session information from the execution day, unavailable at 09:55; these accounts therefore describe retrospective allocation outcomes rather than implementable trading performance (Supplementary Section S17). In the untimed configuration closest to the reference, with three-day holding, F5/W10 and Pearson synchronization, the Top10 account grows to 3.92 times its initial capital: a 15.0% annual compound return after configured commissions and stamp duty. Maximum drawdown is −42.1%, from April 2017 to October 2018. Annual fees amount to about 8.0% of average assets, with annualized two-way turnover of about 129 times average assets. The simulations do not include slippage, market impact or capacity constraints.

Across the 48 archived simulations, compound annual returns range from 12.7% to 33.5% and maximum drawdowns from −59.6% to −14.6%. Paired configurations show how market-breadth timing rules change that risk profile. The one-level and two-level rules use observations available before 09:45 and are applied before the 09:55 trades. They raise annual compound returns by paired medians of 4.1 and 7.3 percentage points, and improve maximum drawdowns by 22.2 and 24.4 percentage points. Zero target-exposure signals occur on approximately 40%–45% of signal dates for scheduled sub-portfolios, while other sub-portfolios may remain invested. These paired comparisons hold the remaining archived framework fixed. Supplementary Section S17 reports the effect of removing the full-session exclusion across the same 48 configurations.

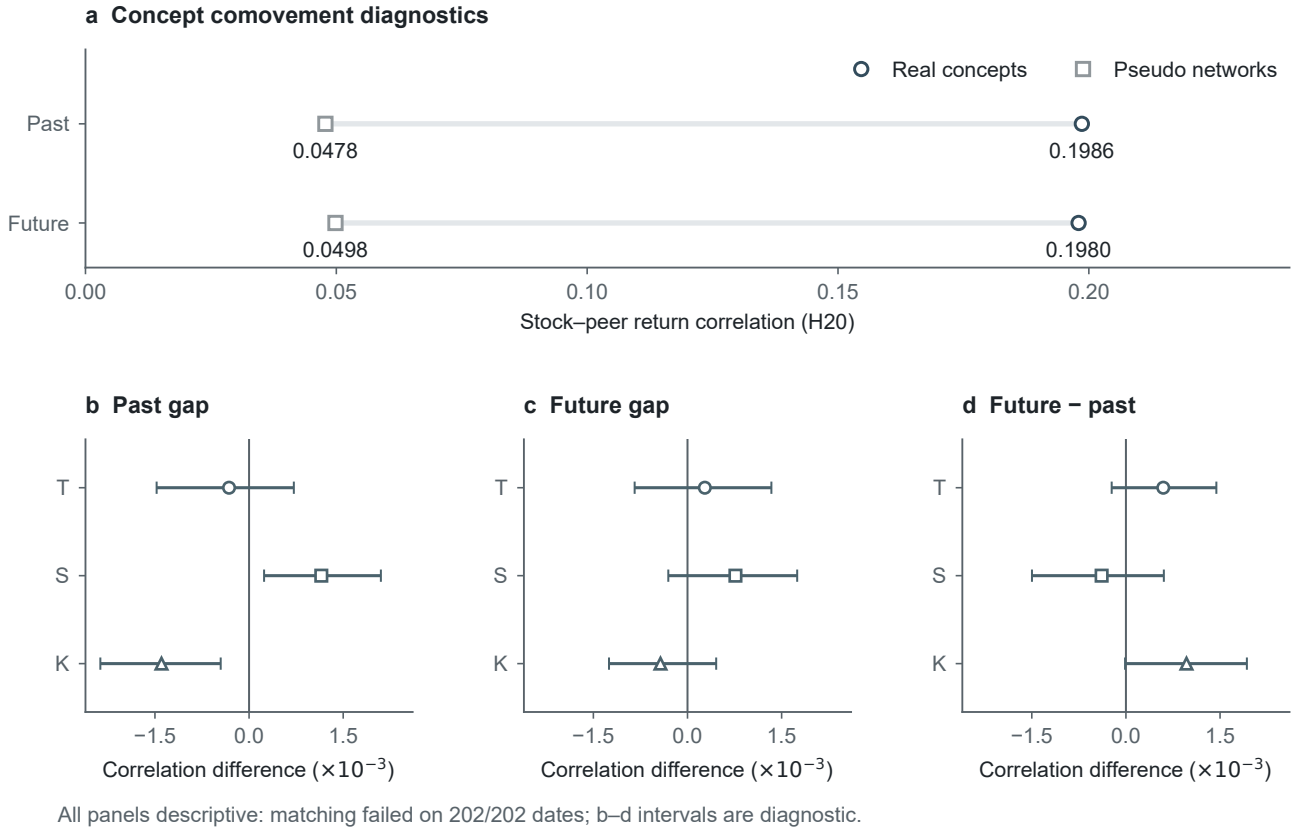


Figure 7. Concept comovement before and after popularity changes. a shows equal-date mean correlations of a focal stock with its real concept-peer portfolio and the average of 50 stratified pseudo networks over the 20 trading days before attention observation and after the signal. Both periods use the same 123,530 stock-days across 202 signal dates. Membership precedes attention observation and focal stocks are excluded; disjoint reference stocks supply industry benchmarks. b–d show slopes per ten-percentage-point increase in the cross-sectional percentile rank of T innovations, five-calendar-day S changes or weekday-standardized K (Supplementary Section S12) for past real-minus-pseudo comovement, future real-minus-pseudo comovement, and their paired difference. Size, degree and liquidity matching fails on all 202 dates, so formal inference is withheld and markers remain open. Whiskers are pointwise diagnostic 95% intervals from 2,000 shared circular block-bootstrap draws of eight sampled dates (40–45 market days). One past-window K comparison has diagnostic Holm9 P=0.0270; it does not pass the separate matching requirement. The future-window and future-minus-past comparisons do not pass diagnostic Holm9. All nine hypotheses remain in the correction family. The last sampled signal is 30 July 2026 and its full 20-day outcome ends on 27 August, within the 31 August observation cutoff.

4 Discussion

The member-selection result clarifies the relation between concept information and short-horizon returns. Existing work links concepts to constituent returns and stock momentum to comovement [1,2]. Here synchronization improves strength-only selection, while alternative joint rankings select nearly the same stocks. When synchronization already enters the fixed prediction model, the additional strength-related block has no confirmed increment. The information comparison, rather than the multiplication formula, is the substantive result: the value of combining the inputs depends on the information already included in the benchmark rule.

Theme-based attention is one candidate explanation for this return pattern. Style-based trading generates comovement and momentum in Barberis and Shleifer's model [8], and stock-search attention is associated with short-run price increases and subsequent reversal in Da et al. [11]. Our concept-level profiles have a related temporal shape across different aggregation levels. However, common thematic news, shared risk and correlated trading demand can also produce return alignment. Wind popularity supplies individual-stock measurements: the joint temporal comparisons support particular participation relationships, while the common-support and control-timing diagnostics examine conditional popularity–return associations in a separate panel. Distinguishing these interpretations requires variation in thematic exposure that is independent of common news and return-driven changes in popularity. Exogenous variation in news positioning, as studied by Fedyk [15], provides one design precedent.

The reconstructed data define the historical scope of these comparisons. Concept and industry records lack certified first-publication and revision histories; accounting fields are excluded because their release chain is unverified. The research pool follows data availability, with survivor coverage and non-random endpoint missingness unresolved. Missing-outcome bounds remain positive under a restricted return range and include zero when that range widens. Supplementary Section S1 gives these bounds, the source corrections and the common 31 August 2026 cutoff. Establishing what information was actually available to investors at the time requires certified records of first publication and subsequent revisions.

The ranking uses no return-fitted weights, and rolling training enforces label maturity. However, later comparison designs were informed by earlier historical results, so the reported mul-

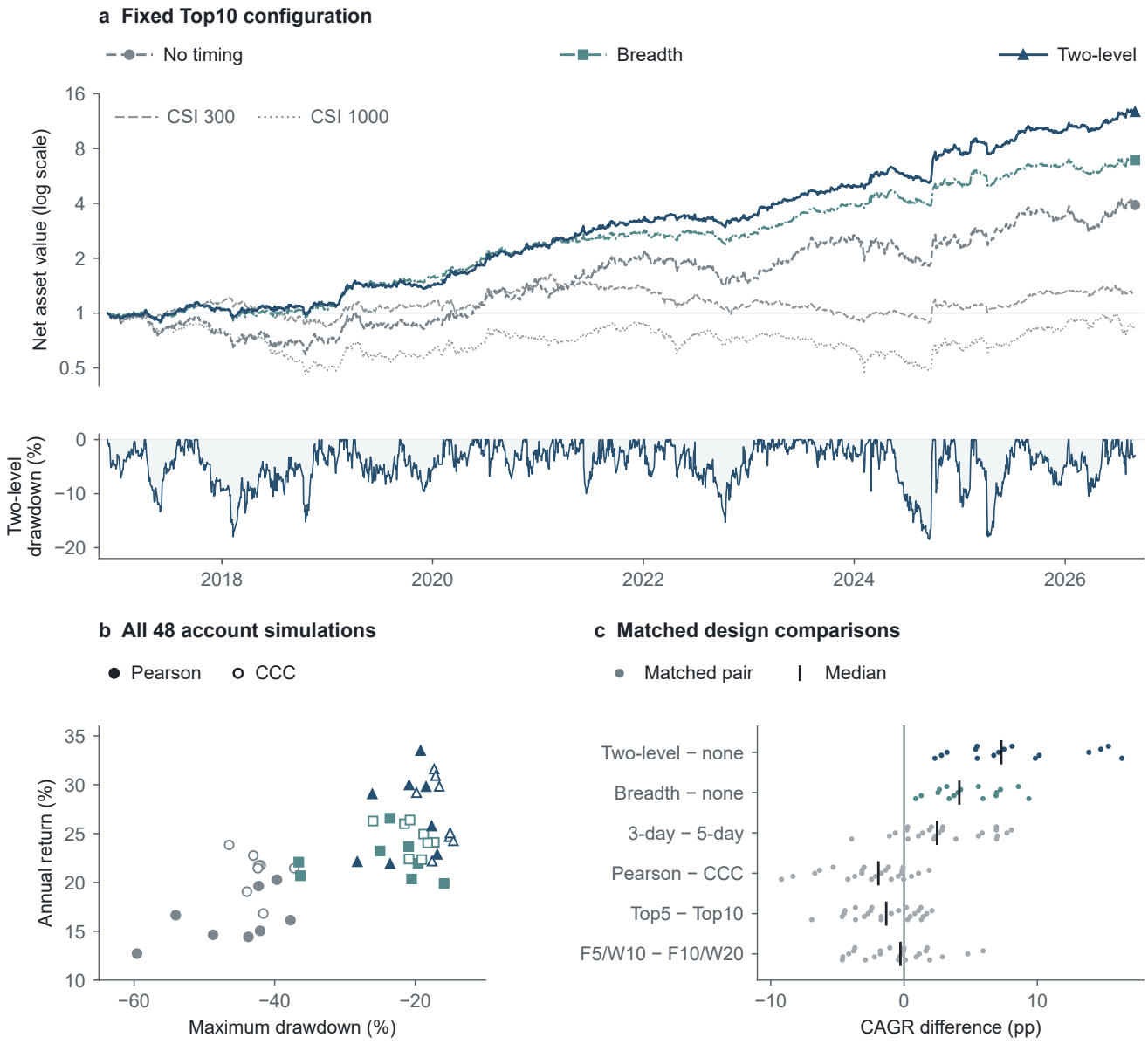


Figure 8. Retrospective account simulations and market-breadth timing. a, Net asset values (log scale) for the fixed Top10, three-day, F5/W10, Pearson configuration. No timing, one-level breadth timing and two-level breadth timing yield annual compound returns of 15.0%, 21.9% and 29.8%, respectively. The single underwater panel shows the two-level account; all daily observations are retained. The complete three-account drawdown comparison is in Supplementary Fig. S13; panel b retains maximum drawdowns for all account simulations. Wealth and CSI 300/CSI 1000 price-index curves begin at one on 1 December 2016 and end on 31 August 2026. b, Compound annual return and maximum drawdown for all 48 account simulations. Circles, squares and triangles denote no timing, one-level breadth timing and two-level breadth timing; filled markers denote Pearson and open markers concordance correlation. Timing colors and symbols follow panel a. c, Annual-return differences between configurations differing in one design element; each dot is a matched pair and each vertical bar is the median across pairs. Accounts start with CNY 30 million and simulate execution at 09:55, using rankings through the preceding close and breadth overlays evaluated at 09:45. Returns include a 1.2-basis-point commission and 0.1% stamp duty on sales; slippage and market impact are not modeled. Index returns exclude dividends (Methods and Supplementary Section S17).

tiplicity adjustments apply within declared comparison families and do not account for the full sequence of earlier specification choices[16,17]. A protocol fixed before new observations arrive would provide a prospective test of persistence. Extending this test to implementable returns requires target formation based on decision-time information and explicit market-impact assumptions. Aligning the account rules with the research lists would connect that evaluation directly to the reference selection.

5 Conclusion

Starting from intertemporal relationships in concept returns, this study examines member selection within overlapping concepts. In this historical A-share sample, stocks aligned with recently strong concepts subsequently earn higher mean short-horizon gross returns. This return continuation is what we call concept momentum: its spread is positive across all 108 parameter combinations and persists after industry–size matching. Synchronization adds to strength-only selection, while several simple joint rankings select nearly the same stocks, indicating that the gain comes from the information in synchro-

nization rather than a particular combination formula. The full concept-feature block improves conventional-model predictions in some ranking and selection tasks, whereas the return increment for the reference selection task is not confirmed. Some temporal associations of recorded popularity with turnover and trading value meet the joint criteria, but neither the adjusted conditional popularity–return associations nor their differences across fitting-sample and control-timing specifications survive correction. Identifying whether investor attention accounts for the member-selection return pattern requires exogenous variation in thematic exposure.

6 Methods

6.1 Data provenance

The price inventory used for the analysis contains 5,907 files and 10,447,754 stock-day records through 31 August 2026. Eligibility depends on historical inputs rather than future outcome availability. Supplementary Section S1 records the source corrections and corresponding recomputations. The archived covariate panel supports recomputation of conditional-attention results (Supplementary Section S1).

Different experiments impose additional coverage and historical-input requirements. The attention and mechanism analyses use the common 31 August cutoff; the sampled conditional-attention panel ends on 27 August 2026. Market-calendar formation windows are used for the main research scores. Earlier observation-row sensitivities retain their stated windows, which may span more market days when observations are missing (Supplementary Sections S3 and S4.1).

6.2 Concept scores and aggregation

For each stock–concept membership, concept returns exclude the focal stock and require at least one valid peer. F compounds returns over L_F market days, requiring all observations and retaining the daily floor of −0.999. W is the Pearson correlation between the focal-stock returns and untruncated peer returns over L_W market days. It requires non-zero variation and the declared number of valid pairs. F5/W10 requires five complete concept-return observations and at least five paired returns within ten market days.

Each quantity is ranked across finite membership records on that date, using the minimum tied rank divided by its finite-record count. Memberships valid for both ranks are then retained. A stock with several memberships contributes several records to this ranking universe. With f and w denoting the resulting ranks, the stock-level scores are

$$\begin{aligned} F_{i,t}^{\text{only}} &= \text{mean}_c\, f_{ic,t}, \\ W_{i,t}^{\text{only}} &= \text{mean}_c\, w_{ic,t}, \\ FW_{i,t} &= \text{mean}_c(f_{ic,t} w_{ic,t}), \\ A_{i,t} &= \tfrac{1}{2}(F_{i,t}^{\text{only}} + W_{i,t}^{\text{only}}), \\ P_{i,t} &= F_{i,t}^{\text{only}} W_{i,t}^{\text{only}}. \end{aligned}$$

The mean and population covariance use the same memberships, giving FW = P + Cov(f, w). This identity isolates the within-stock alignment retained by multiplying before averaging. Integer rank sums and products are compared as exact ratios; ascending stock code resolves mathematical ties. Supplementary Section S3.1 records this numerical refinement and its timing. The six-pair sensitivity reconstructs all scores from the corrected sources under common within-pair eligibility; its windows and support rules appear in Supplementary Section S2.1.

6.3 Outcomes and comparison designs

Daily returns are close/previous close − 1. Adjusted closes compound these returns, and adjusted opens equal adjusted closes multiplied by the same-day open/close ratio. The H-day outcome is adjusted open(t + H + 1)/adjusted open(t + 1) − 1. Both endpoints must be finite and positive on the prescribed market dates; the next available observation cannot replace a missing endpoint. Closing-return diagnostics use close(t + H)/close(t) − 1. Table 1 defines the four timing cells, and Supplementary Section S4 reports their paired contrasts.

Top-N lists are fixed before outcome linkage. Primary list returns average observed outcomes; a separate sensitivity retains denominator N and sets missing returns to zero. The latter does not model suspension or delisting proceeds or define an economic lower bound. Pool-relative spreads subtract the equally weighted research-pool return. Direct contrasts compare frozen lists on the same date. All research returns are gross, complete-H cohort returns, averaged equally across formation dates. Overlapping daily cohorts do not constitute a rebalanced wealth series, and their means are not daily trading profits.

Industry–size comparisons preserve the frozen lists and match their composition using weighted group means. The full daily pool is divided into ten capitalization groups, with all groups retained and selected stocks included in their comparison groups. Industry-only, size-only and joint controls are evaluated alongside 1,000 within-group random portfolios. Actual lists are never replenished for missing outcomes. Exact grouping, paired-date eligibility and missingness rules are given in Supplementary Section S2.4; temporal checks appear in Section S6.

Daily concept-momentum and comovement benchmarks use common historical eligibility and identical endpoints. A monthly mapping uses independent slope-tercile and momentum-decile sorts. These comparisons draw on concept-momentum and comovement studies [1,2], with different market, category and return conventions from the published strategies. Supplementary Section S7 specifies their construction, including the monthly skipped interval and six-month holding period.

6.4 Conditional prediction

To test whether concept features add to conventional information, conventional models contain 19 price, volatility, trading and size characteristics plus a missingness feature. Enhanced models add FW, F-only and W-only, testing the three concept features jointly. Characteristics become within-pool average-tie percentiles minus 0.5, with missing values set to zero. Neither industry nor accounting characteristics enter training. The complete feature definitions and fixed learner settings are given in Supplementary Section S2.2.

Ridge regression and histogram gradient boosting predict each year from 2020 to 2026 using the preceding three calendar years. Training retains every twentieth market-calendar date, with outcomes matured by the preceding year-end and at least

two signal years represented. Each fit consequently uses 23–37 dates rather than every trading day. Paired ordinary and enhanced models use identical rows and settings. Training targets are clipped to [−0.8, 2.0]; evaluation returns are not clipped. Seven years, eight horizons, two learners and two feature sets produce 224 fits.

Predictions cover the common pool, with Top100 lists saved before outcome linkage and Top20 taken as their prefixes. We evaluate date-weighted returns and daily Spearman rank information coefficients (ICs). Industry-conditioned IC correlates prediction and outcome ranks after subtracting industry mean ranks on the same valid subset, without reranking or retraining. Industry labels are not certified point-in-time. A separate H3 sensitivity uses every eligible training date, retaining features, settings and endpoints, and produces 28 additional fits (Supplementary Section S9).

A targeted H3/Top20 comparison holds W in both feature sets and adds F and FW jointly, using all eligible training dates and the fixed boosting settings (14 fits). Maturity and selection rules are unchanged; the stress-state definition and six-test family are specified in Supplementary Section S10. A separate four-model comparison adds W to conventional characteristics and F, with and without past market beta and residual volatility. The risk measures use the preceding 60 trading days, excluding the signal day; seven annual fits of each model give 28 fits. Section S11 defines the three paired contrasts and their joint six-test family.

6.5 Statistical inference

Inference uses paired daily contrasts to retain the common market exposure and overlap between selected lists. Circular moving-block resampling preserves dependence across consecutive dates. The baseline block length is L = max(20, H), with pointwise 95% percentile intervals. Two-sided P values compare the observed absolute mean with centred bootstrap absolute means and use a plus-one correction. Declared block lengths and longer-block sensitivities assess the dependence assumptions (Supplementary Section S2.3). Comparisons with floor(n/L) < 20 remain descriptive and enter correction with P = 1 as a placeholder, not an estimated P value. Monthly comparisons have at most 17 nominal six-month blocks and receive no inferential intervals or P values.

Holm adjustment applies within the declared families in Supplementary Table S3. Resampling counts, module-specific block choices and longer-block sensitivities are consolidated in Supplementary Section S2.3. The prediction family comprises 80 primary contrasts and 16 industry-conditioned IC contrasts. Parameter sensitivity uses separate 540-test complete-outcome and missing-as-zero families. The five timing contrasts share a Holm20 family across two missingness conventions and two block lengths. Adequate temporal support is a prerequisite for formal inference, rather than an indication of significance after correction.

The longer-block analyses were added after the main results. The 192-comparison aggregation grid was fixed before execution after pilot results had been inspected. These comparisons use previously examined history; their family-level corrections do not provide an untouched holdout or cover the full research sequence. Calendar-year, leave-one-year-out and non-overlapping H3-offset comparisons are descriptive; fixed-period tests retain their separate families (Supplementary Section S6).

6.6 Attention and participation

To distinguish persistent popularity from its changes, attention endpoints use T innovations, five-calendar-day S changes and weekday-standardized K, expressed as contemporaneous cross-sectional ranks. The future attention endpoint is t + H + 1; endpoint/current coefficient ratios use identical support. Supplementary Section S12 specifies measurement and reference-set sensitivities. Endpoint inference uses 1,000 circular block draws of 120 trading dates and a Holm24 family; H250 is descriptive with P = 1. HAC estimates are secondary diagnostics.

Expanding-time attention models fit on earlier observations, with outcome labels required to mature before each test fold. AIPW combines propensity weighting with outcome regression[18], and Tree-DML uses flexible regressions to residualize exposures and outcomes[19]. The retained Tree-DML models use role-specific finite and trimmed fitting samples; their original date-HAC intervals and adjusted P values are therefore reported as implementation diagnostics rather than standard orthogonal-score inference. Full-period preprocessing, past-only calibration and the S-coverage window remain separate protocols (Supplementary Figs. S10 and S11 and Table S11). Section S13 specifies the exposure definitions, support and overlap gates, and the timing of controls and concept-peer construction.

The focused γ/H3 diagnostic holds evaluation observations fixed across four specifications and examines fitting support, outcome-label purging, concept controls and the date of the same conventional controls. Earlier historical results were already known when this diagnostic was specified. All 64 nuisance fits use the existing learner settings and four evaluation folds. Date-aggregated score contributions supply diagnostic coefficient and paired-difference intervals at three fixed HAC lengths; Section S13.1 defines the 36- and 27-test families. These conditional-association analyses do not establish a well-defined exogenous exposure or conditional exchangeability.

Participation analyses use trading endpoints at t + H. Their joint criterion requires associations in both directions, temporal asymmetry and negative-control checks. The max-|t| reference covers all 144 directional paths. Prediction of next-session popularity-list visibility provides a related-measure comparison. Supplementary Section S14 reports these designs and the excluded order-size module, whose inputs failed reconciliation.

6.7 Attention and concept comovement

Concept memberships are taken from the latest stored snapshot strictly preceding the attention observation. Peer returns exclude the focal stock, require at least five observed peers per concept and average eligible concepts equally. Correlations require 16 paired observations within each 20-trading-day window. Even-numbered stock codes estimate industry benchmarks and odd-numbered codes supply focal observations, removing indirect focal inclusion through demeaning. This split was introduced after an initial conventional-demeaning analysis, which remains archived.

To assess whether comovement depends on the observed links, fifty fixed-seed membership permutations within industry, size-tercile and degree-tercile strata define one pseudo-network benchmark. Regressions include T/S/K ranks, industry indicators and observed slow-state, momentum, risk, size and liquidity controls. Matching requires at least 60% exchangeable nodes and mean absolute standardized paired distances no greater than 0.10. Failure withholds formal inference while retaining diagnostic estimates. Supplementary Sections S15, S16 and S2.4 report the estimates, gate checks and regression details. The final sampled signal is 30 July 2026; its complete outcome ends on 27 August, within the common cutoff.

6.8 Account simulation design

The separate event-driven framework, select-stock-xhzheng 2.5.0, implements the same two-input idea with three differences: concept returns include the focal stock, F ranks concepts, and minimum-rank ties can extend selection beyond the nominal list size. Membership records and ranking inputs follow the γ information boundary, with execution simulated before 09:55 on the next market day. A separate post-selection filter uses the execution day's full-session one-price upper-limit indicator; this information is unavailable at 09:55 and can change target holdings and sales of existing positions. These backtests therefore evaluate implementation variants rather than execution of the frozen research lists.

All 48 runs cross three- or five-day holding periods, F5/W10 or F10/W20, Pearson or concordance correlation, Top5 or Top10, and three timing settings. Runs begin with CNY 30,000,000 on 1 December 2016 (8 December for F10/W20), exclude Beijing Stock Exchange listings and end on 31 August 2026. Equally capitalized staggered sub-portfolios implement the holding periods. Upper-limit-blocked buys are retried and cancelled if still unfilled at the close; lower-limit-blocked positions are retained and reconsidered. Missing closes are carried forward for valuation, without establishing executable prices.

Commission is 1.2 basis points and stamp duty 0.1% on sales. Slippage, market impact and capacity are not modelled. Daily net values determine annualized returns, maximum drawdowns, yearly returns, invested shares, fees and commission-implied two-way turnover. All runs use the corrected frozen inputs; Supplementary Section S17 gives the execution and timing rules. CSI 300, CSI 500 and CSI 1000 price indices exclude dividends and serve as descriptive references. The backtest comparisons carry no formal inference.

Abbreviations

A, arithmetic combination of the stock-level strength and synchronization scores; AIPW, augmented inverse-probability weighting; CAGR, compound annual growth rate; CCC, concordance correlation coefficient; CSI, China Securities Index; F, concept strength; FW, membership-level product score of concept strength and synchronization; H, holding horizon in trading days; HAC, heteroskedasticity- and autocorrelation-consistent; IC, information coefficient; K, attention stickiness; N, selection size; P, product of the stock-level strength and synchronization scores; PR-AUC, area under the precision–recall curve; S, watchlist popularity; T, total popularity; Tree-DML, tree-based double/debiased machine learning; W, stock–concept synchronization.

Data availability

The training code and experimental-results data for this study are publicly available at https://github.com/xinhao-zheng/concept-momentum-and-investor-attention, together with a concept-membership archive and factor and strategy modules for the trading implementation rules. These modules require the host backtesting framework and separate price inputs; their rules can also be extracted for independent use. Wind popularity measures and other raw vendor data, including licensed price and industry data, are not included.

Declaration of competing interest

The author trades Chinese A-share equities in a personal account, including with rules related to those studied here, and declares no other competing interests.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI Codex and Anthropic Claude in the Cursor editor to review analysis code, run statistical calculations, check the literature, and draft and proofread the manuscript. After using these tools, the author reviewed and edited the content as needed, verified the reported results against the saved computation records and takes full responsibility for the content of the published article.

References

[1] Du, Q., Liang, D., Chen, Z. & Tu, J. Concept links and return momentum. *Journal of Banking & Finance* **134**, 106329 (2022). https://doi.org/10.1016/j.jbankfin.2021.106329

[2] Wahal, S. & Yavuz, M. D. Style investing, comovement and return predictability. *Journal of Financial Economics* **107**, 136–154 (2013). https://doi.org/10.1016/j.jfineco.2012.08.005

[3] Cohen, L. & Frazzini, A. Economic Links and Predictable Returns. *The Journal of Finance* **63**, 1977–2011 (2008). https://doi.org/10.1111/j.1540-6261.2008.01379.x

[4] Ge, S., Li, S. & Zheng, H. Diamond cuts diamond: News co-mention momentum spillover prevails in China. *Journal of Banking & Finance* **171**, 107356 (2025). https://doi.org/10.1016/j.jbankfin.2024.107356

[5] Avramov, D., Ge, S., Li, S. & Linton, O. Dual peer effects and cross-stock predictability. *Journal of Financial Economics* **180**, 104274 (2026). https://doi.org/10.1016/j.jfineco.2026.104274

[6] Jegadeesh, N. & Titman, S. Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency. *The Journal of Finance* **48**, 65–91 (1993). https://doi.org/10.1111/j.1540-6261.1993.tb04702.x

[7] Jegadeesh, N. Evidence of Predictable Behavior of Security Returns. *The Journal of Finance* **45**, 881–898 (1990). https://doi.org/10.1111/j.1540-6261.1990.tb05110.x

[8] Barberis, N. & Shleifer, A. Style investing. *Journal of Financial Economics* **68**, 161–199 (2003). https://doi.org/10.1016/S0304-405X(03)00064-3

[9] Peng, L. & Xiong, W. Investor attention, overconfidence and category learning. *Journal of Financial Economics* **80**, 563–602 (2006). https://doi.org/10.1016/j.jfineco.2005.05.003

[10] Bhui, R. & Jiao, P. Attention Constraints and Learning in Categories. *Management Science* **69**, 5394–5404 (2023). https://doi.org/10.1287/mnsc.2023.4803

[11] Da, Z., Engelberg, J. & Gao, P. In Search of Attention. *The Journal of Finance* **66**, 1461–1499 (2011). https://doi.org/10.1111/j.1540-6261.2011.01679.x

[12] Barber, B. M., Huang, X., Odean, T. & Schwarz, C. Attention-Induced Trading and Returns: Evidence from Robinhood Users. *The Journal of Finance* **77**, 3141–3190 (2022). https://doi.org/10.1111/jofi.13183

[13] Gu, S., Kelly, B. & Xiu, D. Empirical Asset Pricing via Machine Learning. *The Review of Financial Studies* **33**, 2223–2273 (2020). https://doi.org/10.1093/rfs/hhaa009

[14] Gelman, A. & Stern, H. The Difference Between "Significant" and "Not Significant" is not Itself Statistically Significant. *The American Statistician* **60**, 328–331 (2006). https://doi.org/10.1198/000313006X152649

[15] Fedyk, A. Front-Page News: The Effect of News Positioning on Financial Markets. *The Journal of Finance* **79**, 5–33 (2024). https://doi.org/10.1111/jofi.13287

[16] Harvey, C. R., Liu, Y. & Zhu, H. … and the Cross-Section of Expected Returns. *The Review of Financial Studies* **29**, 5–68 (2016). https://doi.org/10.1093/rfs/hhv059

[17] White, H. A Reality Check for Data Snooping. *Econometrica* **68**, 1097–1126 (2000). https://doi.org/10.1111/1468-0262.00152

[18] Bang, H. & Robins, J. M. Doubly Robust Estimation in Missing Data and Causal Inference Models. *Biometrics* **61**, 962–972 (2005). https://doi.org/10.1111/j.1541-0420.2005.00377.x

[19] Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W. & Robins, J. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal* **21**, C1–C68 (2018). https://doi.org/10.1111/ectj.12097

[20] Politis, D. N. & White, H. Automatic Block-Length Selection for the Dependent Bootstrap. *Econometric Reviews* **23**, 53–70 (2004). https://doi.org/10.1081/ETC-120028836

[21] Patton, A., Politis, D. N. & White, H. Correction to "Automatic Block-Length Selection for the Dependent Bootstrap" by D. Politis and H. White. *Econometric Reviews* **28**, 372–375 (2009). https://doi.org/10.1080/07474930802459016

Supplementary information

Concept momentum and investor attention: Evidence from China's A-share market

S1 Historical availability

Research observations and prescribed outcome endpoints are restricted to 31 August 2026 or earlier. The pre-correction price inventory contains 10,447,809 stock-days; the membership and research-calendar inventories contain 82,392,428 records and 2,373 dates, respectively. External checks identified four erroneous closes on 9 June 2025: sh603207, 29.64 to 31.14; sh603341, 38.70 to 39.19; sz301539, 18.33 to 18.43; and sz301662, 79.01 to 80.01. The correct ex-dividend previous-close fields on 10 June are retained. A subsequent review of positive-volume records with missing execution quotes identified a security-code transition, stale daily records and six observations with recoverable quotes. The repaired source links the former code 300114 to 302132, removes 55 invalid economic stock-days and leaves 10,447,754 records in the price inventory. The invalid records comprise one copied IPO record on a suspended date for sh603336 and 54 stale observations after the last trading date under their recorded codes.

All 48 implementation backtests use the same frozen corrected sources, retaining their original protocols and execution settings. Their configurations, account paths and independently compiled summaries are bound to the current report. They remain implementation variants distinct from the research Top20 lists; Section S17 specifies the differences.

Financial files contain publication-date fields, but reporting periods, announcements, revisions and collection times have not been linked into a verified release history. They are therefore not used as strict point-in-time accounting controls. A researcher-defined selection threshold is not a verified exogenous display discontinuity. Fedyk studies exogenous variation in prominent news positioning[15]. The present records supply no comparable source of exogenous exposure.

A complementary sensitivity assigns each missing stock–date return a support interval [−1, U]. The same unknown return enters both the portfolio and any comparison group that contains that stock; it is not varied independently on the two sides. Frozen selections and intended denominators give a linear spread in these unknown returns. Minimizing and maximizing that expression produce support bounds for the equal-date-weighted mean. The pool comparison covers 2,349 mature dates; joint industry–size matching uses its 2,346 paired complete-comparison dates.

With U = 0.20, the pool-relative bounds are [0.0992, 0.5224] percentage points and the joint-matched bounds are [0.0141, 0.3779]. With U = 1.00, they become [−0.0715, 0.6339] and [−0.1360, 0.4701], respectively. These are bounds conditional on the assumed return support, not confidence intervals. The upper endpoints are assumptions, not universal return limits; widening the support admits zero and negative mean spreads.

S2 Detailed research protocols

This section records execution details supporting the main Methods. Existing sections give benchmark construction (S7), attention measurement and temporal models (S12–S13), data provenance (S1), participation analyses (S14), timing comparisons (S4) and implementation backtests (S17).

S2.1 Score windows and comparison grids

The six (L_F, L_W) pairs are (3,5), (3,20), (5,10), (10,20), (20,10) and (20,50). Strength requires all L_F market-day observations. Synchronization requires at least 5, 5, 10 and 25 valid pairs for L_W = 5, 10, 20 and 50, respectively. Each pair's historically eligible pool is intersected with the original common research keys. FW, F-only, W-only, A and P share eligibility and endpoints within a pair. Selection is fixed before future returns are joined. H = 3, 5, 10 and N = 5, 10, 20, 30, 50, 100 define 108 cells, each with five FW-minus-baseline contrasts. The complete-outcome and missing-as-zero versions receive separate 540-test corrections, without replacing selected stocks.

The earlier eight-horizon comparisons use N = 5, 20 and 100 and H = 1, 3, 5, 10, 20, 60, 120 and 250 market days. Common historical eligibility and identical return endpoints are retained within comparisons. The original common pool requires both β and γ scores under market-calendar and observation-row formation conventions. Section S3 reports the γ-only support sensitivity. Exact arithmetic governs rank aggregates and selection, while prices and returns retain double precision.

S2.2 Prediction features and fixed learners

The 19 ordinary characteristics are 1/3/5/10/20/60/120/250-day momentum; negative five-day momentum; 20/60-day sample return standard deviations; current and 20-day mean turnover value; ratios of five- to twenty-day and twenty- to sixty-day mean turnover value, minus one; 20-day mean absolute return/turnover value (Amihud illiquidity); free-float and total capitalization; and the 20-

day mean turnover value/free-float capitalization ratio. Market-calendar windows require exact momentum endpoints and complete rolling observations. Non-positive turnover value or capitalization is missing. After percentile transformation, negative five-day momentum reverses the ordering of five-day momentum; it remains in the original feature list.

Every characteristic becomes its within-pool average-tie percentile minus 0.5, with missing values set to zero. Missing count/19 − 0.5 supplies a twentieth input. The enhanced model transforms γ/calendar FW, F-only and W-only identically, giving 23 inputs. Neither model includes industry or accounting characteristics. All rows receive equal fitting weight. Training dates have global market-calendar indices divisible by 20, and matured outcomes must leave at least two signal years before the prediction year. The main fits therefore contain 23–37 training dates, including 23–24 at H250.

Ridge regression includes an intercept and penalty 100. Histogram gradient boosting uses learning rate 0.06, 40 iterations, at most 15 leaves, at least 100 observations per leaf, L2 penalty 10, 63 bins, no early stopping and seed 20260923. Training targets are clipped to [−0.8, 2.0]; evaluation returns are unchanged. Ordinary and enhanced fits use identical training rows. The annual three-year training window, feature construction and learner settings remain fixed in the full-training-date H3 sensitivity (Section S9).

Industry-conditioned IC first ranks predictions and returns globally within the same valid subset, subtracts industry mean ranks and correlates the residual ranks. It requires known industry and at least five valid stocks per industry; these conditions exclude no additional observations here. Industry labels are not certified point-in-time, and conditioning does not retrain the models.

S2.3 Resampling and multiplicity

For the paired cohort-return analyses below, circular moving-block draws are shared across paired statistics. Intervals are point-wise bootstrap percentiles. Two-sided P values compare absolute observed means with absolute centred bootstrap means and use a plus-one correction. Fewer than 20 nominal complete blocks, floor(n/L) < 20, withholds formal inference while retaining P = 1 in the correction family. Finite-replicate resolution is retained when computing adjusted P values. Automatic block-length selectors provide a data-dependent alternative[20,21]. The present analyses instead use declared lengths and longer-block sensitivities, as consolidated below; Supplementary Table S3 identifies the complete family compositions.

Analysis	Bootstrap replicates	Main block length	Family and longer-block scope
Components, aggregation and daily benchmarks	100,000	max(20,H)	Separate declared families; daily-benchmark longer-block checks cover the four γ/Top20/H3 contrasts
Conditional prediction	10,000	max(20,H)	Holm96; the same family repeats with base blocks of 40 and 60 days
Six-pair parameter sensitivity	100,000	20 dates	Separate Holm540 families by missingness; 40- and 60-date checks cover six pairs × five contrasts at H3/Top20, with Holm30 separately at each length
Paired membership and return timing	20,000	20 and 60 dates	One Holm20 covers five contrasts × two missingness conventions × two block lengths
Industry and size matching	10,000	20 dates	Holm9 covers three list sizes × three matching schemes
Fixed-period H3 contrasts	20,000	20, 40 and 60 dates	A separate Holm4 at each length covers two periods × two benchmarks
Full-training-date H3 sensitivity	10,000	20 and 60 dates	One Holm24 covers twelve contrasts × two block lengths

The prediction family contains eight horizons × two learners × [pool IC + two selection sizes × two missingness conventions] = 80 primary contrasts. Sixteen industry-conditioned IC contrasts bring the total to 96. The full-training-date H3 family instead includes raw IC, industry-conditioned IC, Top20 and Top100 under both missingness conventions for each model. Its protocol was frozen before that computation. Monthly mappings have at most 17 nominal six-month blocks and receive no inferential P values or intervals. Attention and comovement analyses use their distinct inference procedures in Sections S12–S13, S14 and S2.4.

S2.4 Matched controls and comovement regressions

Industry–size matching intersects the frozen research pool with known same-day industry and positive finite float capitalization, retaining all 8,835,000 observations. Corrected-price, double-precision F5/W10 scores are ordered descending, with ascending stock codes resolving ties. Top5 and Top10 are prefixes of the frozen Top20 list. Capitalization uses its stored single-precision values. Stocks are ordered by capitalization ascending and code ascending; zero-based rank i among n candidates receives size group floor(10i/n)+1. All ten groups are retained.

The analytic comparator weights each group's mean available H3 return by k_g/N, where k_g is the selected count in that group. Selected stocks remain in the groups. Each of 1,000 fixed-seed random repetitions draws k_g stocks uniformly without replacement from each group, with the actual selected stocks still eligible. Draws depend on candidate identities and groups, not outcomes. Random-portfolio period-mean distributions use the corresponding paired dates and are conditional references rather than time-series confidence intervals.

H3 outcomes join exact adjusted opens at t + 1 and t + 4, with endpoints no later than 31 August 2026. A primary date requires the observed actual-list mean and every selected group's mean. Joint matching retains 2,348 Top5 dates and 2,346 Top10/Top20 dates, compared with 2,349 under industry-only and size-only matching. Missing selected outcomes number 12, 31 and 66, respectively. A descriptive sensitivity fills missing stock outcomes with zero in both actual lists and groups, retains N and uses all 2,349 mature dates.

Temporal comparisons recompute scores, lists, adjusted-open outcomes and analytic group means from corrected prices. Calendar-year, leave-one-year-out and three disjoint H3 formation-calendar offsets are descriptive. The fixed periods are 2017–2021 and 2022–31 August 2026, with four paired contrasts across pool and joint-group benchmarks. The corresponding block procedures are listed above; gaps, support and results appear in Section S6.

For concept-comovement regressions, even-numbered stock codes estimate industry benchmarks with at least 20 observed reference stocks per industry-day; odd-numbered codes supply focal stocks. Daily cross-sectional regressions jointly include T/S/K ranks, industry indicators and slow-state, momentum, risk, size and liquidity controls timestamped no later than attention observation. Fifty fixed-seed membership permutations within industry, size-tercile and degree-tercile strata define a single pseudo-network benchmark. Coefficients are averaged equally across dates. The 2,000 shared circular draws use blocks of eight sampled dates, corresponding to 40–45 market trading days; 202 dates supply 25 nominal complete blocks. All nine hypotheses remain in their Holm family. The required mean absolute standardized paired distance is no greater than 0.10 for each of size, degree and liquidity, alongside at least 60% exchangeable nodes. Failure closes formal inference without deleting diagnostic estimates (Sections S15 and S16).

S3 Common support and score sensitivity

The score allows separate formation windows L_F and L_W. F5/W10 denotes the reference pair $(5, 10)$ carried over from the original design and retained for component comparisons, rolling prediction models, and H3 timing and matched-control analyses. A separate parameter sensitivity crosses six window pairs, three horizons and six list sizes. The full-training-date check changes training density at H3 with F5/W10 features fixed. These scopes distinguish the analyses throughout the supplement. All observations end by 31 August 2026.

The original common pool requires valid β and γ scores under both formation-window conventions (market-calendar days and valid observation rows), together with valid price benchmarks. Thus γ eligibility partly depends on β validity. Requiring valid γ scores under both window conventions and the necessary historical prices adds 8,045 stock-days on 197 dates. Mean Top20 overlap is 99.84%; FW-minus-F-only and FW-minus-W-only return differences on this γ support are approximately 0.2271 and 0.1555 percentage points. The selected lists therefore remain close to the originals after eligibility is relaxed. List overlaps use all 2,353 signal dates, while H3 return contrasts use 2,349 mature dates (Supplementary Table S2).

The aggregation comparisons retain membership ranks and selected-list sizes (Supplementary Fig. S3). The population-covariance identity was checked on four dates under both β and γ, with maximum error below 10^{-12}. The direct W-only-minus-F-only mean was approximately 0.0723 percentage points, with pointwise interval [−0.0956, 0.2356]. Annual FW-minus-W-only means were approximately −0.0807 and −0.0038 percentage points in 2022 and 2025. The annual values describe variation within the same historical sample.

Assigning zero to missing future outcomes at the original selection denominators yields 64 corrected pool-relative rejections and 80 corrected FW-minus-strength-only rejections across the 108-cell sensitivity grid. No FW-minus-synchronization-only contrast passes the same correction. These results preserve the frozen lists.

FW minus the arithmetic combination A is positive in 75 of 108 grid cells, and FW minus the product of stock-level means P is positive in 55 of 108. Neither set contains a rejection after Holm540 correction. The separate eight-horizon comparison has no corrected aggregation advantage among its 144 temporally supported tests (Supplementary Fig. S3).

S3.1 Parameter-score precision reconciliation

All six specifications are recomputed from the corrected sources in Section S1. The final implementation aggregates integer membership ranks and compares the resulting ratios by integer cross-multiplication, with stock code resolving exact ties. This refinement followed a reproducibility check in which floating-point summation changed the order of mathematically equal component scores. It retains the economic definitions, eligibility, seeds and inference families. Reported F5/W10 FW lists are unchanged from the preceding corrected-source version; affected component, aggregation, parameter and predictive comparisons are recomputed.

S4 A paired separation of membership dates and return clocks

The paired comparison fixes F5/W10, H3 and Top20 and uses corrected-price market-calendar scores through 31 August 2026. The historical-input pool remains defined by valid scores under both membership conventions and both earlier window conventions, without using future outcome availability. It contains 8,835,000 stock–date records on 2,353 signal dates. Daily β and γ lists are fixed by descending score and ascending stock code before future outcomes are linked. Earlier historical results had already been examined; this computational ordering does not create original preregistration or an untouched holdout.

The synchronous β membership snapshot is dated no later than signal day t; the γ snapshot is dated no later than the preceding market day. Both scores use price information through t close. α and β share the synchronous list, while γ and δ share the lagged list. α and δ use adjusted close(t) to close(t+3); β and γ use adjusted open(t+1) to open(t+4). All endpoints are exact global-market-calendar joins for the same stock code, retaining missing values rather than shifting to the next observed row or substituting another selected stock. The price cutoff is 31 August 2026. The close-based cells are diagnostic references: a score using t close does not imply a feasible fill at that same close.

All four cells share the same eligible-pool identities on each date. Each selected-list return is compared with that pool evaluated under the corresponding close or open clock. The 2,349 common mature dates span 22 December 2016 to 25 August 2026, with no calendar gaps and no additional date excluded by the four-cell estimability requirement. There are 8,814,334 mature pool stock–date records. Of 46,980 intended observations per selected list, α, β, γ and δ have 45, 68, 66 and 40 missing outcomes, respectively. Pool outcomes are missing on 9,796 close-clock and 14,234 open-clock records. Complete-observation means therefore have different observed-stock denominators across clocks. A second convention sets missing outcomes to zero while retaining the intended selected and pool denominators; it uses the same dates and neither models delisting proceeds nor guarantees an economic lower bound.

Five paired daily spread contrasts isolate the two membership comparisons (β−γ and α−δ), two clock comparisons (α−β and δ−γ), and their interaction, (α−β)−(δ−γ). The interaction also equals (α−δ)−(β−γ), an identity verified numerically. Equal-date means are evaluated using 20,000 paired circular moving-block replicates at L=20 and L=60, with bootstrap starts shared across contrasts and missingness conventions within each block length. Pointwise 95% intervals use percentiles. Two-sided P values compare the observed absolute mean with centred bootstrap absolute means and include a plus-one correction. One Holm20 family covers five contrasts, two missingness conventions and two block lengths. The two lengths provide 117 and 39 nominal complete blocks; these counts are support criteria, not a claim of independent blocks.

Under complete observations and L=20, α−β is 0.040421 percentage points (pointwise 95% interval 0.018988–0.061803; Holm20 P=0.005000), and δ−γ is 0.038018 (0.016461–0.059703; P=0.007650). The corresponding L=60 adjusted P values are 0.007650 and 0.016249. The missing-as-zero clock contrasts are 0.040988 and 0.038565 percentage points and also pass under both lengths. Both membership contrasts and the interaction remain unconfirmed throughout, with adjusted P=1. The eight passing rows in the twenty-row family represent two related clock contrasts evaluated under two missingness and two dependence assumptions. They do not represent eight independent findings. Nonrejection of the membership contrasts is not an equivalence test.

The γ/open cell uses the same corrected-price score, list and outcome definitions as the main specification on all 2,349 mature dates. Its source manifest preserves the calculation version and paired endpoints. Earlier spot checks of three stock–date windows against the raw adjustment chain remain archived as checks of the previous snapshot. This experiment uses market-calendar scoring and differs from the observation-row sensitivity below. The concept-level analysis in Supplementary Fig. S1 shares the γ information boundary while comparing concept portfolios rather than selected-stock returns.

S4.1 Observation-row formation-window sensitivity

A separate timing sensitivity constructs F5/W10 from valid stock–concept observation rows under the same β and γ membership conventions. Its common pool is the intersection of finite positive scores, fixed before outcome linkage. The grid covers eight horizons and Top20/Top30, with separate Holm80 families for complete observations and missing-as-zero returns. Outcomes use exact market-calendar endpoints no later than 31 August 2026; inference uses date-HAC lag max(H−1,5). At H3/Top20, the four cell spreads on 2,364 common dates from 1 December 2016 to 25 August 2026 are 0.3153, 0.2747, 0.2748 and 0.3135 percentage points for α, β, γ and δ. The α−β clock contrast is 0.04061 percentage points (pointwise 95% interval 0.01709 to 0.06412; Holm80 P=0.04071); δ−γ is 0.03866 (0.01535 to 0.06198; P=0.06458). The two membership contrasts and their interaction have adjusted P=1. With missing returns set to zero, the two clock contrasts are 0.04121 and 0.03926, with adjusted P=0.03386 and 0.05347, respectively. The clock-contrast point estimates are therefore similar under both membership conventions; under this broader correction, the synchronous-membership clock contrast passes, whereas the lagged-membership contrast does not. The formation window, pool, dependence estimator and correction family differ from the market-calendar comparison; these related analyses are reported separately.

S5 Relation to earlier analyses

Earlier work compared concept and industry rankings, then introduced stock–concept synchronization and distinguished entry into a candidate pool from additional ranking within it. Raw IC and standardized IC-z answer different questions; standardization does

not remove overlapping memberships. Permutations, industry/capitalization adjustment and matching examine score relationships and competing explanations rather than exogenous treatments. Earlier entry-spread and attention event-matching analyses retained material imbalance. They provide background to the research sequence; the formal quantitative results reported here are drawn from the current analyses restricted to the stated cutoff.

An information coefficient computed against future returns becomes available only after those outcomes mature. Retrospectively selecting its peak cannot supply an ex ante timing signal. The implementation overlays instead use market breadth observed before execution (Section S17).

Later comparisons, developed after inspection of earlier results, address aggregation on common samples, attention measurement, daily benchmarks, conditional prediction, monthly mapping and timing. They estimate different quantities: selected-list returns, properties of rank-based proxies and covariate-adjusted return associations. Their samples and outcomes remain separate, and shared observations are treated as related evidence rather than independent replications.

Supplementary Fig. S1 compares concept-level return associations across formation and holding horizons under α, β and γ. On each market day u, the concept return is the equal-weight mean of finite member-stock returns, requiring at least one observed stock. α and β use the latest recorded membership snapshot dated no later than u; γ uses the latest snapshot dated no later than the preceding market day, u−1. Formation strength compounds F complete market-day concept returns through t in all three designs. Thus γ delays membership while retaining formation returns through t, as in the stock-level design. In α, the future outcome compounds daily concept returns over t+1 to t+H, with memberships updated on each day. In β, it averages individual-stock adjusted opening returns from t+1 to t+H+1 within the eligible concept basket fixed at t. The γ basket uses membership dated no later than t−1, retains members with an observed stock return at t and uses the same opening-return interval. The α–β comparison therefore changes portfolio construction as well as the return clock. These are concept portfolios without a focal stock; the leave-one-out construction applies to the stock-level selection tests.

Past formation eligibility is determined without future outcomes. The α index requires a valid concept return on every day of its H-day outcome. In β and γ, stocks lacking either prescribed opening endpoint have missing returns; the primary outcome averages observed returns within the basket fixed at the information date. Each design uses its own valid concepts and dates within an F/H cell, with at least 20 concepts and nonconstant ranks for each daily correlation. All source observations and prescribed endpoints are no later than 31 August 2026, and holding periods are not shortened. Supplementary Data S9 also reports results on the same concepts and dates across the three designs, a fixed-member closing-return sensitivity for α, and fixed-denominator missing-as-zero sensitivities for β and γ.

For each date t, we calculate the cross-concept Spearman correlation IC_t and denote its value clipped to [−0.999999, 0.999999] by $\widetilde{\mathrm{IC}}_t$. The scaled value is $z_t = \operatorname{atanh}(\widetilde{\mathrm{IC}}_t)\sqrt{N_t - 3}$, where N_t is the number of concepts entering that day's correlation. The figure reports equal-date means of IC_t and z_t. This count-based scaling is descriptive and does not account for dependence arising from overlapping memberships or repeated observations over time. Supplementary Data S9 reports all 486 signed values, the date support of each cell and the common-support sensitivity. Among the 25 cells with F and H no greater than ten days, mean raw IC is positive in every α cell and in 21 cells under each of β and γ. For the 12 cells with F of at least five days and H of 60 or 120 days, it is negative in 10 α cells and all 12 cells under each of β and γ. These counts describe direction rather than statistical significance.

S6 Reconstruction, annual variation and fixed-period sensitivity

The source-corrected analysis retains common-pool eligibility rules and joins all 8,814,334 mature stock–date records to exact H3 endpoints. Scores, selections, daily group means and paired spreads are recomputed under the same definitions. Across 2,349 mature dates, 66 of 46,980 intended Top20 outcomes remain missing. Earlier frozen-score and raw-source replay checks are archived separately from the current corrected-price estimates. Section S1 states the data-provenance scope of these computational checks.

Annual means are reported descriptively, with 2016 and 2026 explicitly identified as partial years (Supplementary Fig. S2 and Table S6). Pool-relative and joint industry–size spreads average 0.3364 and 0.1919 percentage points in 2017–2021, and 0.2270 and 0.1426 in 2022 through 31 August 2026. All four comparisons pass their respective Holm4 families with 20- and 40-date blocks; the later pool comparison is close to the significance threshold, with adjusted P values of 0.0343 and 0.0415. At block length 60, the later period has 18 nominal complete blocks, falling short of the sample-support requirement, so formal inference is not reported. Different block lengths do not supply independent findings. Dropping missing paired dates leaves two calendar gaps in the early joint-group series. Omitting any single calendar year leaves the pool-relative mean between 0.2496 and 0.3238 percentage points and the joint industry–size-matched mean between 0.1537 and 0.1911 percentage points.

A contemporaneous benchmark restricts the H3 Top20 comparisons to the common window of the attention analyses. The first eligible signal is 25 March 2022, after all three T/S/K fields become observable under the two-calendar-day attention lag; outcomes must mature by 31 August 2026. The resulting 1,072 dates end on 25 August 2026. FW exceeds the pool by 0.218 percentage points (pointwise 95% interval 0.010 to 0.440), F-only by 0.264 (0.083 to 0.444), W-only by 0.120 (−0.081 to 0.337), and joint industry–size peers by 0.131 (0.031 to 0.235), using 20,000 circular 20-date-block draws. Field coverage and information timing determine the window; these intervals are descriptive and enter no correction family.

The 2026 partial year contains 156 signal dates through 25 August. Its pool-relative and joint industry–size means are 0.6040 and 0.2783 percentage points, respectively (Table S6). The nine complete calendar years are unaffected by the final observation cutoff.

Selecting each of the three market-calendar offsets separately yields non-overlapping H3 holding intervals within an offset. Pool-relative means range from 0.2759 to 0.2955 percentage points; joint-group means range from 0.1573 to 0.1742. These are descriptive historical contrasts. Adjacent cohorts may still share broader market regimes, and offsets are not independent replications.

S7 Daily and monthly benchmark implementation

CM5 and CM20 compound leave-one-out concept returns over five and twenty market days, requiring at least 5 and 15 observations, respectively, and valid ten- and sixty-day comovement slopes. Compounding retains the −0.999 daily floor; slope estimation does not truncate returns. BC10 and BC60 average membership-level OLS slopes with intercepts, requiring at least 5 and 40 paired returns and demeaned sums of squares above 10^{-15} for both series. Unlike W10, these statistics are slopes rather than correlations. Order k = 1, …, n follows descending mean slope and ascending stock code. Groups are defined by floor[3(k − 1)/n]; the highest-slope group is prioritized, with descending five- or twenty-day stock momentum within groups. Its size always exceeds 100, so every Top-N list comes from that group. This is a slope-priority sort, not an independent three-by-ten sort.

The daily-benchmark bootstrap uses base seed 230923. Of 384 comparisons, 288 meet time support and 216 pass Holm correction; 96 long-horizon comparisons are descriptive. All point differences are positive. The four benchmarks were developed after earlier research and define the scope of this comparison. In their stricter common pool, main-specification FW gross return is 0.4027% and its pool-relative spread is 0.2814 percentage points. Missing future returns retain their positions in the fixed selections, and complete-endpoint means use the observed outcomes in this pool.

For formation month m, monthly stock momentum uses complete months m − 5 to m, requiring every market-day return to be valid and above −1. Membership-level OLS uses daily observations in months m − 2 to m, at least 20 valid pairs and both demeaned sums of squares above 10^{-15}. Slopes are averaged arithmetically within stocks. Eligible memberships satisfy γ at formation; historical concept returns use each historical day's membership rather than backfilled formation-date membership. A membership need not exist on every estimation day. Independent slope terciles and momentum deciles use deterministic code-based ties, pools of at least 300 stocks and at least ten stocks per intersection. July 2017 is excluded before outcome evaluation because its smallest cell contains eight stocks. Across the 110 retained formation months from May 2017 to July 2026, the smallest pool contains 1,419 stocks and the smallest intersection 13.

Scores and lists are fixed at month-end, skip m + 1, enter at the first adjusted open in m + 2 and exit at the first adjusted open in m + 8. Neither FW nor the slope-priority BC comparator is updated in the skipped month. There are 103 mature formation months from May 2017 to December 2025, all exiting by 31 August 2026. The outcome is a six-month cohort return, not a monthly return series of overlapping portfolios; annualized returns and Sharpe ratios are therefore not reported. Overall endpoint coverage is 99.424%, falling to 76.923% in the worst month–cell. Missing-as-zero FW-minus-pool spreads are −0.041 and +0.215 percentage points for Top20 and Top100, versus complete-endpoint spreads of −0.082 and +0.215. Monthly comparisons are reported descriptively under both conventions (Supplementary Fig. S5).

In the monthly independent sort, the high-minus-low comovement difference in winner-minus-loser six-month returns is −0.782 percentage points. The 103 mature formation months provide at most 17 nominal six-month blocks, so these values are not used for statistical inference. Some daily adapted rules themselves have low mean returns; FW's advantage over these constructions therefore cannot be extrapolated to an advantage over the original published strategies.

S8 Model support and longer-block sensitivity

Ordinary and enhanced models share training rows, prediction pools and learner settings. Parameters were not searched against the reported results. After the source corrections, all 224 models were refitted across 56 year–horizon cells. The model bootstrap uses seed 20260923. H120 and H250 supply 12 and 5 nominal complete blocks, respectively, below the sample-support requirement for formal inference. Industry conditioning leaves the valid stock–date set unchanged, as checked across 62,215 date–horizon–method counts under the same subset rule. Ordinary ridge and boosting H3 pool ICs are 0.055210 and 0.053621, versus 0.037238 for FW; these level differences were not separately tested. The prediction loss is not tailored to extreme Top20 selection.

At H3, the concept-feature increments in daily rank correlation are 0.001656 for ridge regression and 0.000392 for boosting. The Top20 return increments are 0.057239 and 0.030821 percentage points, respectively. All four Holm-adjusted P values equal 1. Negative long-horizon estimates with inadequate temporal support are reported descriptively.

For ridge H5/Top100, the mean increment is 0.139702 percentage points, with a pointwise 95% interval of [0.074263, 0.206752] and Holm P = 0.009599. The adjusted P values under 40- and 60-day base blocks are 0.009599 and 0.028497. Assigning zero to missing outcomes at the original list denominator gives 0.139632 percentage points. The H5 Top20 comparison has Holm P = 0.054595 despite a positive pointwise interval.

A post hoc paired diagnostic directly compares list widths on the same 1,609 H5 ridge evaluation dates, from 2 January 2020 to 21 August 2026. Let D_N be the enhanced-minus-ordinary mean return of each frozen Top-N list. The mean D_100 minus D_20 is −0.044035 percentage points. Its pointwise 95% intervals are [−0.130721, 0.039462], [−0.129415, 0.040815] and [−0.129814, 0.040157] for blocks of 20, 40 and 60 dates. The corresponding fixed-denominator missing-as-zero mean is −0.043939, with intervals [−0.131640, 0.040094], [−0.130453, 0.041583] and [−0.131125, 0.041516]. All six Holm-adjusted P values are 1. The design uses 100,000 circular moving-block draws per length, shared across paired quantities, seed 2026100823, centred two-sided P values and a plus-one correction. The six-test family covers both missingness conventions and all three block lengths; it is separate from the original Holm96. Saved predictions and selections are unchanged, and no learner is refitted. An independent exact-endpoint reconstruction from the frozen price cache reproduces the saved daily list returns within 1.5e-8 in return units. The last H5 entry is 24 August and the exit is 31 August 2026. Because this diagnostic was specified after the original width-specific estimates were known, it is not an untouched holdout test. Its results do not support a larger increment for the broader list or establish equivalence between widths.

In the original 10,000-replicate analysis, no primary Top20 or unconditioned pool-IC increment passes Holm96. The five passing cells represent the same ridge H5/Top100 contrast under two missingness conventions and three industry-conditioned IC increments, not five independent findings. The ridge H5/Top100 mean is unchanged under 40- and 60-day base blocks, with pointwise intervals [0.069332, 0.208331] and [0.064716, 0.209258] percentage points. For industry-conditioned IC, ridge H5 adjusted P values become 0.134987/0.224978, and boosting H20 values become 0.027897/0.074393; ridge H1 passes under both longer blocks, whereas boosting H20 passes at 40 days but not at 60 days. Ridge Top20 means at H60/H120/H250 are approximately −0.351/−0.253/−0.878 percentage points. The latter two lack time support, and none provides formal negative confirmation. These fixed models test a conditional increment without exhausting competing information sets.

A subsequent simulation-precision sensitivity retains all forecasts and the full 96-test family. With 1,000,000 circular block draws per horizon and base length (20, 40 and 60 dates), centred two-sided plus-one P values and seed sequences [2026100824, H, L], ridge H5/Top20 complete-endpoint Holm P values are 0.046592, 0.057776 and 0.094116, respectively. Under the missing-as-zero convention, the value is 0.051570 at the 20-date base length. Mapping the complete-endpoint simulation tail-count intervals through Holm gives a sensitivity range of [0.042557, 0.050712] at that base length. These ranges describe finite simulation error; they are neither effect intervals nor simultaneous confidence intervals. The original Top20 significance decision therefore changes with block length, missingness convention and simulation precision; a Top20 increment robust to block length has not been established. H5/Top100 has adjusted P values of 0.002444, 0.010434 and 0.021056 at the three base lengths; the H3 primary-task conclusion is unchanged. Each base length provides a sensitivity analysis of the same 96-test family, retaining placeholders for non-estimable cells, rather than a newly pooled family of 288 confirmatory tests. This post hoc protocol and its fixed simulation budget were frozen before computation. They changed neither forecasts, outcomes nor observed effects, and cannot remove prior research selection. The original figures and preceding numerical summaries retain their provenance from the original 10,000 draws.

S9 Full-training-date H3 results

The full-training-date sensitivity fits 28 models over the seven annual evaluation windows, using 722–727 training dates in each fit and the same 1,611 mature evaluation dates (2 January 2020–25 August 2026). The maximum training-label endpoint is verified to precede each evaluation year. Fixed ridge and boosting settings, all ordinary features and the three added concept scores are unchanged. This increases the number of training dates while retaining the model settings and target.

Top20 concept-minus-ordinary spreads are 0.0992 percentage points for ridge and 0.0101 for boosting. Neither passes Holm24, nor do Top100 spreads or raw pool ICs (Supplementary Fig. S7 and Table S8). Boosting's industry-conditioned IC increment is 0.005414, with pointwise intervals [0.002725, 0.008111] at length 20 and [0.002388, 0.008413] at length 60. Adjusted P values are 0.0024 and 0.0230. Both dependence specifications support the same conditional-ranking increment at H3. The dense-training exercise is confined to this horizon.

S10 Strength-related information conditional on synchronization

This targeted diagnostic (E1) asks whether the F and FW feature block improves H3/Top20 selection given W and conventional predictors. The baseline includes 19 ordinary characteristics, their shared missing-count summary and W (21 inputs); the enhanced model adds F and FW (23 inputs). The estimand is the joint predictive increment of these two additions. We retain γ/calendar F5/W10 scores, the feature transformations and fixed histogram-boosting settings in Section S2.2, including no early stopping. Each year from 2020 to 2026 uses all eligible dates in the preceding three calendar years, with training outcomes matured by the previous year-end and before fitting. The models share training and prediction rows, yielding 14 fits. All annual predictions and Top20 lists are fixed before evaluation outcomes are joined. H3 returns use the exact adjusted opens at t + 1 and t + 4; 1,611 formation dates span 2 January 2020–25 August 2026, and all outcome endpoints are no later than 31 August 2026.

The specified endpoints are the equal-date-weighted augmented-minus-baseline return difference over all dates, its mean in a stress state, and the stress-minus-nonstress difference. Stress is defined at formation-day close by the CSI300 trailing 20-trading-day

sample standard deviation of simple close-to-close returns exceeding the 90th percentile of that measure during the preceding three calendar years. Each annual threshold uses market observations through the previous 31 December; it requires no future H3 labels. Rows after 31 August 2026 are excluded before constructing index returns or volatility. The stress state contains 174 dates across five evaluation years but occupies only 17 nonoverlapping 20-day blocks anchored at the first evaluation date. It fails the fixed support requirement of at least 120 stress dates, four years and 20 occupied blocks. Both stress endpoints are therefore not estimable for inference; their four block-length cells receive no intervals or P values.

The six-cell family combines three endpoints and block lengths L20/L60. Paired circular moving-block bootstrap uses 100,000 replicates and seed 20261008, with centred two-sided P values and a plus-one correction. Four inestimable cells retain correction placeholders of one. Intervals are pointwise 95% percentiles; a positive-increment claim requires Holm6 confirmation under both block lengths. Resampling conditions on the fitted prediction path, without refitting the learners. This newly fixed comparison uses previously studied history and does not create an untouched holdout or establish an individual F effect.

The full-period increment is −0.0225 percentage points and is not confirmed under either block length (Table S9). Missing outcomes affect 355 of 64,440 intended stock slots (0.5509%); lists are unchanged, and all 1,611 dates retain observed means for both models. Assigning missing outcomes zero on the same lists gives a descriptive difference of −0.0218 percentage points. Annual estimates vary in sign; excluding 2024 changes the full-period mean to +0.0148 percentage points. Neither the negative full-period point estimate nor the failure to reject establishes a stable disadvantage or equivalence.

S11 Synchronization conditional on observed market risk

A simple projection clarifies the interpretation of synchronization. Write $r_i = \beta_{ic} f_c + \varepsilon_i$, with $\mathrm{Cov}(f_c, \varepsilon_i) = 0$, finite second moments and positive variances for f_c and r_i. Then

$$\mathrm{Corr}(r_i, f_c) = \frac{\beta_{ic}\sigma_f}{\sqrt{\beta_{ic}^2\sigma_f^2 + \sigma_{\varepsilon,i}^2}}.$$

Correlation depends on common exposure and residual variation. For positive exposure and positive residual variance, increasing exposure or reducing residual variance raises correlation. Across stocks, a higher correlation need not imply a larger exposure coefficient. This projection motivates testing a risk-conditioned increment; it neither prescribes the aggregation rule nor identifies attention as the source of persistence. The measured W remains a short-window sample statistic over overlapping concepts.

The diagnostic fixes H3/Top20 and four feature sets. Model A contains the 19 conventional characteristics, their original missing-count summary, F, and a shared missing-count summary for two risk measures. B adds W to A; C adds market beta and residual volatility to A; D adds W to C. Risk measures use paired stock and CSI 300 returns over t−60 through t−1, require at least 45 valid pairs, and include a regression intercept. Residual volatility is the residual standard deviation with n−2 degrees of freedom. The single-market residual can retain industry and concept risk. The existing within-pool rank transformation and common missing-value encoding retain the full prediction pool.

The four models share annual three-calendar-year training windows, all eligible training dates, matured labels and the fixed boosting settings in Section S2.2. Seven prediction years, 2020–2026, give 28 fits. Predictions and Top20 lists are sealed before evaluation outcomes are joined. The 1,611 evaluation dates end on 25 August 2026, with all exits by 31 August. Three paired contrasts—B−A, D−C and (D−C)−(B−A)—share 100,000 circular moving-block draws at L20 and L60, seed 20261009. The six tests share Holm correction; intervals are pointwise 95% percentiles conditional on the fitted predictions. D−C is the primary contrast. The specification was frozen before these results were calculated, using previously studied history.

The W increment is 0.0742 percentage points before adding the two risk measures and 0.0032 afterwards. Neither increment nor their direct change is confirmed by the six-test family (Table S10). The reduction in the point estimate is insufficient to establish that market risk explains or absorbs W's contribution. Missing outcomes affect 744 of 128,880 intended model–stock–date slots; all four models retain observed outcomes on every evaluation date. The descriptive fixed-denominator, missing-as-zero means are 0.0728, 0.0012 and −0.0715 percentage points for the three contrasts.

S12 Attention measurement and ranking reference sets

The endpoint analysis in main Fig. 6 describes the temporal profiles of the three recorded fields. We use future endpoint levels so that current attention does not enter both the predictor and a future-minus-current outcome. On identical support, S's future-endpoint regression coefficient is approximately 103–119% of its contemporaneous coefficient over 1–120 trading days. The corresponding K ratio is about 10–11% at days 1–3, declining to 5.1% at day 20 and 1.9% at day 120. T endpoint responses are about −7.5% and −7.4% of the contemporaneous response at days 1 and 3, with endpoint-family Holm-adjusted P values of 0.3866 and 0.4555. The positive day-120 T coefficient is about 5.1% of the contemporaneous coefficient and passes correction (P = 0.0300). These ratios describe persistence in the recorded fields, not the retention of individual users.

The attention-measurement analysis uses Wind's stock-popular-concept-detail fields for total popularity (T), watchlist popularity (S) and stickiness (K). The jointly valid trading-day panel contains 5,682,319 stock-days from 23 March 2022 to 31 August 2026. Extraction retains 15,743,951 of 15,871,852 raw records after excluding 127,901 records beyond the cutoff. On 30 April 2026 no stocks have jointly valid T/S/K values; the date remains missing. These vendor fields define the measurement system studied here. Their interpretation differs from evidence based on Google searches[11].

All three fields use a common daily percentile-rank transformation. At H3, current ranks correlate with their own future ranks by approximately T = 0.8918, S = 0.9998 and K = 0.4542. Replacing current levels with five-trading-day rank changes gives 0.0824, −0.0801 and 0.0101. These quantities distinguish persistence in rank levels from prediction using rank changes. Their target is future attention rank, not future stock return, and neither directly observes individual attention. To isolate reference-set sensitivity, we rerank raw current and past values within the stocks present both today and five trading days earlier, then subtract the two common-set ranks. Only this explanatory change is replaced; the future target is unchanged. Stocks must survive in the common set, so this transformation does not remove selection.

For S at H3, 1,067 paired dates yield mean correlations of −0.080145 under the original definition and +0.026331 under the common-set definition (Supplementary Fig. S8). The paired difference is 0.106476. Pointwise 95% intervals are [0.081422, 0.133429] with 20-day blocks and [0.073230, 0.143213] with 40-day blocks; each has Holm24 P = 0.000240. The family contains three fields × eight horizons, corrected separately by block specification; H60 and longer are descriptive. Requiring defined correlations for all three fields under both definitions removes no further dates. The sign reversal identifies sensitivity of the attention-rank association to its ranking reference set.

For H = 1, 3, 5, 10, 20, 60, 120 and 250, paired reranking uses 1,069, 1,067, 1,065, 1,060, 1,050, 1,010, 952 and 822 dates. All begin on 30 March 2022 and end, respectively, on 28 August 2026, 26 August 2026, 24 August 2026, 17 August 2026, 3 August 2026, 5 June 2026, 9 March 2026 and 19 August 2025, so that every future rank is observed by 31 August 2026. H3 has 1,068 endpoint-mature dates before requiring at least 30 valid future common stocks, which removes one date. Missing observations are not zero-filled. The composition analysis contains 1,071 dates and 5,648,305 common stock-days.

$$D_{\text{full}} = D_{\text{common}} + E_{\text{reference}}.$$

The reference term is the exact residual between the two rank-change definitions. For S, its mean daily cross-sectional standard deviation is 0.000228, compared with 0.008236 for common-set change. Strictly opposite non-zero signs occur for an average of 7.101% of all common stocks; a larger absolute reference term occurs for 15.118%. The two attention-change Top20 and Top100 lists overlap by 99.556% and 99.323% (Supplementary Fig. S9). Perturbations around zero can therefore coexist with stable extreme attention-change lists. These lists differ from the FW portfolios evaluated for returns. Integer-rank and rational-number checks reproduce stock-level and variance identities within 3×10^{-16}.

The common five-trading-day transformation differs from earlier fresh measures: historically standardized one-calendar-day change for T, five-calendar-day log change for S, and same-weekday historical robust standardization for K, followed by cross-sectional ranking. For the one-calendar-day T change and five-calendar-day S change, source records must have consecutive calendar dates across the lag; even when both endpoint levels exist, a missing intermediate date makes the change unavailable. K's same-weekday standardization uses its separate historical reference.

The earlier Δ20/H1 comparison of S minus K now uses the same 1,040 dates under 20- and 40-day blocks, because its 1,054 available dates are truncated to complete blocks. The mean is −0.056901. Holm96 P values are approximately 0.026 and 0.481, from unadjusted P values of approximately 0.00066 and 0.01233. On identical dates, the conclusion therefore depends on the assumed dependence length. The complete-block prefix ends with the 10 August 2026 signal and 11 August outcome.

An additional post-audit circular moving-block sensitivity retains all 1,054 dates through the 28 August signal and 31 August outcome. It gives a mean of −0.057794 and Holm96 P values of 0.022 and 0.345 at 20- and 40-day blocks, respectively. Each block specification re-estimates the full 96-comparison family with 100,000 draws. This supplementary resampling changes both tail inclusion and block construction; it preserves the qualitative dependence-length sensitivity without replacing the primary estimates. These transformations and inferential sensitivities should not be merged into a single attention-decay estimate.

S13 Temporal preprocessing in causal machine learning

This analysis estimates the association between recorded attention and subsequent panel-relative returns after flexible adjustment for observed characteristics. It does not estimate mediation of the FW selection spread. The conditional-attention panel contains 448,660 stock-days, 474 sampled dates and 5,220 stocks in each of α, β and γ. Signal dates run from 1 December 2016 to 27 August 2026. Every fifth eligible market date is sampled; stock-days are then selected by the fixed hash rule on common finite, positive β/γ score support. The F5/W10 network uses valid stock–concept observation rows: five returns for F and at least five valid pairs within the latest ten observations for W. Membership records are dated no later than t under β and no later than the preceding market date under γ; both retain price information through t. α uses the β network. Own and peer attention records are dated no later than t minus two calendar days. Delayed attention measurement and delayed membership information impose two separate timing constraints.

Peer exposure is the winning concept's share of raw market-wide popularity, with overlap allocated by membership degree. The focal stock's allocated contribution is removed from the concept numerator and its raw popularity from the market denominator. It is not the mean popularity of the peers. The winning concept is selected by the highest signal-day F5/W10 membership score, after the naturally lagged attention observation; lagged membership records do not make that selection a pre-exposure network. This branch differs from the main FW score, which averages products across memberships.

The first 40% of sampled dates form the initial training period; the remaining dates form six expanding-window test folds under the fixed 40% plus six 10% partition rule. Their test ranges are 22 October 2020–28 September 2021, 12 October 2021–23 September 2022, 30 September 2022–12 September 2023, 19 September 2023–9 September 2024, 18 September 2024–29 August 2025 and 5 September 2025–27 August 2026. Training uses earlier signal dates and excludes labels whose endpoints have not matured strictly before the test start. All inputs and complete-H outcomes are restricted to 31 August 2026. The target is the complete-H return minus the same-design, same-date panel mean, calculated before state grouping and model-validity filtering. α uses C(t) to C(t+H); β and γ use O(t+1) to O(t+H+1). The target is a panel-relative return, not an index excess return.

Tree-DML residualizes continuous own-stock and leave-one-out category-peer exposures for T, S and K and the return outcome, using the residualization construction in the DML framework[19]. Own T is a historically standardized one-calendar-day change in log(1 + T); own S and K are observation-date percentile ranks of their log levels. These quantities then undergo the stated historical calibration and scaling. The past-calibrated protocol estimates exclusion thresholds at the 1st and 99th percentiles using only dates before 22 October 2020; observations outside them are removed, not winsorized. Remaining doses are demeaned within date and scaled by a fixed historical population standard deviation. Calibration requires at least 1,000 finite rows and 20 dates; otherwise estimation is deferred. S lacks sufficient early calibration data and does not borrow later information. AIPW uses an augmented inverse-probability-weighted contrast[18] and compares states defined by within-date percentile ranks of the change in T and the log levels of S and K: low ≤1/3, high ≥2/3, with middle observations excluded. The joint comparison requires all three fields high or all three low. Continuous doses, rank changes and these states are distinct exposures, not interchangeable attention shocks. In the retained Tree-DML implementation, exposure and outcome regressions are fitted on their own finite and trimmed supports, then evaluated through paired residuals on their intersection. They need not estimate the two conditional means under the same target distribution. The retained coefficients and date-HAC calculations are residual-association diagnostics; their reproducibility does not establish the usual orthogonal-score inference conditions. Section S13.1 reports a common-support diagnostic for own T/S/K at γ/H3.

Controls comprise 26 numerical characteristics and Shenwan level-1 industry: one-day return; 5/20/60-day momentum; log free-float and total capitalization; log current and 20-day mean turnover value; 20-day turnover value/capitalization and Amihud illiquidity; 60-day CSI 300 beta, idiosyncratic volatility and drawdown from the high; current market return; 20-day suspension and zero-turnover counts; 20-day institutional, large-order and retail net inflow/turnover; CSI 300/500/1000 membership; F5/W10 score; concept degree; network membership count; and peer count. Non-finite numeric inputs use native missing-value handling. Industry vocabulary is learned from actual fitting rows and unseen categories are missing. Some signal-day controls follow the naturally lagged attention observation, so not all can be interpreted as pretreatment confounders. Flow variables are vendor-defined order-size proxies, not verified investor identities or an exhaustive decomposition of turnover.

Nuisance regressions and propensity models use histogram boosting with learning rate 0.05, 31 leaves, at least 100 rows per leaf, L2 = 1 and 80 fixed iterations, without random internal early stopping. DML deterministically retains at most 30,000 training rows per role; AIPW retains at most 15,000 per state and restores presampling state proportions through propensity-model weights. DML nuisance fits require at least 1,000 training and 100 test rows. Reported diagnostics require at least 10,000 usable rows, 100 test dates and all six completed folds. AIPW additionally requires at least 1,000 high and low observations in every training and test fold, with at most 20% of pooled test propensities clipped to [0.05, 0.95]. These thresholds assess estimability and overlap, not unconfoundedness.

DML divides the sum of exposure–outcome residual products by the sum of squared exposure residuals. AIPW averages pseudo-outcomes combining state-specific outcome predictions and inverse-propensity-weighted residuals. Both weight stock-day contributions rather than daily coefficients equally. Diagnostic standard errors aggregate scores by date and apply HAC lag max(1, ceil(H/5)) sampled dates; two-sided tests use a normal approximation and intervals are pointwise 95%. DML coefficients are complete-H percentage points per fixed historical standard deviation; AIPW coefficients are high-minus-low percentage points. Six Holm24 families separate own and peer exposures within each design; three Holm32 families cover four states × eight horizons within each design. Deferred cells retain P = 1 as a correction placeholder, with coefficient estimates withheld. Supplementary Table S11 gives implementation support and diagnostic Holm and secondary BH counts; Supplementary Figs. S10 and S11 display estimates together with their horizon-specific support.

Under past-only calibration and fixed training, none of the 84 estimable continuous-exposure coefficients or 42 estimable state contrasts passes its diagnostic Holm family. For K high minus low at H20, β gives −0.474102 percentage points, pointwise 95% interval [−0.814250, −0.133954] and Holm32 P = 0.201555; γ gives −0.454133, interval [−0.796372, −0.111894] and P = 0.297651. Each uses 215,670 stock-days and 281 evaluation dates. The T and K cells are estimable through H120. H250 has no mature test outcomes in the final fold, whose first signal is 5 September 2025, and therefore fails the requirement that all test folds contribute. S also lacks early calibration or training support.

The global-preprocessing diagnostic has one Holm-significant AIPW contrast: α, K high minus low at H5, −0.228626 percentage points (Holm32 P = 0.024395). Its preprocessing uses information beyond the training period, so it is reported separately from the past-only conditional comparisons. The analyses share observations and estimate conditional associations; causal interpretation additionally requires the identification assumptions stated in the main text.

The S-coverage analysis starts on 30 March 2022, the first sampled date with finite S. Calibration uses dates before 30 September 2022, the first test-fold start preceded by at least 1,000 finite S rows and 20 sampled dates; before the six-exposure intersection, this coverage rule has 26,936 finite own-S rows on 25 dates per design. The ensuing dose calibration uses 26,846 common rows on those 25 dates. The four subsequent folds evaluate 30 September 2022–27 August 2026. The S-coverage branch first restricts the panel to the intersection of all six finite raw own and peer exposures: 262,877 stock-days on 215 dates per design, including calibration. Subsequent exposure-specific trimming and outcome availability determine the role-specific fitting and evaluation supports. The same models, seeds and support gates apply, with the complete-fold requirement evaluated over these four folds.

All six continuous exposures are estimable at H1–H120 in each design, giving 126 of 144 Tree-DML cells on 166–190 evaluation dates; no coefficient passes the diagnostic Holm24 correction (smallest adjusted P = 0.316). H120 has sufficient mature training labels before the first test fold. H250 has none before that fold and also lacks mature outcomes in the final test fold. AIPW remains estimable for T and K at H1–H60 (36 of 96 cells), with none passing Holm32. At H120, the first fold fails the requirement of at least 1,000 training observations in each high and low state group. S and joint-state AIPW contrasts fail the overlap requirement even at H1–H60: pooled test propensities are clipped in approximately 42–45% and 51–53% of observations, respectively (Supplementary Fig. S11c). Thus longer field coverage permits estimation of continuous S exposures, but a high-versus-low S comparison meeting the support requirements has not been established. This coverage-defined sensitivity shares data with the primary protocol and is neither preregistered nor an independent test.

S13.1 Common support and control timing

A focused diagnostic (N1) examines whether the residual-association estimates change when auxiliary-model support and the timing of controls are aligned. The design was fixed before this diagnostic was run, after the earlier historical results had been examined. It retains γ, H3, own T/S/K, the S-coverage window and four expanding evaluation folds. The exposure definitions, original historical trimming thresholds, within-date demeaning and fixed historical standard deviations are unchanged. The outcome remains the same γ three-day panel-relative return. The final evaluation signal is 20 August 2026, with its return endpoint on 26 August 2026.

Four specifications share 217,831 evaluation stock-days across 185 dates. M0 retains the original role-specific fitting supports and evaluates the resulting residuals on this common sample. M1 restricts both nuisance regressions to the same eligible training rows and requires H3 outcomes to mature before the test fold. M2 removes four concept controls—F5/W10 score, concept degree, network membership count and peer count—while retaining 22 conventional characteristics and industry at their signal-day values. M3 dates those same 22 characteristics and industry strictly before the recorded attention observation. M1−M0 therefore changes both support and label purging; M2−M1 changes the concept-control block; M3−M2 changes the timing of an otherwise fixed control set. These comparisons do not decompose causal pathways.

Pre-attention records are matched before outcome values are loaded. For each stock, the latest retained covariate record must precede the natural attention-observation date by no more than seven calendar days; its feature and source-trade timestamps must not exceed that record's date. Of 262,877 candidate stock-days, 257,880 pass this match and 4,997 do not. The common evaluation sample additionally requires all three calibrated own exposures and H3 outcomes to be finite and to enter completed test folds. The retained panel supplies historical records, without certifying first-publication vintages. M0–M3 retain the existing boosting settings, deterministic training cap and fit-only industry encoding, giving 64 nuisance fits. Sixteen additional fits of the original implementation verify the M0 residuals and are not a separate scientific comparison.

Each exposure has its own residual-projection coefficient, calculated as the ratio of the exposure–outcome residual product sum to the exposure-residual sum of squares. These are single-exposure projections, not joint effects of T, S and K. Coefficient units are full three-day return percentage points per corresponding fixed historical standard deviation; the fields have different exposure definitions and scales. Diagnostic standard errors aggregate score contributions by date. Paired specification differences subtract the two estimators' contributions on the same dates. Three fixed HAC lags—1, 8 and 12 sampled dates—cover short and longer dependence assumptions. A Holm27 family contains three adjacent specification differences × three fields × three lags; a separate Holm36 family contains four specifications × three fields × three lags. All intervals are pointwise 95% score-based diagnostics conditional on the fitted nuisance functions and the stated date-dependence specification. They omit uncertainty from re-estimating the nuisance functions; aligned support alone does not establish nuisance convergence or coverage in this dependent panel.

Table S12 reports the twelve coefficients. No test passes either diagnostic family: the smallest adjusted P values are 0.100605 for specification differences and 0.057180 for coefficients. Figure S12 displays the nine paired differences at HAC8; the complete 27 contrasts and 36 coefficient calculations accompany the source data. Point estimates change when the controls move before attention, but the paired tests do not establish specification differences under the declared correction. Nonrejection does not establish equivalence, zero attention effects or valid orthogonal-score coverage. The diagnostic addresses observable support and

timing choices within one γ/H3 branch; exposure assignment, persistent popularity and unobserved common news remain separate identification questions.

S14 Attention measures, list visibility and participation

Current attention information improves prediction of continued appearance on supplier-labelled one-hour and 24-hour popularity lists at the next market session. Compared with strictly historical slow-state models, adding current T/S/K increased the out-of-time area under the precision–recall curve (PR-AUC). The increments were 0.001274 for retention on the one-hour list (95% interval, 0.000496–0.002037) and 0.000933 on the 24-hour list (0.000539–0.001340). Both headline-family Holm8-adjusted P values were 0.015984, and the increments were positive across all eight available folds.

The increment in top-100 recall for new entries on the one-hour list is 0.005235 (pointwise 95% interval, 0.001053–0.009669; Holm8-adjusted P = 0.083916), with positive increments in six folds, but it does not survive family correction. Its code-rotation increment is −0.002613 (95% interval, −0.005029 to −0.000245); the interval is not contained within the prespecified ±0.0025 equivalence band, whereas both retention controls satisfy that bound. Among the eight primary comparisons, the two retention comparisons meet all implemented primary criteria. These results provide a related-measure test of persistent list visibility rather than independent validation of latent attention.

The one-hour and 24-hour labels denote list construction windows; both predictions target the next market session. Complete current and next-session Top100 anchors and complete current and past 5-, 20- and 60-session T/S/K vectors are required. Each headline uses 797 evaluation dates in eight folds. The last evaluated one-hour signal is 27 July 2026, with its outcome on 28 July; the corresponding 24-hour dates are 20 and 21 August. The one-hour candidate panel extends to 13 August, but later candidates fail the historical-input gate: a missing 30 April source record interrupts the consecutive-calendar-day S-change window and propagates through the 60-session lag; incomplete list anchors exclude additional dates. All inputs and outcomes remain within the 31 August cutoff.

The last partial fold contains 15 and 16 evaluation dates, respectively. Excluding it in a diagnostic reaggregation leaves the same two retention comparisons passing the implemented Holm8, fold-sign and observed-versus-rotation criteria. After excluding that fold, the one-hour new-entry increment is 0.004424 (pointwise 95% interval, 0.000540–0.008337; Holm8-adjusted P = 0.191808), with positive increments in five of seven folds. The available records also do not establish official platform identity, exposure levels or independence between upstream sources.

Eight of 72 attention–trading–horizon combinations meet the joint reporting requirements: T with trading value at H5 and turnover at H5/H10; S with trading value at H3/H5 and turnover at H20; and K with turnover at H5/H10. All involve participation at 3–20 days. Adding signal-day trading controls reduces absolute candidate coefficients to a median of 68.2% of their original values; this is not a mediated share. Three of 24 attention-to-return paths pass the max-|t| test against the complete 144-path maximum: T at H3 (positive, P=0.045954) and S at H1/H3 (negative, P=0.007992/0.022977). None satisfies the full bidirectional criterion. The primary maximum-t correction and the separately reported Holm144 correction use different reference procedures and are not interchangeable.

Stock-identity permutations operate within signal-date × industry × size-quintile × liquidity-quintile strata. Within each stratum, stocks are ordered by a fixed seeded hash and cyclically mapped to the next stock; singleton strata do not contribute a permuted contrast. The historical-misalignment control replaces attention in the attention-to-trading path with its same-stock value from 60 trading days earlier. In the trading-to-attention path, the trading observation is moved a further 60 trading days back from its H-session lag before the attention observation date. The joint criterion requires the 90% coefficient intervals for both controls to fall within the predefined ±0.02 equivalence band in both directions, alongside the directional and asymmetry tests.

The order-size module specifies 96 comparisons across three attention measures, two order-size outcomes, endpoint and path-mean responses, and eight horizons. It requires the sums of buying and selling amount buckets to reconcile with total turnover with relative errors no greater than 10^{-6}. Among 5,495,232 otherwise finite stock-days, 12,903 (0.235%) exceed this tolerance, above the allowed 0.1%. Discrepancies are concentrated in July–August 2026 (12,753 of 228,778 stock-days, 5.574%). The full window therefore does not meet the study's prespecified reconciliation standard, and its formal estimates are not reported.

This finding does not establish whether the discrepancies arise from source errors or differences in the definitions of the bucket and total-amount fields.

The causal-machine-learning flow controls come from a separate vendor-classified source. Those order-size categories remain proxies: their investor identities and completeness as a partition of turnover are unverified. The endpoint, trading-feedback and comovement controls do not use these amount buckets.

S15 Numerical detail for the attention-to-comovement diagnostic

On 123,530 complete stock-days across 202 dates, equal-date mean correlations with real concept peers and the pseudo-network benchmark are 0.19861 and 0.04782 in the earlier window and 0.19796 and 0.04983 in the future window. For a ten-percentage-

point increase in the cross-sectional percentile ranks of T innovations, five-calendar-day S changes and weekday-standardized K (Section S12), future real-minus-pseudo slopes are 0.000276, 0.000764 and −0.000431. Future-minus-past slopes are 0.000595, −0.000387 and 0.000965, with diagnostic Holm9 P values of 0.8906, 1 and 0.3533. The earlier-window K slope is −0.001396 (pointwise diagnostic 95% interval −0.002370 to −0.000452; Holm9 P=0.0270). It is the only diagnostic comparison to cross 0.05, and it does not meet the matching requirement. All 202 dates fail the required paired size, degree and liquidity balance criteria. Formal inference is therefore withheld for the entire network comparison.

S16 Additional source and mechanism diagnostics

S16.1 Raw-source and historical-vintage checks

Before the correction of four source closing prices, a deterministic raw-source audit covered 20 complete selected observations from each calendar year from 2016 to 2026, plus all 66 selected observations with missing endpoints, giving 286 stock–date checks. Adjusted open prices were reconstructed directly from the raw daily open, close, and previous-close records. All source files were available in the fixed snapshot, the 220 complete returns agreed exactly with the cached returns, and all 66 missingness classifications agreed. The same 286 stock–date checks reproduced the cached concept-membership sets from the raw records without disagreement. Across 2,372 calendar mappings, no membership snapshot carried a recorded date later than the preceding market day.

These checks establish agreement among the available price, membership and calendar records, but consistency of recorded dates is not equivalent to point-in-time provenance. First-publication dates, earlier versions, historical backfilling and survivor coverage require the corresponding historical source archive. The targeted corrections in Section S1 update the available source version without supplying that archive.

S16.2 Why the pseudo-network matching gate failed

The gate audit reproduced the failure classification on all 202 primary split-reference dates. Block-based exchangeability ranged from 0.9876 to 0.9941, exceeding the 0.60 requirement throughout. In contrast, the mean absolute standardized distances between original and permuted nodes exceeded 0.10 for every date and every gated attribute. The median distances were 0.4365 for size, 0.3895 for degree, and 0.7732 for liquidity; their respective ranges were 0.4261–0.4470, 0.3401–0.4040, and 0.6493–0.8841. These quantities measure average absolute paired-node distances, not standardized differences between aggregate covariate means. Permutation within industry and broad size and degree terciles preserves coarse composition but does not ensure close individual matches; liquidity was evaluated by the gate but was not a permutation stratum. The failed gate therefore cannot be interpreted as evidence for the absence of an attention mechanism.

A covariate-only extension applied the existing strict-caliper diagnostic to all 202 saved graph dates, from 25 May 2022 to 23 July 2026. The extension was specified after reviewing the existing gate failures and before calculating these diagnostics. It retained the original industry, size-tercile and degree-tercile strata and required distinct candidate partners to lie within 0.10 population standard deviations on each of size, degree and liquidity. In the current 5,236-stock universe, the fraction with any eligible partner ranged from 6.86% to 12.82%, with a median of 9.36%; no date reached 60%. These fractions are upper bounds on candidate coverage under the stronger per-pair rule, not evidence that a complete permutation exists.

A deterministic disjoint-swap construction covered 6.61%–11.76%, with a median of 8.77%, providing a feasible lower bound rather than an optimal assignment. The original coarse-stratum exchangeable fractions were reproduced on every date. The diagnostic used covariates only and calculated no replacement return outcomes.

Because the original gate constrains mean distances rather than every pair, these results do not prove that all assignments satisfying the original gate are infeasible. No replacement network passed the original validation procedure, and no threshold was relaxed.

S17 Retrospective account simulations

The implementation grid is run in a separate stock-selection framework (select-stock-xhzheng 2.5.0). All 48 runs were repeated using the frozen corrected inputs documented in Section S1, with their original scientific configurations unchanged. The daily net values, selection records and configurations are retained in the replication archive together with the framework code; the archived outputs are byte-identical to the framework outputs. The score follows the research construction with different details: concept daily returns are equal-weighted over all member stocks, including the focal stock; F is ranked across concepts and W across stock–concept pairs; memberships come from the preceding market day; and ties take the minimum rank, so a list can exceed its nominal size on tied dates. The research lists instead use leave-one-out concept returns and rank both quantities across membership records. These backtests therefore evaluate an implementation variant of the idea rather than the frozen research lists.

Each backtest starts with CNY 30,000,000 and excludes Beijing Stock Exchange listings. Daily rankings use information through the preceding close. After selection, the archived framework also removes target stocks using a one-price upper-limit flag defined

by the execution day's minimum price being at or above its upper limit. Holding periods of three or five days are implemented as three or five equally capitalized sub-portfolios rebalanced on staggered days. Buy orders blocked by the upper price limit are retried at subsequent simulated price points that day and cancelled if still unfilled at the close; positions blocked by the lower limit are retained, with sale attempts repeated at subsequent simulated price points. Missing daily closes are carried forward for valuation; this convention does not imply that a carried-forward price is executable. Commission is fixed at 1.2 basis points and stamp duty at 0.1% on sales, above the statutory rate of 0.05% after 28 August 2023. Slippage and market impact are not modelled, and capacity is not assessed.

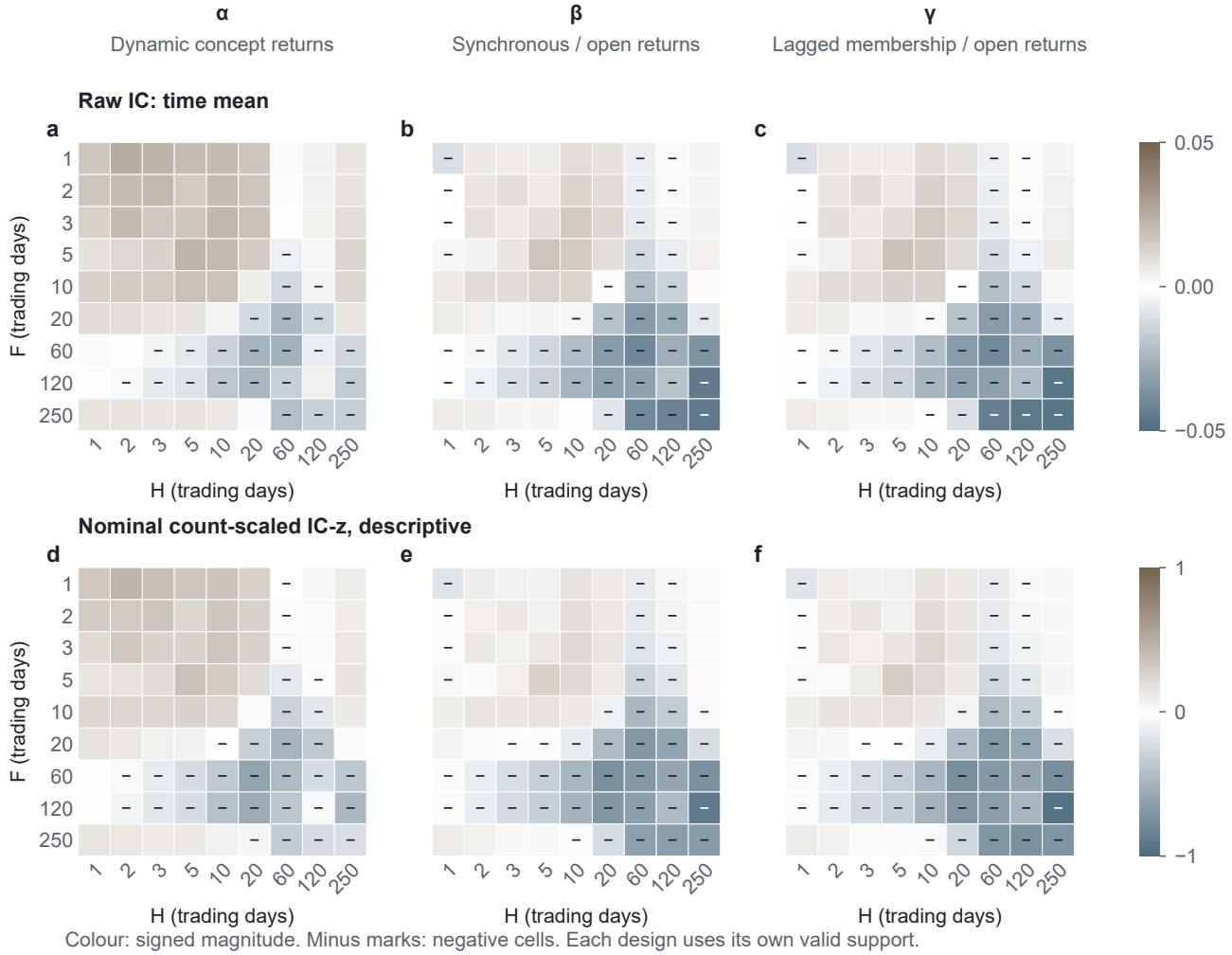
Both timing overlays first rank candidates using information through the preceding close, then form market-breadth signals and simulate execution at their specified times. The one-level and two-level rules set the target exposure of the sub-portfolio scheduled for rebalancing to zero when a specified proportion of candidates within a specified top-ranked fraction have prices below a specified price. Their parameters are framework and trading settings and were not searched in this study. Because the earlier parameter-selection process was not recorded, the timing results should be interpreted as exploratory retrospective evidence conditional on the target filter described above.

A single-gate sensitivity removed the full-session limit-up exclusion across all 48 configurations, holding ranking and weighting rules, fees and dynamic price-limit handling fixed. Relative to the archived paths, the change in annualized return ranged from −0.138 to +0.340 percentage points (median +0.003); 24 changes were positive, 19 negative and five were unchanged at the reported precision. Maximum-drawdown changes ranged from −0.309 to +0.007 percentage points (median 0.000), with drawdowns expressed as negative values. The remaining next-session state filters and quote-availability assumptions were retained, so this comparison isolates the exclusion's effect on the reported accounts. S3 Dataset reports the 48 paired outcomes and numerical tolerances.

Statistics are recomputed from the daily net values rather than taken from framework summaries. Across the 48 runs, untimed compound annual returns range 12.7%–23.8%, with maximum drawdowns of −59.6%–−37.2%. Breadth-timed runs return 19.9%–26.6% a year (drawdowns −36.6%–−15.9%) and two-level-timed runs 21.9%–33.5% (−28.2%–−14.6%). Untimed runs keep approximately 96%–97% of assets invested on average; annualized two-way turnover is 71–132 times average assets, and annual fees are 4.4%–8.2% of average assets. Timed runs issue zero target-exposure signals on approximately 40%–45% of signal dates, applying to scheduled sub-portfolios rather than the entire account. Commission-implied annualized two-way turnover is approximately 86%–114% of the median for runs with the same holding period and timing. It is calculated as total commission divided by the commission rate, mean assets and actual elapsed years; the peer ratio is a trading-intensity diagnostic, not a ratio of raw cumulative traded values.

The Top5 and Top10 list sizes lie within the research grid (Fig. 2c), and the staggered sub-portfolios mirror its overlapping daily cohorts. The account grid covers two window pairs and these two list sizes. It contains no one-to-one implementation of the Top20 research lists, same-framework W-only or simple-aggregation benchmarks, or the other four window pairs. Attribution of account performance to the score is therefore confined to the comparisons actually implemented.

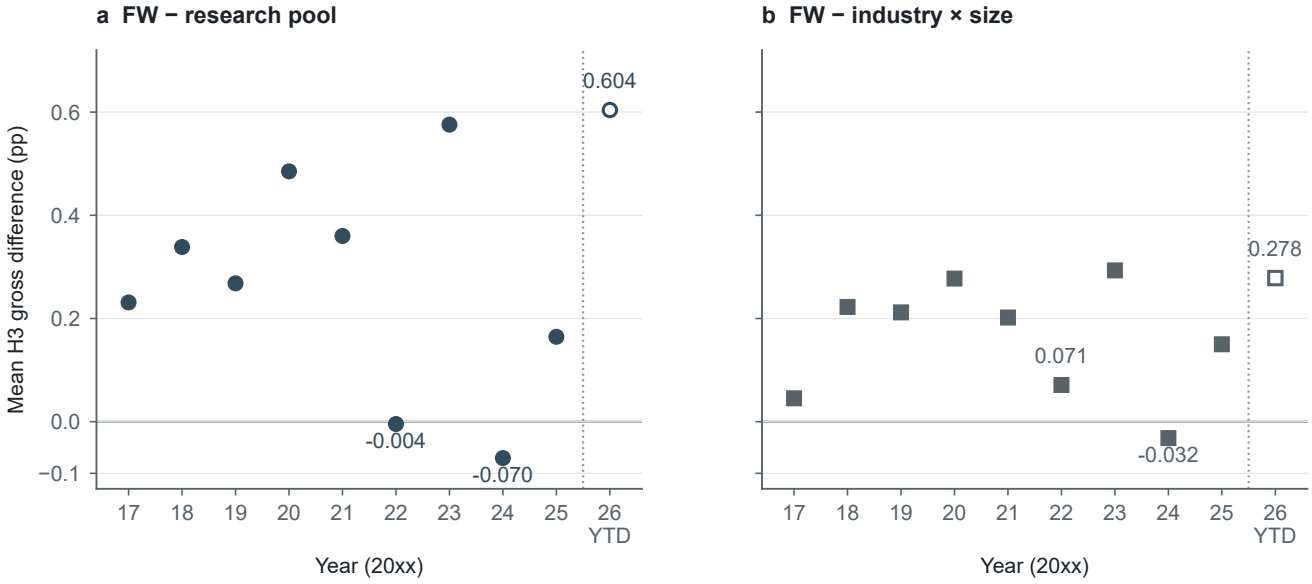
Supplementary Figure S1



Colour: signed magnitude. Minus marks: negative cells. Each design uses its own valid support.

Concept-level return associations across formation and holding horizons. Columns show α, β and γ. Formation returns extend through t in all three designs. At each formation day u, γ uses membership dated no later than the preceding market day u−1 and stock returns observed on u. In α, the outcome compounds daily concept returns over t+1 to t+H, with memberships updated each day. In β and γ, the outcome averages individual-stock open(t+1)-to-open(t+H+1) returns in baskets fixed using membership dated no later than t and t−1, respectively. Thus α versus β changes portfolio construction as well as the return clock. Each design uses its own valid concepts and mature dates, with at least 20 concepts per daily correlation. All observations and endpoints are no later than 31 August 2026. Rows display equal-date means of daily Spearman IC and descriptively scaled IC-z. Symmetric scales are shared across columns: blue denotes negative values, white zero and muted bronze positive values. A short minus mark identifies every strictly negative cell independently of colour. Supplementary Section S5 defines the scaling; Supplementary Data S9 supplies all 486 signed values, date support and common-support and outcome-construction sensitivities.

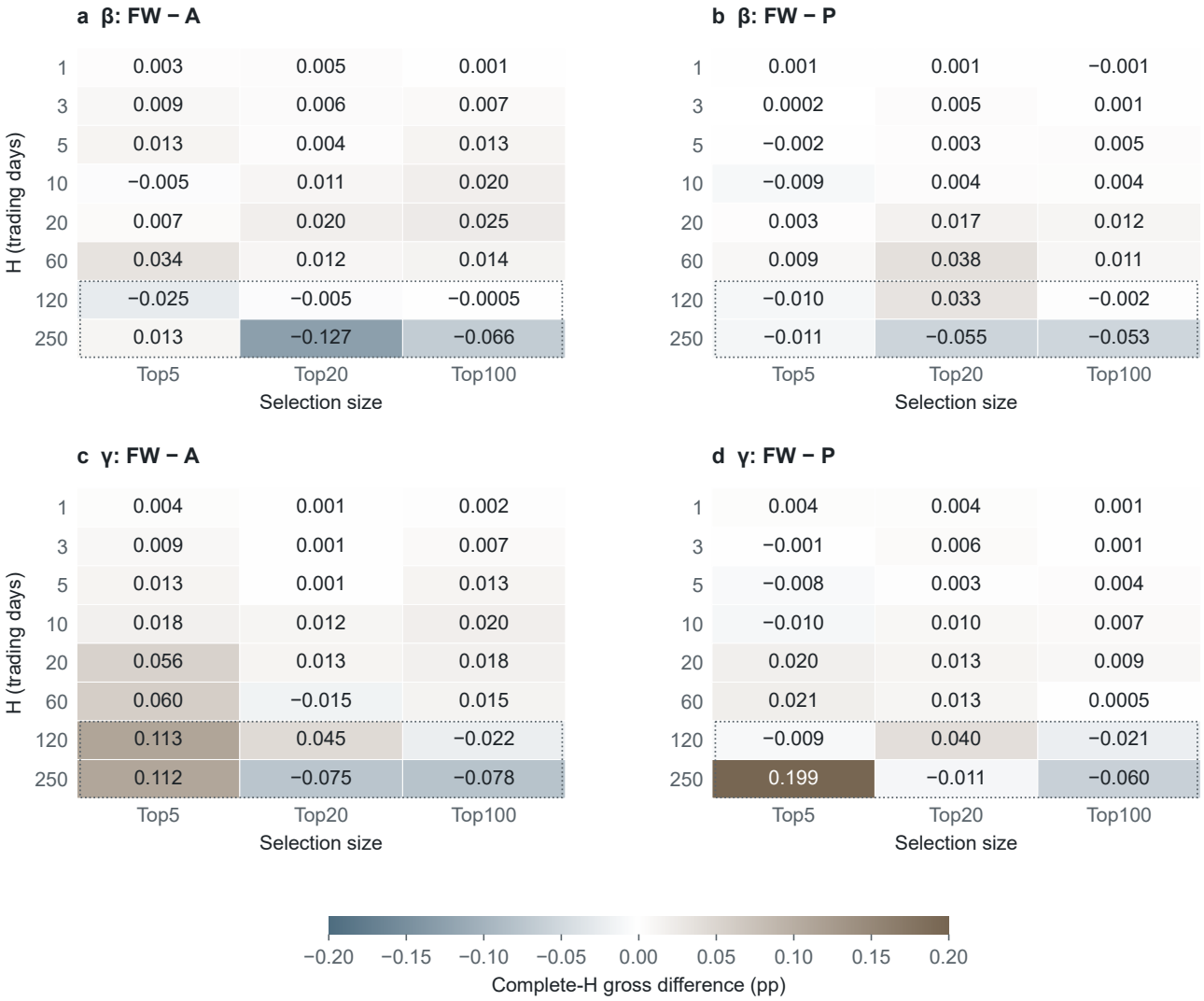
Supplementary Figure S2



Equal-date annual means; no confidence intervals. 2026 YTD: through 25 August (156 dates).

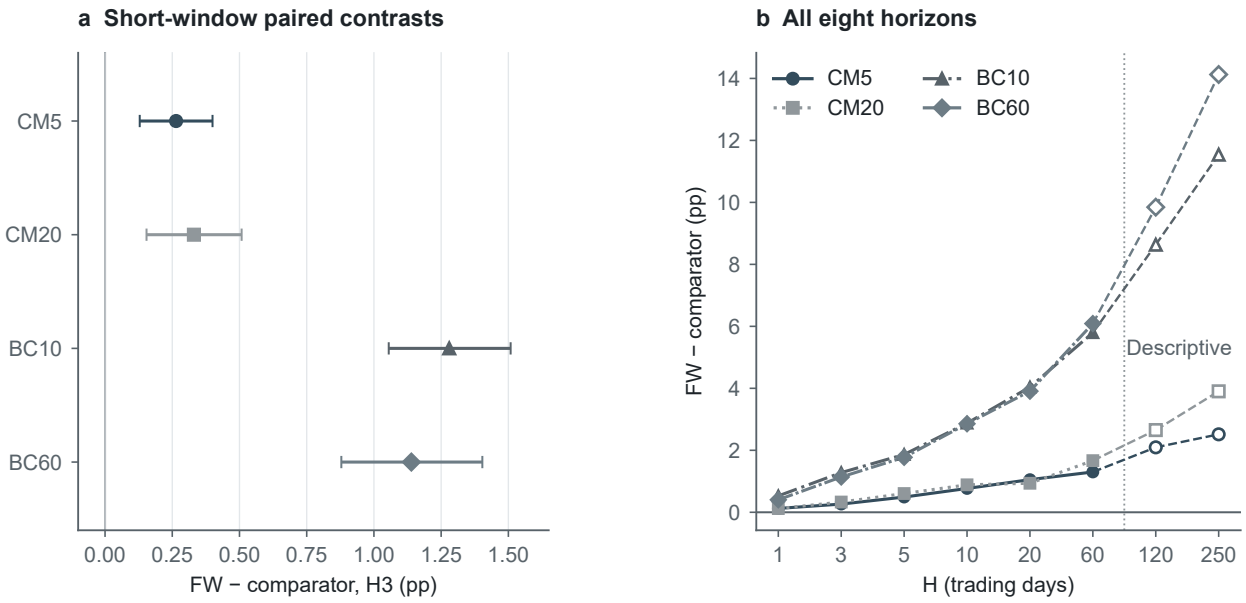
Year-by-year historical sensitivity of the reference rule. a, F5/W10, Top20, H3 complete-case gross spreads over the historically eligible research pool. b, Spreads over industry-by-size-matched group means. Each point is an equal-date annual mean in percentage points, using each comparator's paired support. The nine complete years 2017–2025 have 242–244 pool dates and 242–244 joint-matched dates per year. Pool means are positive in seven of nine complete years and joint-matched means in eight; both are negative in 2024. The 2026 open symbols represent 156 signal dates through 25 August, with outcomes mature by the 31 August 2026 observation cutoff, and are not full-year estimates. The seven-date 2016 fragment is omitted. No uncertainty or significance is assigned to the yearly means. These partitions reuse the historical research sample and measure temporal sensitivity, not independent validation.

Supplementary Figure S3



Same-information aggregation across the horizon grid. a,b, Same-day membership β; c,d, lagged membership γ. Left and right panels compare FW with arithmetic mean A and product of means P, respectively; within each panel, rows are holding horizons and columns are Top5/20/100. All eight horizons and 96 complete-endpoint means are retained. Numbers are signed percentage-point differences, generally shown to three decimal places; near-zero nonzero values use an additional decimal place. All panels share one zero-centred scale with blue negative and muted bronze positive. One dotted outline per panel flags insufficient time support at H120/H250. These are means, not significance scores. Observed-endpoint and missing-as-zero variants together form the 192-test family; 144 tests meet temporal support and none passes Holm correction. Endpoints end on 31 August 2026. Saved pointwise intervals for every test remain in source data.

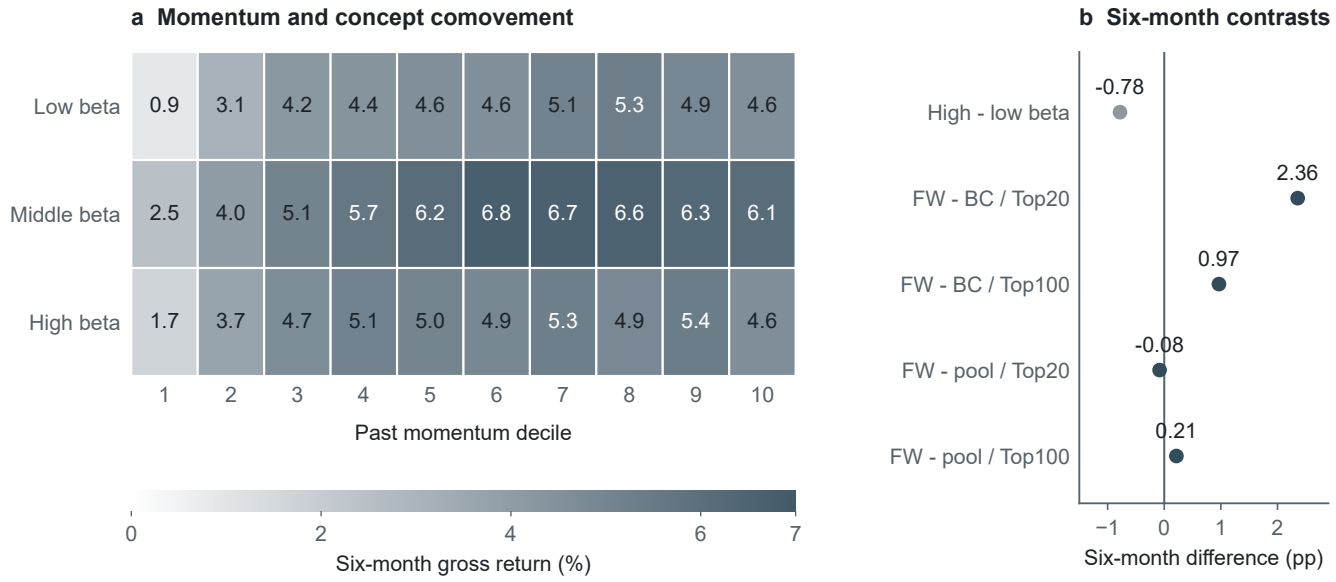
Supplementary Figure S4



Literature-inspired adaptations. Open long-horizon estimates are descriptive only.

Daily adapted comparators test a different claim from same-information aggregation. a, FW minus CM5, CM20, BC10 and BC60 under γ, Top20 and H3; n=2,329 paired dates with endpoints on or before 31 August 2026. Points and whiskers are mean gross-return differences and pointwise 95% intervals. Holm384 P values are 0.01620, 0.04056, 0.00384 and 0.00384, respectively. b, The same comparisons across all eight horizons, in complete-H percentage points. Marker shapes and supported-segment line patterns identify comparators. Filled markers denote time support, not corrected significance; open H120/H250 points and dashed continuations retain descriptive values. Each comparison uses its common eligible pool. These are literature-inspired daily adaptations, not exact reproductions of the published portfolios.

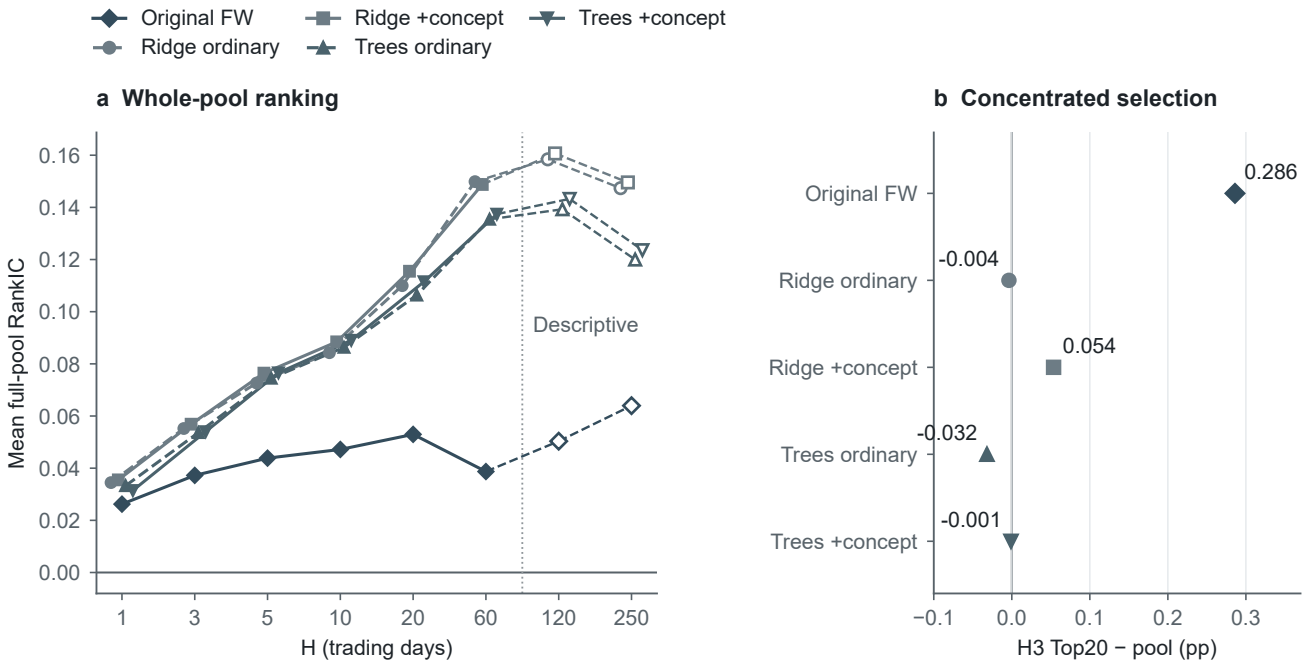
Supplementary Figure S5



103 mature formation months; one-month skip and six-month holding.

Monthly momentum and concept comovement. a, Mean six-month cumulative gross returns across ten past-momentum deciles, from losers to winners, within low-, middle- and high-comovement groups. All 30 observed cells are shown in their economic order, without interpolation, using a common sequential scale from 0 to 7%. b, The high-minus-low-comovement difference in winner-minus-loser spreads and FW-minus-comparator means for Top20 and Top100. The summaries use 103 mature formation months from May 2017, one skipped month and six-month holding, with all exits on or before 31 August 2026. This gives at most 17 nominal six-month blocks; the displayed means carry no inference. Units are percentages in a and percentage points in b. Beta denotes the OLS comovement slope; the monthly experiment uses lagged concept membership (design gamma). The long-short summaries describe return contrasts rather than implemented account paths.

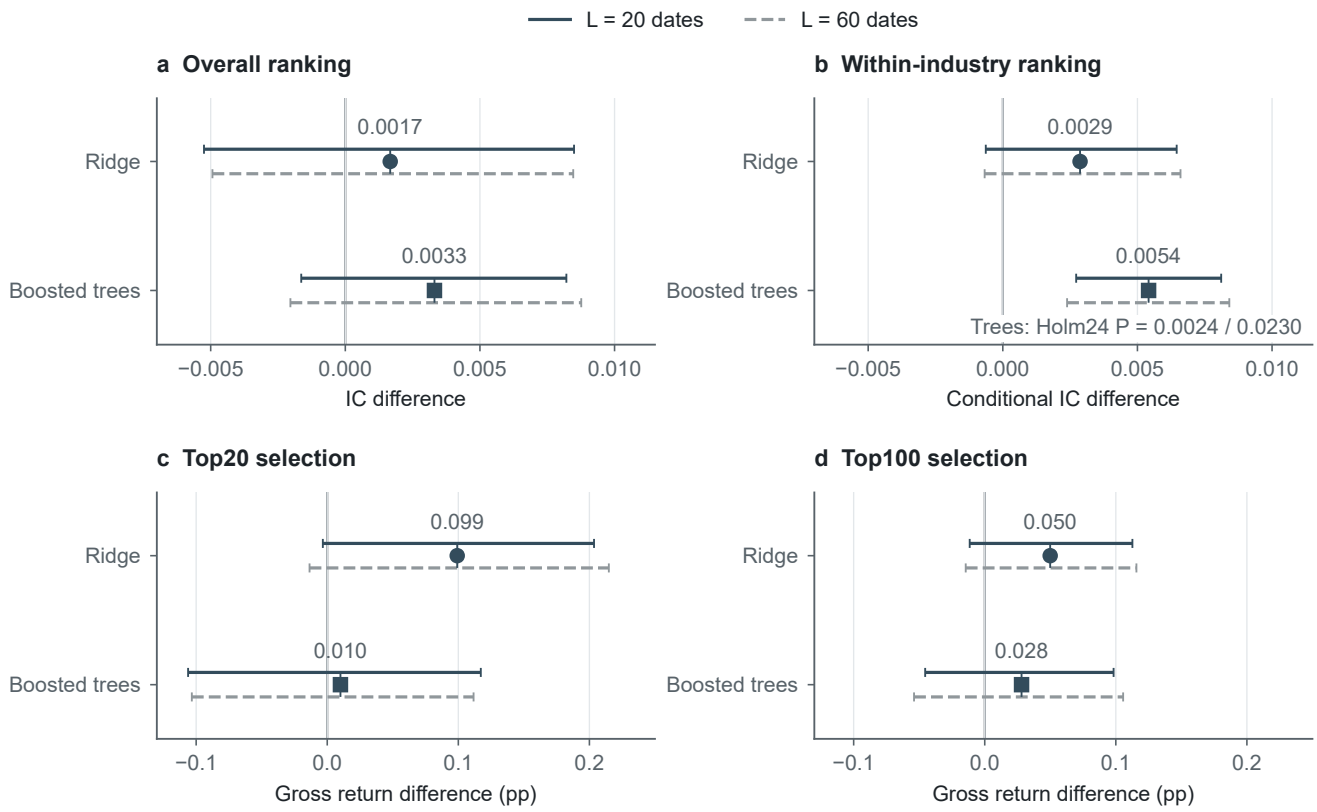
Supplementary Figure S6



Descriptive model levels on common evaluation dates.

Whole-pool ranking and concentrated selection are different tasks. a, Mean full-pool rank correlations for original FW, conventional ridge and boosted-tree models and their concept-enhanced counterparts, using 2020–2026 rolling predictions evaluated through 31 August 2026. H120/H250 are retained as open points with descriptive continuations. b, H3 Top20 gross spreads over the same research pool, n=1,611 mature dates, in percentage points. The plotted model levels are descriptive and have no between-model uncertainty intervals; formal enhanced-minus-conventional tests are reported separately. FW in b is evaluated on the 1,611-date model sample, not the 2,349-date historical rule sample. Similar rounded values do not imply identical samples.

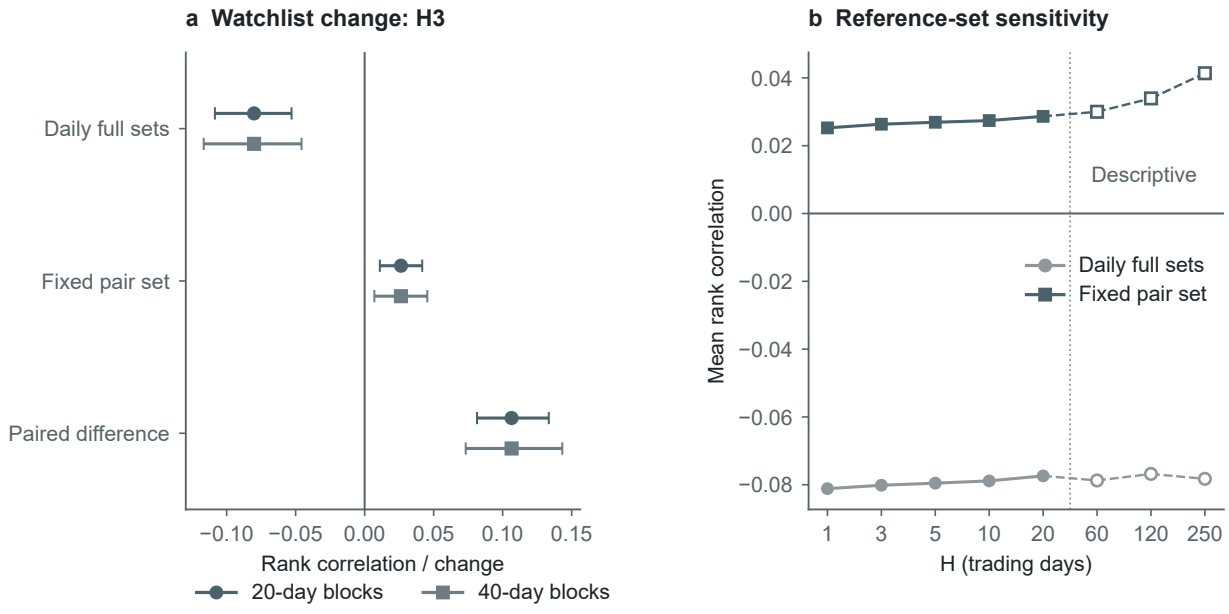
Supplementary Figure S7



One estimate per model; two pointwise 95% block-bootstrap intervals. Historical sensitivity.

Dense-training H3 model sensitivity. The same conventional and concept-enhanced model specifications are refitted on all eligible training dates, with 722–727 dates in each fit, 28 rolling fits, and 1,611 common evaluation dates from 2 January 2020 to 25 August 2026. Panels show enhanced-minus-conventional differences for a, full-pool IC; b, within-industry IC; c, complete-case Top20 gross return; and d, complete-case Top100 gross return. Each circle or square represents one model estimate. The two horizontally aligned interval rows are pointwise 95% paired circular moving-block bootstrap intervals with L=20 and L=60 dates and 10,000 replicates; their repeated point estimate is connected, not counted as independent evidence. A single Holm24 family covers both block lengths, both models and all six metrics, including two missing-as-zero return variants reported in the accompanying table. Boosted-tree within-industry IC passes that family (estimate +0.005414, adjusted P=0.0024 and 0.0230 under the two related block-length checks). Raw IC and all return increments remain unconfirmed. This is a historical training-density and inference sensitivity on the same data, not independent validation.

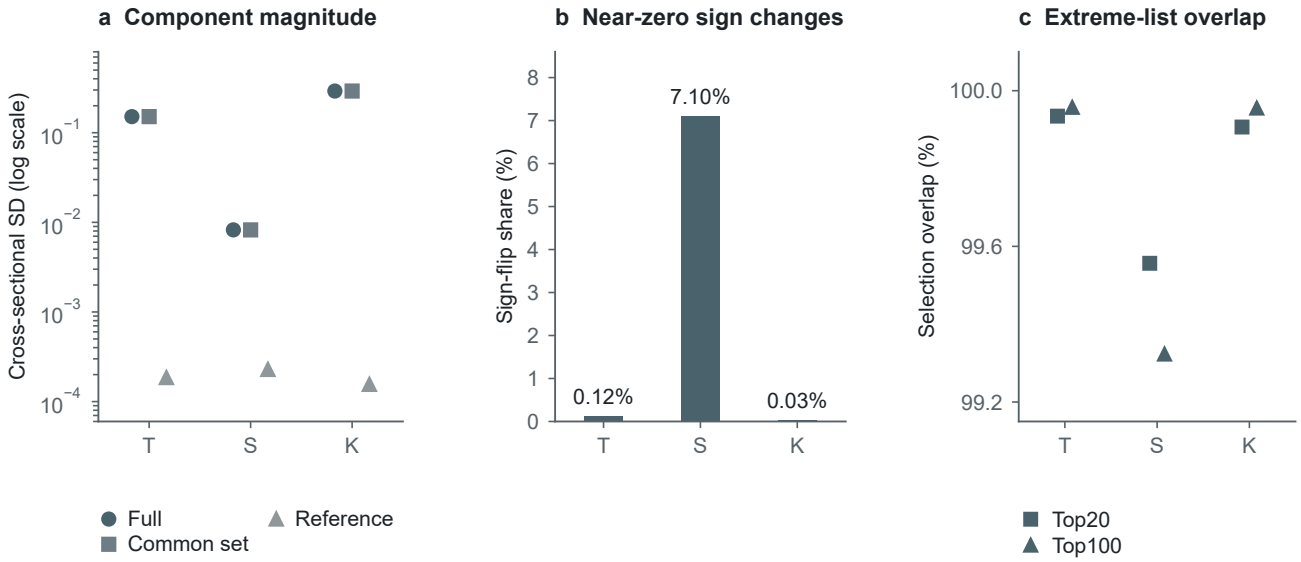
Supplementary Figure S8



Same stock–dates and future ranks; attention-rank measurement.

A fixed reference set changes an attention-rank association. a, Mean H3 watchlist-change rank correlations under original date-specific universes, the fixed common set, and their paired difference, using 1,067 paired dates whose future ranks fall on or before 31 August 2026. Circles and squares show 20- and 40-day block estimates with pointwise 95% intervals. b, Mean correlations across all eight horizons under the two reference definitions; open H60/H120/H250 points indicate insufficient temporal support and are shown descriptively. The target future rank is unchanged, and the underlying watchlist-change measure spans five days. These are correlations and correlation differences, not returns; the intervals are not simultaneous.

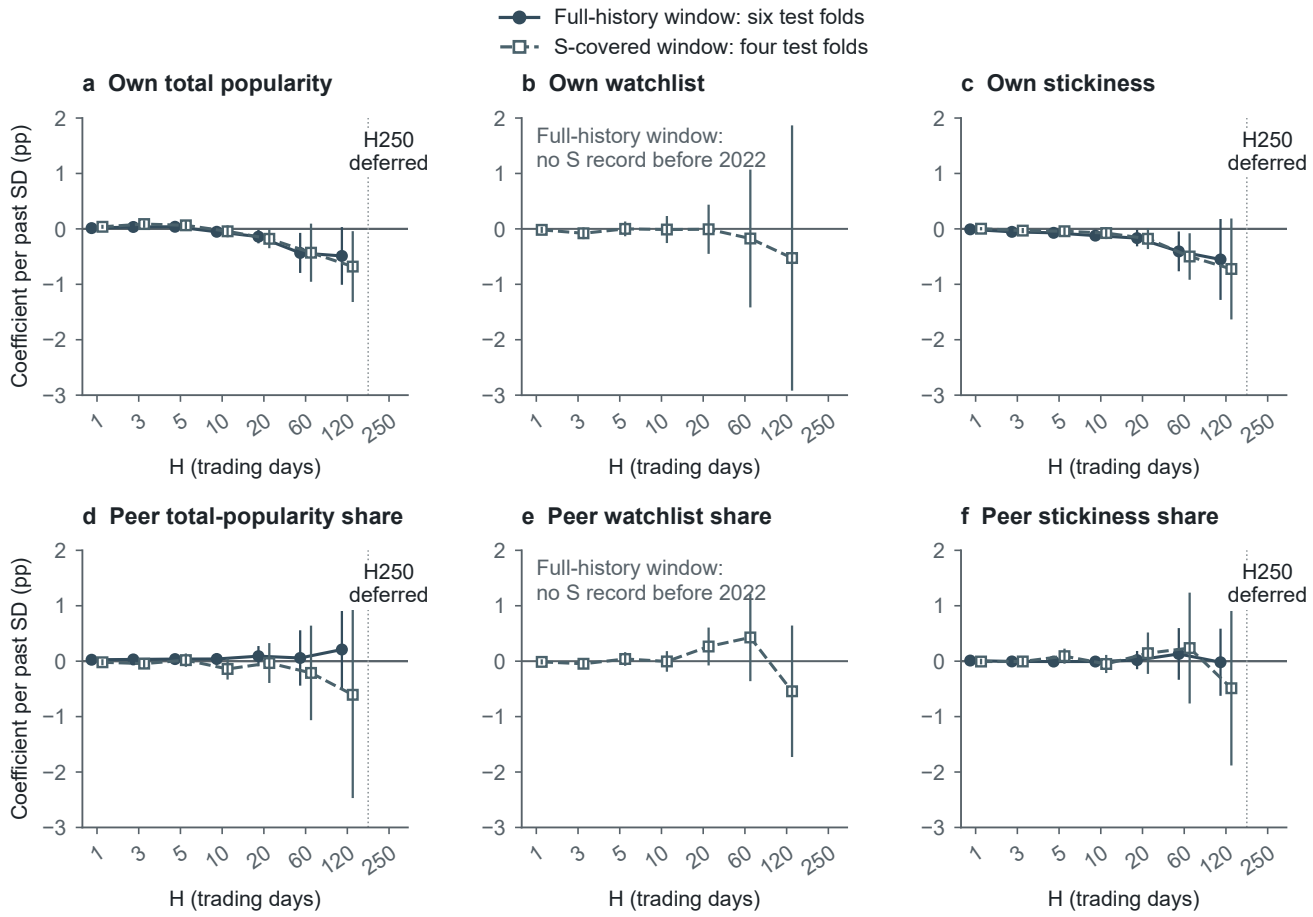
Supplementary Figure S9



Descriptive means over 1,071 dates. a: log scale. c: axis starts at 99.15%.

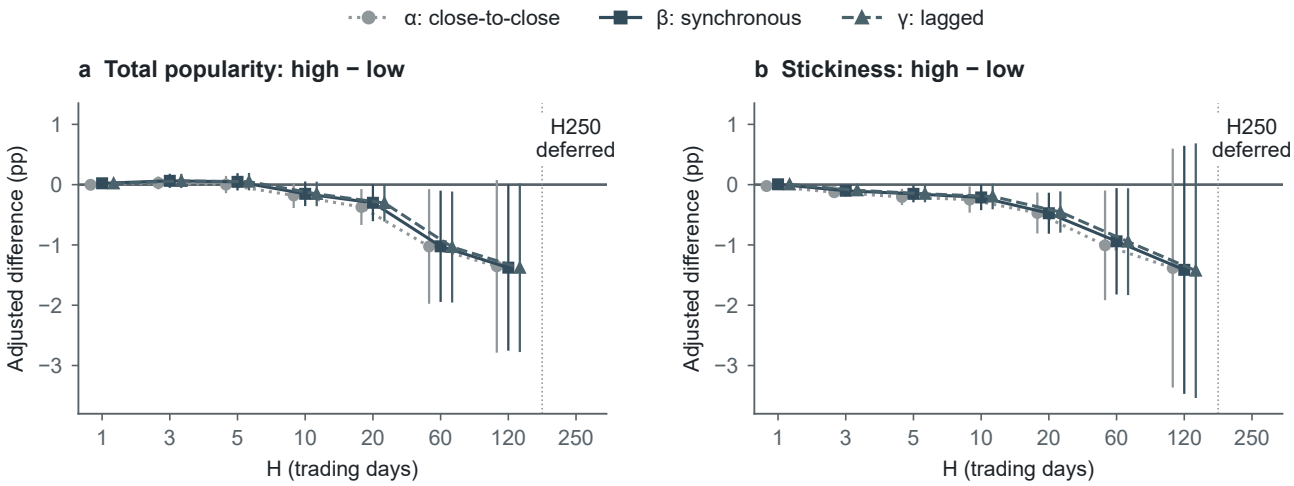
Small reference effects coexist with stable extreme lists. a, Mean daily cross-sectional standard deviations of full attention-rank changes, common-set changes and reference remainders, with a logarithmic y-axis. b, The fraction of all common stocks whose non-zero changes have opposite signs under the two definitions; zero changes are not counted as flips. c, Top20/Top100 overlaps for the two change definitions, with a clearly truncated y-axis beginning at 99.15%. All panels are descriptive date-weighted summaries for 1,071 dates ending on 31 August 2026. T/S/K denote total popularity, watchlist popularity and stickiness. The reference remainder follows from an algebraic identity rather than a causal adjustment or variance-explained decomposition. These lists are ranked by popularity changes and differ from the FW lists used for return evaluation; their overlap does not measure returns.

Supplementary Figure S10



Residual-association diagnostics under lagged membership γ. a–c show own total popularity, watchlist popularity and stickiness; d–f show leave-one-out concept-peer shares. Filled circles use the full-history window, with calibration before 22 October 2020 and six expanding test folds; 28 of 48 displayed cells are estimable. In the sampled panel, the watchlist field S first appears on 30 March 2022, after that calibration period. Open squares use the S-covered window, with calibration from 30 March 2022 through the day before 30 September 2022 and four test folds from that date to 27 August 2026. The S-covered window first intersects finite raw own and peer exposures; exposure-specific trimming then yields role-specific nuisance and evaluation supports. In this window, 42 of 48 displayed cells are estimable, including 14 own/peer S cells. Horizons without sufficient mature training or test labels are left unplotted, including H250, which does not reach the final test fold before the cutoff. All outcomes must mature by 31 August 2026. Points and whiskers are residual-projection coefficients and pointwise 95% date-HAC diagnostic intervals, in complete-H return percentage points per fixed past exposure standard deviation. No coefficient in either window passes its diagnostic 24-test Holm family. Source data retain all 288 design-exposure-horizon cells across both windows. Role-specific nuisance supports do not establish the usual orthogonal-score inference guarantee. Section S13.1 examines aligned support and control timing on a separate γ/H3 own-exposure sample. Deferred estimates are not zero effects.

Supplementary Figure S11



c States without a formal estimate

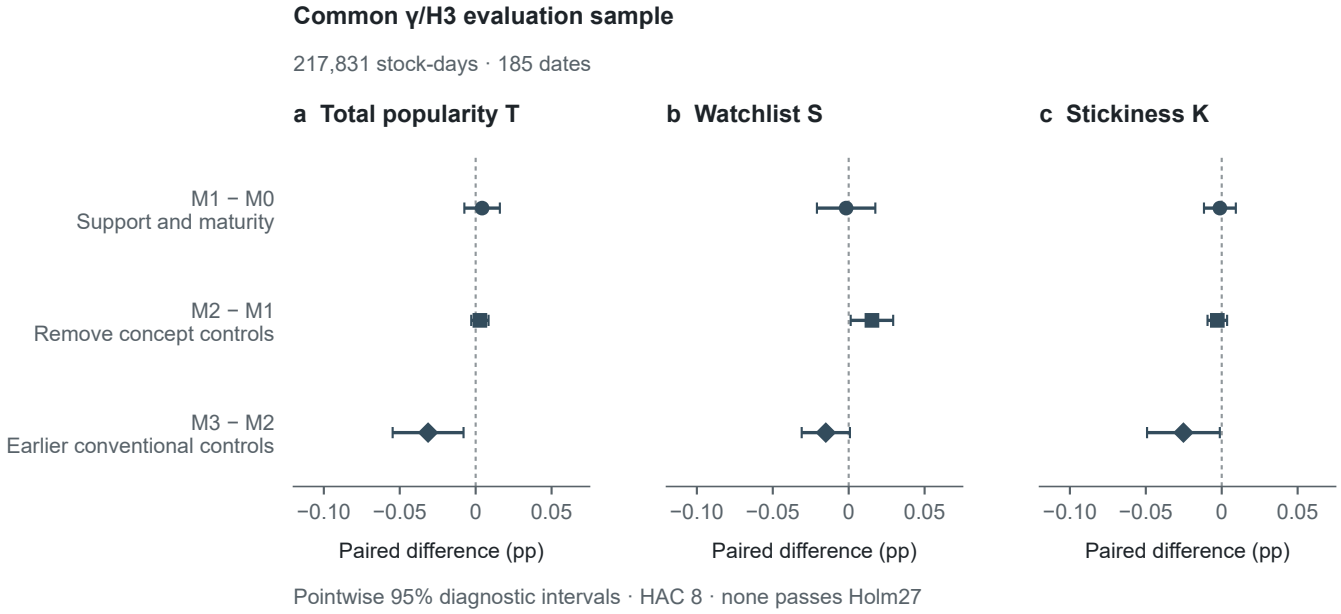
State contrast	Full-history window	S-covered window
Watchlist S: high − low	42–45% clipped; early training lacks S	42–45% clipped
Joint T/S/K: all high − all low	51–53% clipped; early training lacks S	51–53% clipped

Share of test propensities clipped to [0.05, 0.95], pooled over H1–H60 and α/β/γ.
Overlap criterion: at most 20% clipped; these contrasts exceed it.

Pointwise 95% date-HAC intervals. Symbols identify specifications, not corrected significance.

Adjusted high–low associations and states without a formal estimate. Circles, squares and triangles identify α, β and γ with distinct line patterns and horizontal offsets. a,b show estimable high-minus-low return differences for total popularity T and stickiness K in the full-history window. Units are complete-H percentage points; whiskers are pointwise 95% date-HAC intervals. 42 of 96 design-state-horizon cells are estimable. H250 lacks mature observations in the final test fold under the 31 August 2026 cutoff. Symbols identify designs and do not encode corrected significance. No state contrast in either window passes its 32-test Holm family. c reports the pooled share of clipped propensities at H1–H60 for watchlist S and the joint T/S/K state. These contrasts fail overlap in both windows; early full-history training folds also lack S. The S-covered window calibrates from 30 March 2022 through the day before 30 September 2022 and tests four folds from that date to 27 August 2026; 36 of 96 cells are estimable. Supplementary Table S11 reports support across all horizons. Both windows use fixed 80-iteration training and fit-fold industry encoding. These are related observational estimates, not identified causal effects.

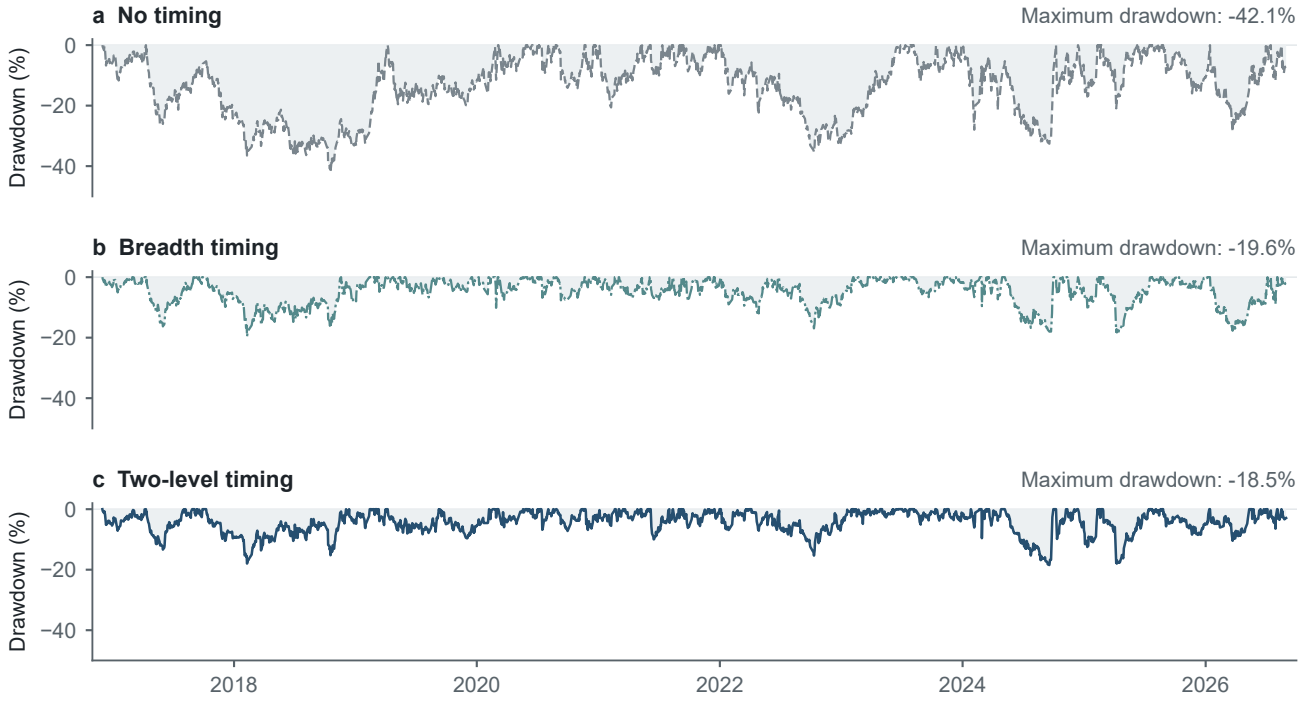
Supplementary Figure S12



Specification contrasts in conditional attention diagnostics. Panels a–c report own T, S and K separately; each compares M1−M0 (aligned auxiliary support and outcome-label purging), M2−M1 (removal of four concept controls) and M3−M2 (the same 22 conventional characteristics and industry moved before the recorded attention observation). Points and whiskers are paired coefficient differences and pointwise 95% score-based date-HAC diagnostic intervals at lag 8 sampled dates, using the same 217,831 γ/H3 stock-days and 185 dates. Units are complete-H3 return percentage points per the corresponding fixed historical exposure standard deviation. The fields have distinct definitions and scales; magnitudes across panels are not relative attention effects. The zero line denotes no change in a residual projection. The Holm27 family includes all nine contrasts at lags 1, 8 and 12; none passes. Intervals condition on the fitted nuisance predictions and do not establish standard orthogonal-score coverage. M1−M0 changes support and purging together. These contrasts measure specification sensitivity, not causal pathways or equivalence. Table S12 gives the underlying coefficients; the accompanying source data contain all 27 contrasts and 36 coefficient tests. Outcomes mature by 31 August 2026.

The complete drawdown histories of the three fixed configurations in Fig. 8a are shown in Supplementary Fig. S13.

Supplementary Figure S13



Complete drawdown histories for the fixed configuration in Fig. 8a. a–c, No timing, one-level breadth timing and two-level breadth timing, respectively, for Top10, three-day holding, F5/W10 and Pearson synchronization. All three panels retain every daily account observation and share dates and a −50% to 2% scale. Drawdown is net asset value divided by its running maximum minus one. Accounts run from 1 December 2016 to 31 August 2026, including commissions and stamp duty under the implementation settings in Supplementary Section S17. Slippage and market impact are not modeled. The main figure shows the two-level underwater path at a larger scale; all three maximum drawdowns remain in the complete 48-account comparison in Fig. 8b. The paths describe retrospective accounts whose target filter uses intraday information that must be obtained during actual trading (Supplementary Section S17).

Supplementary Table S1 | Sample description by year

Year	Signal dates	Pool stocks per date	Distinct stocks	Concepts per date	Median concept size	Memberships per stock	Median float cap (CNY bn)
2016 (partial)	7	1,740	1,783	148	21	2.0 (2)	7.31
2017	244	1,896	2,312	153	22	2.1 (2)	6.57
2018	243	2,300	2,593	166	25	2.5 (2)	4.48
2019	244	2,731	3,022	183	29.5	2.9 (2)	4.40
2020	243	3,160	3,504	206	36	3.5 (3)	4.65
2021	243	3,732	4,021	231	41	3.9 (3)	4.54
2022	242	4,301	4,614	295	45	4.8 (4)	4.35
2023	242	4,761	4,925	340	50	6.2 (5)	4.39
2024	242	4,994	5,075	370	59.5	7.9 (6)	3.88
2025	243	5,085	5,167	387	73	10.1 (8)	5.25
2026 (to 31 August)	160	5,140	5,212	385	83.5	11.3 (9)	6.02
All	2,353	3,755	5,228	266	—	6.1 (4)	4.73

Pool statistics use the frozen common research pool; pool stocks per date are equal-date means. Concepts per date count distinct labels in daily membership snapshots; median concept size is the median across snapshot dates of the within-date median number of member stocks per concept. Memberships per stock are scored lagged memberships of pool stocks, shown as mean (median) across stock-days. Float capitalization is the pool median in CNY billions. The first-level industry classification has 28 industries through 2020 and 31 from 2021. In the partial 2016 row, pool statistics cover seven trading dates from 22 to 30 December; concept-snapshot statistics cover 38 recorded snapshot dates from 24 November to 31 December. The 2026 row ends on 31 August. The snapshots contain 404 distinct concept labels in total.

Supplementary Table S2 | Mature signal dates by horizon

H	Historical rules	Daily adaptations	Rolling models	Last mature signal
1	2,351	2,331	1,613	2026-08-27
3	2,349	2,329	1,611	2026-08-25
5	2,347	2,327	1,609	2026-08-21
10	2,342	2,322	1,604	2026-08-14
20	2,332	2,312	1,594	2026-07-31
60	2,292	2,272	1,554	2026-06-04
120	2,232	2,212	1,494	2026-03-06
250	2,102	2,082	1,364	2025-08-18

Counts are mature paired signal dates, not independent stock observations. The three series begin on 22 December 2016, 20 January 2017 and 2 January 2020. Daily adaptations use the γ common pool. The observation cutoff is 31 August 2026, and every exit falls on or before that date. Selection overlap in Fig. 4c does not require outcome maturity and uses all 2,353 frozen dates; Fig. 4a,b,d use 2,349 mature dates.

Supplementary Table S3 | Multiplicity families and temporal support

Family	Composition	Total	Supported
Historical rules	2 labels × 2 formation-window conventions × 3 N × 8 H × 2 missingness × 5 controls	960	720
Same-information aggregation	2 labels × 3 N × 8 H × 2 missingness × 2 controls	192	144
Daily adaptations	2 labels × 3 N × 8 H × 2 missingness × 4 controls	384	288
Model increment	80 primary comparisons + 16 industry-conditioned IC	96	72
Six-pair parameter sensitivity (complete)	6 F/W × 3 H × 6 N × 5 controls	540	540
Six-pair parameter sensitivity (missing = 0)	Same grid; separately corrected sensitivity	540	540
Industry and size controls	3 N × 3 group benchmarks at H3	9	9
Full-training-date H3 sensitivity	2 models × 6 metrics × 2 block lengths	24	24
Strength block given W (E1)	3 endpoints × 2 block lengths; four stress cells not estimable	6	2
W conditional on market risk (S11)	3 paired contrasts × 2 block lengths	6	6
Paired membership and return timing	5 contrasts × 2 missingness conventions × 2 block lengths	20	20
Fixed-period H3 sensitivity	2 periods × 2 benchmarks; separately per block length	4	4 at L20/L40; 2 at L60
Paired H5 ridge list-width diagnostic	2 missingness conventions × 3 block lengths	6	6
Comovement bridge	3 attention fields × 3 temporal contrasts	9	0 formal

Supplementary Table S3 (continued)

Family	Composition	Total	Supported
Attention endpoints	3 measures × 8 H; block bootstrap	24	21
List visibility (headline)	2 list windows × 4 tasks	8	8
Attention–trading maximum-t	3 measures × 3 trading outcomes × 8 H × 2 directions	144	144
Reference-set reranking	3 fields × 8 H, per block specification	24	15
Attention rank-change differences	4 transformations × 8 H × 3 field pairs	96	60
Tree-DML temporal audit	6 families (own/peer × α/β/γ), each 3 exposures × 8 H	144	84
AIPW temporal audit	3 families (one per design), each 4 states × 8 H	96	42
S-coverage window: Tree-DML	Same structure, separate families	144	126
S-coverage window: AIPW	Same structure, separate families	96	36
N1 paired specification differences (S13.1)	3 adjacent contrasts × 3 own fields × 3 HAC lengths	27	27 diagnostic
N1 residual-projection coefficients (S13.1)	4 specifications × 3 own fields × 3 HAC lengths	36	36 diagnostic
Longer-block parameter checks	6 F/W × 5 controls at H3/Top20, per block length	30	30
Observation-row timing (S4.1)	5 contrasts × 8 H × 2 N, per missingness convention	80	80
Order-size composition	3 measures × 2 outcomes × 2 estimands × 8 H	96	0: turnover reconciliation not satisfied

Supported counts do not denote significant discoveries; two of the eight list-visibility comparisons meet the implemented joint headline criteria. Holm adjustment is applied separately to each declared family and does not control error over the full research history. Intervals are pointwise unless stated otherwise.

Marker semantics are specified per figure: stars in Fig. 2b encode Holm540 significance; filled points in the earlier horizon plots encode temporal support, not corrected significance. Monthly mappings receive no significance tests. The bridge fails its matching requirement; all nine tests are diagnostic only. Unsupported tests remain in their correction families with P = 1. Longer-block parameter-sensitivity checks use a separate 30-test H3/Top20 family at each block length. Family sizes recur across modules; in the text each family is named with its module. The joint feedback criterion and the order-size reporting requirements are rules, not Holm families.

The temporal-audit rows refer to the retained past-calibrated implementation diagnostics; full-period-preprocessing diagnostics are listed separately in Table S11. Failure to reconcile the amount fields is not an estimate of a zero effect.

Supplementary Table S4 | Current F5/W10 H3 Top20 timing cells

Cell	Recorded membership date	Score prices through	Return clock	Top20 return (%)	Pool return (%)	Top20 − pool (pp)	Observed selected / intended
α	≤ t	t close	close(t) → close(t+3)	0.426989	0.109995	0.316994	46,935 / 46,980
β	≤ t	t close	open(t+1) → open(t+4)	0.390197	0.113624	0.276573	46,912 / 46,980
γ	≤ preceding market day	t close	open(t+1) → open(t+4)	0.396856	0.113624	0.283232	46,914 / 46,980
δ (internal reference)	≤ preceding market day	t close	close(t) → close(t+3)	0.431245	0.109995	0.321250	46,940 / 46,980

Complete-observation, equal-date means on 2,349 common dates (22 December 2016–25 August 2026). Top20 and pool returns are percentages; their differences are percentage points (pp). The close references are not demonstrations of same-close execution. Cell means are descriptive; formal inference is attached to the paired contrasts in Supplementary Table S5.

Supplementary Table S5 | Paired timing contrasts and their shared Holm20 correction

Estimates and interval bounds are complete-H3 percentage points. C means complete observations; Z means missing outcomes set to zero at fixed denominators. All rows use 2,349 dates and 20,000 replicates. Intervals are pointwise rather than simultaneous; one Holm20 family includes every row.

Contrast	Missingness	Block L	Estimate (pp)	Pointwise 95% interval (pp)	Holm20 P
β − γ	C	20	−0.006659	[−0.023406, 0.010009]	1.000000
α − δ	C	20	−0.004256	[−0.020830, 0.012197]	1.000000
α − β	C	20	0.040421	[0.018988, 0.061803]	0.005000
δ − γ	C	20	0.038018	[0.016461, 0.059703]	0.007650
interaction	C	20	0.002403	[−0.003185, 0.008025]	1.000000
β − γ	Z	20	−0.006790	[−0.023389, 0.009829]	1.000000
α − δ	Z	20	−0.004367	[−0.020953, 0.012044]	1.000000
α − β	Z	20	0.040988	[0.019635, 0.062247]	0.005000
δ − γ	Z	20	0.038565	[0.017011, 0.060177]	0.007200
interaction	Z	20	0.002423	[−0.003098, 0.008049]	1.000000
β − γ	C	60	−0.006659	[−0.023528, 0.009632]	1.000000
α − δ	C	60	−0.004256	[−0.021383, 0.012449]	1.000000
α − β	C	60	0.040421	[0.018189, 0.062627]	0.007650
δ − γ	C	60	0.038018	[0.015457, 0.060430]	0.016249
interaction	C	60	0.002403	[−0.003714, 0.008576]	1.000000
β − γ	Z	60	−0.006790	[−0.023624, 0.009420]	1.000000
α − δ	Z	60	−0.004367	[−0.021527, 0.012328]	1.000000
α − β	Z	60	0.040988	[0.018816, 0.063014]	0.007650
δ − γ	Z	60	0.038565	[0.015977, 0.061029]	0.015399
interaction	Z	60	0.002423	[−0.003696, 0.008568]	1.000000

Supplementary Table S6 | Annual H3 Top20 selection spreads

Year	Pool dates	FW − pool (pp)	Joint dates	FW − joint (pp)
2016 (partial)	7	0.0666	7	0.1397
2017	244	0.2311	242	0.0454
2018	243	0.3386	242	0.2224
2019	244	0.2681	244	0.2119
2020	243	0.4852	243	0.2774
2021	243	0.3599	243	0.2019
2022	242	−0.0045	242	0.0711
2023	242	0.5757	242	0.2932
2024	242	−0.0704	242	−0.0317
2025	243	0.1645	243	0.1501
2026 (partial)	156	0.6040	156	0.2783

All estimates are descriptive means of complete three-day gross-return differences. Years 2016 and 2026 cover only the observed part of the year; the 2026 row ends with the 25 August 2026 signal, whose exit is on 31 August 2026. Joint comparisons match industry and daily capitalization decile and retain dates with all required group means. No annual significance search was performed.

Supplementary Table S7 | Direct contrasts in the main specification

Comparison	Mean difference	Pointwise 95% interval	Holm P
FW − F-only	0.2288	[0.1213, 0.3361]	0.02772 (960)
FW − W-only	0.1565	[0.0331, 0.2834]	1.0000 (960)
FW − A	0.0006	[−0.0104, 0.0127]	1.0000 (192)
FW − P	0.0060	[−0.0028, 0.0161]	1.0000 (192)
FW − CM5	0.2642	[0.1290, 0.3997]	0.01620 (384)
FW − CM20	0.3306	[0.1543, 0.5080]	0.04056 (384)
FW − BC10	1.2800	[1.0551, 1.5084]	0.00384 (384)
FW − BC60	1.1392	[0.8788, 1.4032]	0.00384 (384)

All differences are complete-H3 gross return percentage points, averaged equally over signal dates, not daily profits. γ, market-calendar formation windows, the next-open return clock, F5/W10 and Top20 are fixed. The first four rows use 2,349 mature dates (22 December 2016–25 August 2026); the last four use a stricter five-rule common pool and 2,329 dates (20 January 2017–25 August 2026). Lists are frozen and missing outcomes do not cause replacement. Intervals use 100,000 paired circular block resamples, block length 20 days. Parentheses identify the separate correction families, not a common 960+192+384 correction. CM is raw concept-return averaging; BC prioritizes regression slopes and then stock momentum. These daily adaptations are not the original published strategies.

Supplementary Table S8 | Full-training-date H3 paired increments

Block length 20.

Model	Outcome	Estimate	Pointwise 95% interval	Holm24 P
Ridge	Raw IC	0.001668	[−0.005248, 0.008495]	1.0000
Ridge	Industry IC	0.002869	[−0.000632, 0.006458]	1.0000
Ridge	Top20 observed	0.099154	[−0.003358, 0.203532]	1.0000
Ridge	Top20 zero	0.096657	[−0.003990, 0.198127]	1.0000
Ridge	Top100 observed	0.049920	[−0.011478, 0.112653]	1.0000
Ridge	Top100 zero	0.050248	[−0.011050, 0.113152]	1.0000
Boosting	Raw IC	0.003311	[−0.001637, 0.008211]	1.0000
Boosting	Industry IC	0.005414	[0.002725, 0.008111]	0.0024
Boosting	Top20 observed	0.010073	[−0.106148, 0.117153]	1.0000
Boosting	Top20 zero	0.009685	[−0.106431, 0.116002]	1.0000
Boosting	Top100 observed	0.028080	[−0.045440, 0.098302]	1.0000
Boosting	Top100 zero	0.028447	[−0.045095, 0.098503]	1.0000

Block length 60.

Model	Outcome	Estimate	Pointwise 95% interval	Holm24 P
Ridge	Raw IC	0.001668	[−0.004934, 0.008466]	1.0000
Ridge	Industry IC	0.002869	[−0.000675, 0.006600]	1.0000
Ridge	Top20 observed	0.099154	[−0.013499, 0.214830]	1.0000
Ridge	Top20 zero	0.096657	[−0.013519, 0.209075]	1.0000
Ridge	Top100 observed	0.049920	[−0.014507, 0.115713]	1.0000
Ridge	Top100 zero	0.050248	[−0.013976, 0.115785]	1.0000
Boosting	Raw IC	0.003311	[−0.002039, 0.008763]	1.0000
Boosting	Industry IC	0.005414	[0.002388, 0.008413]	0.0230
Boosting	Top20 observed	0.010073	[−0.103309, 0.111629]	1.0000
Boosting	Top20 zero	0.009685	[−0.102621, 0.110814]	1.0000
Boosting	Top100 observed	0.028080	[−0.053933, 0.105585]	1.0000
Boosting	Top100 zero	0.028447	[−0.053340, 0.105746]	1.0000

All rows compare enhanced minus ordinary versions of the same model on 1,611 dates (2 January 2020–25 August 2026). Return units are complete-three-day percentage points; IC units are correlations. Zero denotes a fixed-denominator missing-outcome convention. Holm24 includes both block lengths and all twelve contrasts. The two passing rows measure the same industry-conditioned boosting contrast.

Supplementary Table S9 | F and FW added jointly to W and ordinary predictors

Endpoint	Block	Mean difference	Pointwise 95% interval	Raw P	Holm6 P
All dates	20	−0.0225	[−0.1131, 0.0660]	0.6211	1.0000
All dates	60	−0.0225	[−0.1165, 0.0700]	0.6358	1.0000
Stress	20	NE	—	—	—
Stress	60	NE	—	—	—
Stress − nonstress	20	NE	—	—	—
Stress − nonstress	60	NE	—	—	—

Differences are augmented minus baseline Top20 mean gross returns in percentage points per full H3 period, averaged equally over formation dates. Primary means use observed outcomes on fixed lists. NE denotes not estimable because the stress-support gate failed; correction placeholders of one are not estimated P values. Pointwise intervals are unadjusted, and L20/L60 are dependence sensitivities rather than independent replications. This six-test family does not replace the earlier component or prediction families.

Supplementary Table S10 | Synchronization increments with and without observed market risk

Contrast	Block	Mean difference	Pointwise 95% interval	Raw P	Holm6 P
W increment: B−A	20	0.0742	[−0.0034, 0.1547]	0.0659	0.3373
Risk-conditioned W: D−C	20	0.0032	[−0.0790, 0.0871]	0.9405	1.0000
Direct change: (D−C)−(B−A)	20	−0.0711	[−0.1566, 0.0082]	0.0903	0.3373
W increment: B−A	60	0.0742	[−0.0007, 0.1518]	0.0562	0.3373
Risk-conditioned W: D−C	60	0.0032	[−0.0756, 0.0871]	0.9398	1.0000
Direct change: (D−C)−(B−A)	60	−0.0711	[−0.1524, 0.0049]	0.0755	0.3373

Differences are percentage points per complete H3 period, averaged equally over the same 1,611 formation dates. All four models include the same missing-risk summary. Pointwise intervals and family-adjusted P values answer different inferential questions. This predictive comparison is conditional on the specified feature sets and single-market proxy; it is not a causal risk decomposition.

Supplementary Table S11 | Support and correction in the temporal audit

Workflow and method	Planned	Estimable	Deferred	Holm	Secondary BH
Global preprocessing diagnostic: Tree-DML	144	84	60	0	6
Past calibration, fixed 80 rounds: Tree-DML	144	84	60	0	0
Global preprocessing diagnostic: AIPW	96	42	54	1	2
Training-fold encoding, fixed 80 rounds: AIPW	96	42	54	0	0
S-coverage window: Tree-DML	144	126	18	0	0
S-coverage window: AIPW	96	36	60	0	0

DML uses six separate 24-test families: own and peer exposures under each of α, β and γ. AIPW uses three separate 32-test families, one per design, each with four states and eight horizons. Counts are not based on single 144- or 96-test corrections. Deferred estimates remain blank and enter correction with placeholder $P = 1$. The Holm and secondary BH columns count diagnostic adjusted P or $q < 0.05$. Global preprocessing is diagnostic; the past-information and S-coverage protocols provide temporally calibrated conditional comparisons. Role-specific nuisance fitting supports in the retained DML implementation do not establish the standard orthogonal-score inference guarantee (Section S13). Signals end on 27 August 2026 and outcomes must mature by 31 August 2026. H250 has no mature final-fold test outcomes. The S-coverage branch calibrates from 30 March 2022 to before 30 September 2022 and evaluates four folds from 30 September 2022 on common finite T/S/K exposures. Its DML estimates extend through H120; AIPW at H120 fails the first-fold training-size gate within the high and low groups, and S/joint-state comparisons fail overlap. All branches share observations and are not independent replications.

Supplementary Table S12 | Own-popularity residual projections on a common γ/H3 sample

Specification	T	S	K
M0 · Original role-specific supports	+0.087976	−0.088081	−0.042007
M1 · Aligned support and label purging	+0.092221	−0.089768	−0.043204
M2 · Conventional controls at the signal date	+0.095007	−0.074457	−0.046100
M3 · Same controls before attention	+0.063678	−0.089530	−0.071321

Coefficients use the same 217,831 evaluation stock-days and 185 dates. Each column is a separate exposure projection, in complete-H3 return percentage points per its original historical standard deviation. M1 changes both auxiliary support and outcome-label purging; M2 removes four concept controls; M3 changes the date of the same remaining 22 characteristics and industry. Coefficient values do not change with the diagnostic HAC lag. The complete 36 coefficient tests and 27 paired-contrast tests include lags 1, 8 and 12 in their respective Holm families; none passes. Different exposure scales preclude interpreting cross-column magnitudes as relative attention effects.

Supplementary Table S13 | Implementation backtest grid

Holding	F/W	Similarity	N	Timing	CAGR (%)	Max DD (%)	Sharpe	Fees (%/yr)
3 days	5/10	Pearson	10	None	15.0	−42.1	0.63	8.0
3 days	5/10	Pearson	10	Breadth	21.9	−19.6	1.15	4.8
3 days	5/10	Pearson	10	Two-level	29.8	−18.5	1.44	5.1
3 days	5/10	Pearson	5	None	14.6	−48.8	0.60	8.2
3 days	5/10	Pearson	5	Breadth	23.2	−25.0	1.13	4.9
3 days	5/10	Pearson	5	Two-level	30.0	−20.9	1.35	5.2
3 days	5/10	CCC	10	None	21.7	−42.3	0.87	7.8
3 days	5/10	CCC	10	Breadth	24.9	−18.8	1.37	4.7
3 days	5/10	CCC	10	Two-level	29.8	−16.6	1.54	5.1
3 days	5/10	CCC	5	None	23.8	−46.5	0.89	8.1
3 days	5/10	CCC	5	Breadth	26.4	−20.7	1.35	4.8
3 days	5/10	CCC	5	Two-level	30.9	−17.1	1.49	5.2
3 days	10/20	Pearson	10	None	19.6	−42.3	0.74	6.3
3 days	10/20	Pearson	10	Breadth	26.6	−23.6	1.28	4.2
3 days	10/20	Pearson	10	Two-level	33.5	−19.2	1.52	4.6
3 days	10/20	Pearson	5	None	12.7	−59.6	0.53	6.5
3 days	10/20	Pearson	5	Breadth	22.1	−36.6	1.03	4.3
3 days	10/20	Pearson	5	Two-level	29.1	−26.1	1.25	4.6
3 days	10/20	CCC	10	None	21.8	−42.0	0.83	6.1
3 days	10/20	CCC	10	Breadth	26.0	−21.5	1.35	4.2
3 days	10/20	CCC	10	Two-level	31.6	−17.3	1.53	4.5
3 days	10/20	CCC	5	None	19.0	−43.9	0.72	6.4
3 days	10/20	CCC	5	Breadth	26.3	−26.0	1.26	4.3
3 days	10/20	CCC	5	Two-level	29.2	−19.8	1.32	4.6

Supplementary Table S13 (continued)

Holding	F/W	Similarity	N	Timing	CAGR (%)	Max DD (%)	Sharpe	Fees (%/yr)
5 days	5/10	Pearson	10	None	16.1	−37.8	0.67	5.4
5 days	5/10	Pearson	10	Breadth	19.9	−15.9	1.12	3.1
5 days	5/10	Pearson	10	Two-level	22.9	−16.9	1.22	3.3
5 days	5/10	Pearson	5	None	14.4	−43.7	0.60	5.5
5 days	5/10	Pearson	5	Breadth	20.3	−20.5	1.08	3.1
5 days	5/10	Pearson	5	Two-level	21.9	−23.6	1.11	3.3
5 days	5/10	CCC	10	None	21.4	−42.4	0.87	5.4
5 days	5/10	CCC	10	Breadth	22.3	−19.1	1.32	3.0
5 days	5/10	CCC	10	Two-level	24.3	−14.6	1.36	3.3
5 days	5/10	CCC	5	None	22.8	−43.0	0.87	5.4
5 days	5/10	CCC	5	Breadth	24.0	−18.2	1.33	3.1
5 days	5/10	CCC	5	Two-level	25.1	−15.0	1.32	3.3
5 days	10/20	Pearson	10	None	20.3	−39.7	0.77	4.5
5 days	10/20	Pearson	10	Breadth	23.7	−20.9	1.23	2.7
5 days	10/20	Pearson	10	Two-level	25.8	−17.7	1.28	3.0
5 days	10/20	Pearson	5	None	16.6	−54.1	0.64	4.6
5 days	10/20	Pearson	5	Breadth	20.7	−36.3	1.03	2.8
5 days	10/20	Pearson	5	Two-level	22.1	−28.2	1.05	3.0
5 days	10/20	CCC	10	None	21.5	−37.2	0.84	4.4
5 days	10/20	CCC	10	Breadth	24.1	−17.3	1.34	2.7
5 days	10/20	CCC	10	Two-level	24.7	−15.2	1.30	3.0
5 days	10/20	CCC	5	None	16.8	−41.6	0.67	4.6
5 days	10/20	CCC	5	Breadth	22.4	−20.9	1.18	2.7
5 days	10/20	CCC	5	Two-level	22.2	−17.6	1.12	3.0

Each row is an archived retrospective simulation with the execution-day full-session target filter (Section S17). Runs extend from 1 December 2016 (8 December for F10/W20) to 31 August 2026 with CNY 30,000,000, three or five staggered sub-portfolios, simulated execution at 09:55, a 1.2-basis-point commission, 0.1% stamp duty on sales and no slippage. N is the list size. CAGR uses the actual start and end dates, and maximum drawdowns (Max DD) are recomputed from daily net values. Sharpe ratios use a zero risk-free rate. Fees are commissions and stamp duty divided by mean assets, per year; mean exposure, commission-implied turnover and final net values of every run are in the source data. Breadth and two-level denote the one-level and two-level breadth timing rules (Supplementary Section S17). No inference is attached.

Supplementary Table S14 | Calendar-year returns of the reference configurations

Year	Top10 none	Top10 breadth	Top10 two-level	Top5 none	Top5 breadth	Top5 two-level	CSI 300	CSI 500	CSI 1000
2016 (December)	−6.1	−3.3	−5.4	−3.7	−3.2	−3.8	−7.2	−5.4	−6.2
2017	−17.4	8.5	9.8	−24.8	5.9	11.2	21.8	−0.2	−17.4
2018	−12.1	−0.6	5.9	−15.9	−3.7	6.1	−25.3	−33.3	−36.9
2019	31.7	51.0	29.1	29.1	52.4	30.4	36.1	26.4	25.7
2020	66.0	51.7	64.5	92.2	67.0	74.3	27.2	20.9	19.4
2021	40.9	16.5	40.5	28.1	12.0	31.5	−5.2	15.6	20.5
2022	−29.9	−6.7	−1.0	−42.3	−16.6	−11.6	−21.6	−20.3	−21.6
2023	82.3	52.6	55.1	109.7	70.8	76.3	−11.4	−7.4	−6.3
2024	−9.4	30.4	55.1	−5.6	37.6	57.4	14.7	5.5	1.2
2025	39.8	27.4	29.4	46.2	37.4	33.0	17.7	30.4	27.5
2026 (to 31 August)	15.4	5.0	25.7	17.3	0.8	16.5	−0.1	6.5	2.3

The archived accounts include the execution-day full-session target filter described in Section S17. Returns are expressed as percentages and calculated from daily net values after fees for the three-day holding-period, F5/W10, Pearson configurations. The 2016 row covers December and the 2026 row ends on 31 August. Index returns are price returns without dividends. Annual values are descriptive.

References

[1] Du, Q., Liang, D., Chen, Z. & Tu, J. Concept links and return momentum. *Journal of Banking & Finance* **134**, 106329 (2022). https://doi.org/10.1016/j.jbankfin.2021.106329

[2] Wahal, S. & Yavuz, M. D. Style investing, comovement and return predictability. *Journal of Financial Economics* **107**, 136–154 (2013). https://doi.org/10.1016/j.jfineco.2012.08.005

[3] Cohen, L. & Frazzini, A. Economic Links and Predictable Returns. *The Journal of Finance* **63**, 1977–2011 (2008). https://doi.org/10.1111/j.1540-6261.2008.01379.x

[4] Ge, S., Li, S. & Zheng, H. Diamond cuts diamond: News co-mention momentum spillover prevails in China. *Journal of Banking & Finance* **171**, 107356 (2025). https://doi.org/10.1016/j.jbankfin.2024.107356

[5] Avramov, D., Ge, S., Li, S. & Linton, O. Dual peer effects and cross-stock predictability. *Journal of Financial Economics* **180**, 104274 (2026). https://doi.org/10.1016/j.jfineco.2026.104274

[6] Jegadeesh, N. & Titman, S. Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency. *The Journal of Finance* **48**, 65–91 (1993). https://doi.org/10.1111/j.1540-6261.1993.tb04702.x

[7] Jegadeesh, N. Evidence of Predictable Behavior of Security Returns. *The Journal of Finance* **45**, 881–898 (1990). https://doi.org/10.1111/j.1540-6261.1990.tb05110.x

[8] Barberis, N. & Shleifer, A. Style investing. *Journal of Financial Economics* **68**, 161–199 (2003). https://doi.org/10.1016/S0304-405X(03)00064-3

[9] Peng, L. & Xiong, W. Investor attention, overconfidence and category learning. *Journal of Financial Economics* **80**, 563–602 (2006). https://doi.org/10.1016/j.jfineco.2005.05.003

[10] Bhui, R. & Jiao, P. Attention Constraints and Learning in Categories. *Management Science* **69**, 5394–5404 (2023). https://doi.org/10.1287/mnsc.2023.4803

[11] Da, Z., Engelberg, J. & Gao, P. In Search of Attention. *The Journal of Finance* **66**, 1461–1499 (2011). https://doi.org/10.1111/j.1540-6261.2011.01679.x

[12] Barber, B. M., Huang, X., Odean, T. & Schwarz, C. Attention-Induced Trading and Returns: Evidence from Robinhood Users. *The Journal of Finance* **77**, 3141–3190 (2022). https://doi.org/10.1111/jofi.13183

[13] Gu, S., Kelly, B. & Xiu, D. Empirical Asset Pricing via Machine Learning. *The Review of Financial Studies* **33**, 2223–2273 (2020). https://doi.org/10.1093/rfs/hhaa009

[14] Gelman, A. & Stern, H. The Difference Between "Significant" and "Not Significant" is not Itself Statistically Significant. *The American Statistician* **60**, 328–331 (2006). https://doi.org/10.1198/000313006X152649

[15] Fedyk, A. Front-Page News: The Effect of News Positioning on Financial Markets. *The Journal of Finance* **79**, 5–33 (2024). https://doi.org/10.1111/jofi.13287

[16] Harvey, C. R., Liu, Y. & Zhu, H. … and the Cross-Section of Expected Returns. *The Review of Financial Studies* **29**, 5–68 (2016). https://doi.org/10.1093/rfs/hhv059

[17] White, H. A Reality Check for Data Snooping. *Econometrica* **68**, 1097–1126 (2000). https://doi.org/10.1111/1468-0262.00152

[18] Bang, H. & Robins, J. M. Doubly Robust Estimation in Missing Data and Causal Inference Models. *Biometrics* **61**, 962–972 (2005). https://doi.org/10.1111/j.1541-0420.2005.00377.x

[19] Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W. & Robins, J. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal* **21**, C1–C68 (2018). https://doi.org/10.1111/ectj.12097

[20] Politis, D. N. & White, H. Automatic Block-Length Selection for the Dependent Bootstrap. *Econometric Reviews* **23**, 53–70 (2004). https://doi.org/10.1081/ETC-120028836

[21] Patton, A., Politis, D. N. & White, H. Correction to "Automatic Block-Length Selection for the Dependent Bootstrap" by D. Politis and H. White. *Econometric Reviews* **28**, 372–375 (2009). https://doi.org/10.1080/07474930802459016